



Ministry of Higher Education and Scientific Research
Middle Technical University
Electrical Engineering Technical College
Department of Cybersecurity Technology Engineering



General Physics

First Stage

By

Lecturer: Dr. Thamir. Al-hamadany



*Ministry of Higher Education and Scientific Research
Middle Technical University
Electrical Engineering Technical College
Department of Cybersecurity Technology Engineering*



General Physics

First Stage

Lecture One

By

Lecturer: Dr. Thamir. Al-hamadany

1. Introduction Physical Quantities, Unites and Measurement

Like all other sciences, physics is based on experimental observations and quantitative measurements. The main objectives of physics are to identify a limited number of fundamental laws that govern natural phenomena and use them to develop theories that can predict the results of future experiments. The fundamental laws used in developing theories are expressed in the language of mathematics, the tool that provides a bridge between theory and experiment. In 1960, an international committee established a set of standards for the fundamental quantities of science. It is called the SI (System International), and its units of **length**, **mass**, and **time** are the **meter**, **kilogram**, and **second**, respectively. Other **SI** standards established by the committee are those for **temperature (the kelvin)**, **electric current (the ampere)**, **luminous intensity (the candela)**, and the **amount of substance (the mole)**, as shown below in Table (1-1).

Quantity	SI Unit (symbol)
Length	Meter (m)
Mass	Kilogram (Kg)
Time	Second (s)
Temperature	Kelvin (K)
Electric Current	Ampere (A)
Luminous Intensity	Candela (Cd)
Amount of Substance	Mole (mol)

In mechanics, three fundamental quantities are used: **Length, Mass, Time** All other quantities in mechanics can be expressed in terms of the three fundamental quantities. Derived quantities can be expressed as a mathematical combination of fundamental quantities. **Examples:**

Area: A product of two lengths.

Speed: A ratio of a length to a time interval.

Density: A ratio of mass to volume.

1.2 Prefixes: Prefixes correspond to powers of 10. Each prefix has a specific name and has a specific abbreviation. The prefixes can be used with any basic units. They are multipliers of the basic unit.

Examples: $1 \text{ mm} = 10^{-3} \text{ m}$ & $1 \text{ mg} = 10^{-3} \text{ g}$

Table (1-2): Prefixes for powers of ten.

Power	Prefix	Abbreviation	Power	Prefix	Abbreviation
10^{-24}	yocto	y	10^3	kilo	k
10^{-21}	zepto	z	10^6	mega	M
10^{-18}	atto	a	10^9	giga	G
10^{-15}	femto	f	10^{12}	tera	T
10^{-12}	pico	p	10^{15}	peta	P
10^{-9}	nano	n	10^{18}	exa	E
10^{-6}	micro	μ	10^{21}	zetta	Z
10^{-3}	milli	m	10^{24}	yotta	Y
10^{-2}	centi	c			
10^{-1}	deci	d			

1.3 Dimensions and Units

Each dimension can have many actual units as shown in Table (1-3).

Quantity	Area (<i>A</i>)	Volume (<i>V</i>)	Speed (<i>v</i>)	Acceleration (<i>a</i>)
Dimensions	L^2	L^3	L/T	L/T^2
SI units	m^2	m^3	m/s	m/s^2
U.S. customary units	ft^2	ft^3	ft/s	ft/s^2

Quantity	SI Unit		Dimension
velocity	m/s	ms^{-1}	LT^{-1}
acceleration	m/s^2	ms^{-2}	LT^{-2}
force	N $kg\ m/s^2$	$kg\ ms^{-2}$	MLT^{-2}
energy (or work)	Joule J N m, $kg\ m^2/s^2$	$kg\ m^2s^{-2}$	ML^2T^{-2}
power	Watt W N m/s $kg\ m^2/s^3$	Nms^{-1} $kg\ m^2s^{-3}$	ML^2T^{-3}
pressure (or stress)	Pascal P, N/m^2 , $kg/m/s^2$	Nm^{-2} $kg\ m^{-1}s^{-2}$	$ML^{-1}T^{-2}$
density	kg/m^3	$kg\ m^{-3}$	ML^{-3}

1.4 Dimensional Analysis

The word **dimension** has a special meaning in physics. It denotes the physical nature of a quantity. Whether a distance is assured in units of feet or meters or fathoms, it is still a distance. We say its dimension is **length**. The symbols we use to specify the dimensions of **Length, Mass, and Time** are **L, M, and T**, respectively. We shall often use brackets [] to denote the dimensions of a physical quantity. For example, the symbol we use for **speed** is **v**, and in our notation the dimensions of speed are written $[\mathbf{v}] = \mathbf{L}/\mathbf{T}$. As another example, the dimensions of **area A** are $[\mathbf{A}] = \mathbf{L}^2$. The dimensions and units of **area, volume, speed, and acceleration** are listed in Table 1.3. **Dimensional analysis** makes use of the fact that dimensions can be treated as algebraic quantities. For example, quantities can be added or **subtracted** only if they have the same dimensions. Furthermore, the terms on both sides of an equation must have the same dimensions. By following these simple rules, you can use dimensional analysis to help determine whether an expression has the correct form. The **relationship can be correct only if** the dimensions on both sides of the equation are the same. To illustrate this procedure, suppose you wish to derive an equation for the **position x** of a car at a time **t** if the car starts from rest and moves with constant acceleration **a**. we shall find that the correct expression is $\mathbf{x} = \frac{1}{2} \mathbf{a} \mathbf{t}^2$. Let us use dimensional analysis to check the validity of this expression. The quantity **x** on the left side has the dimension of length. For the equation to be dimensionally, correct, the quantity on the right side must also have the dimension of length. We can perform a dimensional check by substituting the dimensions for acceleration, $\mathbf{L}/\mathbf{T}^2$ (Table 1.3), and time, **T**, into the equation. That is, the dimensional form of the equation $\mathbf{x} = \frac{1}{2} \mathbf{a} \mathbf{t}^2$ is

$$\mathbf{L} = \frac{\mathbf{L}}{\mathbf{T}^2} \cdot \mathbf{T}^2 = \mathbf{L}$$

The dimensions of time cancel as shown, leaving the dimension of length on the right hand

Side so:

- The equation is dimensionally correct.
- There are no dimensions for the constant.

1.5 Conversion of Units

Sometimes it is necessary to convert units from one measurement system to another, or to convert within a system, for example, from kilometers to meters. Equalities between SI and U.S. customary units of length are as follows:

$$1 \text{ mile} = 1\,609 \text{ m} = 1.609 \text{ km} \quad 1 \text{ ft} = 0.304\,8 \text{ m} = 30.48 \text{ cm}$$

$$1 \text{ m} = 39.37 \text{ in.} = 3.281 \text{ ft} \quad 1 \text{ in.} = 0.025\,4 \text{ m} = 2.54 \text{ cm (exactly)}$$

Area

$$1 \text{ m}^2 = 10^4 \text{ cm}^2 = 10.76 \text{ ft}^2$$

$$1 \text{ ft}^2 = 0.092\,9 \text{ m}^2 = 144 \text{ in.}^2$$

$$1 \text{ in.}^2 = 6.452 \text{ cm}^2$$

Volume

$$1 \text{ m}^3 = 10^6 \text{ cm}^3 = 6.102 \times 10^4 \text{ in.}^3$$

$$1 \text{ ft}^3 = 1\,728 \text{ in.}^3 = 2.83 \times 10^{-2} \text{ m}^3$$

$$1 \text{ L} = 1\,000 \text{ cm}^3 = 1.057\,6 \text{ qt} = 0.035\,3 \text{ ft}^3$$

$$1 \text{ ft}^3 = 7.481 \text{ gal} = 28.32 \text{ L} = 2.832 \times 10^{-2} \text{ m}^3$$

$$1 \text{ gal} = 3.786 \text{ L} = 231 \text{ in.}^3$$

Mass

$$1\,000 \text{ kg} = 1 \text{ t (metric ton)}$$

$$1 \text{ slug} = 14.59 \text{ kg}$$

$$1 \text{ u} = 1.66 \times 10^{-27} \text{ kg} = 931.5 \text{ MeV}/c^2$$

Length

$$1 \text{ in.} = 2.54 \text{ cm (exact)}$$

$$1 \text{ m} = 39.37 \text{ in.} = 3.281 \text{ ft}$$

$$1 \text{ ft} = 0.304\,8 \text{ m}$$

$$12 \text{ in.} = 1 \text{ ft}$$

$$3 \text{ ft} = 1 \text{ yd}$$

$$1 \text{ yd} = 0.914\,4 \text{ m}$$

$$1 \text{ km} = 0.621 \text{ mi}$$

$$1 \text{ mi} = 1.609 \text{ km}$$

$$1 \text{ mi} = 5\,280 \text{ ft}$$

$$1 \mu\text{m} = 10^{-6} \text{ m} = 10^3 \text{ nm}$$

$$1 \text{ lightyear} = 9.461 \times 10^{15} \text{ m}$$

Force

$$1 \text{ N} = 0.2248 \text{ lb}$$

$$1 \text{ lb} = 4.448 \text{ N}$$

Velocity

$$1 \text{ mi/h} = 1.47 \text{ ft/s} = 0.447 \text{ m/s} = 1.61 \text{ km/h}$$

$$1 \text{ m/s} = 100 \text{ cm/s} = 3.281 \text{ ft/s}$$

$$1 \text{ mi/min} = 60 \text{ mi/h} = 88 \text{ ft/s}$$

Acceleration

$$1 \text{ m/s}^2 = 3.28 \text{ ft/s}^2 = 100 \text{ cm/s}^2$$

$$1 \text{ ft/s}^2 = 0.3048 \text{ m/s}^2 = 30.48 \text{ cm/s}^2$$

Pressure

$$1 \text{ bar} = 10^5 \text{ N/m}^2 = 14.50 \text{ lb/in.}^2$$

$$1 \text{ atm} = 760 \text{ mm Hg} = 76.0 \text{ cm Hg}$$

$$1 \text{ atm} = 14.7 \text{ lb/in.}^2 = 1.013 \times 10^5 \text{ N/m}^2$$

$$1 \text{ Pa} = 1 \text{ N/m}^2 = 1.45 \times 10^{-4} \text{ lb/in.}^2$$

Time

$$1 \text{ yr} = 365 \text{ days} = 3.16 \times 10^7 \text{ s}$$

$$1 \text{ day} = 24 \text{ h} = 1.44 \times 10^3 \text{ min} = 8.64 \times 10^4 \text{ s}$$

Energy

$$1 \text{ J} = 0.738 \text{ ft} \cdot \text{lb}$$

$$1 \text{ cal} = 4.186 \text{ J}$$

$$1 \text{ Btu} = 252 \text{ cal} = 1.054 \times 10^3 \text{ J}$$

$$1 \text{ eV} = 1.602 \times 10^{-19} \text{ J}$$

$$1 \text{ kWh} = 3.60 \times 10^6 \text{ J}$$

Power

$$1 \text{ hp} = 550 \text{ ft} \cdot \text{lb/s} = 0.746 \text{ kW}$$

$$1 \text{ W} = 1 \text{ J/s} = 0.738 \text{ ft} \cdot \text{lb/s}$$

$$1 \text{ Btu/h} = 0.293 \text{ W}$$

Quiz 1. True or False: Dimensional analysis can give you the numerical value of constants of proportionality that may appear in an algebraic expression.

Summary

The three fundamental physical quantities of mechanics are **length, mass, and time**, which in the SI system have the units meters (m), kilograms (kg), and seconds (s), respectively. Prefixes indicating various powers of ten are used with these three basic units. The **density** of a substance is defined as its mass per unit volume: $\rho = \frac{M}{V}$. Different substances have different densities mainly because of differences in their atomic masses and atomic arrangements. The method of dimensional analysis is very powerful in solving physics problems. Dimensions can be treated as algebraic quantities. By making estimates and performing order-of-

magnitude calculations, you should be able to approximate the answer to a problem when there is not enough information available to completely specify an exact solution. When you compute a result from several measured numbers, each of which has a certain accuracy, you should give the result with the correct number of significant figures. When multiplying several quantities, the number of significant figures in the final answer is the same as the number of significant figures in the quantity having the lowest number of significant figures. The same rule applies to division. When numbers are added or subtracted, the number of decimal places in the result should equal the smallest number of decimal places of any term in the sum.

Length: The meter is defined as the distance traveled by light during a precisely specified time interval.

Time: The second is defined in terms of the oscillations of light emitted by an atomic (cesium-133) source. Accurate time signals are sent worldwide by radio signals keyed to atomic clocks in standardizing laboratories.

Mass: The kilogram is defined in terms of a platinum–iridium standard mass kept near Paris. For measurements on an atomic scale, the atomic mass unit, defined in terms of the atom carbon-12, is usually used.

Density The density ρ of a material is the mass per unit volume:

Note: 2π radians = 360°

QUESTIONS

1. What is the SI unit of surface tension?
2. The temperature of two bodies measured by a thermometer are $t_1 = 20^\circ\text{C} \pm 0.5^\circ\text{C}$ and $t_2 = 50^\circ\text{C} \pm 0.5^\circ\text{C}$. The maximum absolute error in temperature difference is:
(a) $\pm 1^\circ\text{C}$ (b) $\pm 2^\circ\text{C}$ (c) $\pm 3^\circ\text{C}$ (d) $\pm 4^\circ$.

3. The following observations were taken for determining surface tension T of water by capillary method. Diameter of capillary, $d = 1.25 \times 10^{-2}$ m, rise of water, $h = 1.45 \times 10^{-2}$ m. Using $g = 9.80 \text{ m/s}^2$ and the simplified relation $T = \frac{r h g}{2} \times 10^3 \text{ N/m}^2$, the possible error in surface tension is closest to:
- (a) 1.5% (b) 2.4%
(c) 10% (d) 0.15%
4. In a simple pendulum, experiment for determination of acceleration due to gravity (g), time taken for 20 oscillations is measured by using a watch of 1 second least count. The mean value of time taken comes out to be 30 s. The length of pendulum is measured by using a meter scale of least count 1 mm and the value obtained 55.0 cm. The percentage error in the determination of g is close to:
- (a) 0.7% (b) 6.8% (c) 3.5% (d) 0.2%
5. The time period T of simple pendulum depends upon length l of the pendulum and Gravitational acceleration. The formula for time period of simple pendulum is given by:
- (a) $T = 2\pi \sqrt{\frac{l}{g}}$ (b) $T = 2\pi \sqrt{\frac{g}{l}}$ (c) $T = \frac{1}{2\pi} \sqrt{l g}$ (d) $T = \frac{2\pi l}{g}$
6. The volume of a wallet is 8.50 in.^3 . Convert this value to m^3 , using the definition $1 \text{ in.} = 2.54 \text{ cm}$.



Ministry of Higher Education and Scientific Research
Middle Technical University
Electrical Engineering Technical College
Department of Cybersecurity Technology Engineering



General Physics

Lecture Two

Kinematics and Dynamics

First Stage

By

Lecturer: Dr. Thamir. Al-hamadany

2. Kinematics and Dynamics

The motion of objects; baseballs, automobiles, joggers, and even the Sun and Moon is an obvious part of everyday life. It was not until the sixteenth and seventeenth centuries that our modern understanding of motion was established. Many individuals contributed to this understanding, particularly Galileo Galilei (1564–1642) and Isaac Newton (1642–1727). The study of the motion of objects, and the related concepts of force and energy, form the field called mechanics. Mechanics is customarily divided into two parts: **Kinematics** is one of the branch of Mechanics. It describes the motion of objects without looking at the cause of the motion, and **dynamics**, which deals with force and why objects move as they do.

2.1 Reference Frames and Displacement

Any measurement of position, distance, or speed must be made with respect to a **reference frame** or **frame of reference**. For example, while you are on a train traveling at 80 km/h, suppose a person walks past you toward the front of the train at a speed of, say 5 km/h (Fig. 2.1). This 5 km/h is the person's speed with respect to the train as frame of reference. With respect to the ground, that person is moving at a speed of $80 \text{ km/h} + 5 \text{ km/h} = 85 \text{ km/h}$. It is always important to specify the frame of reference when stating a speed. In everyday life, we usually mean “with respect to the Earth” without even thinking about it, but the reference frame must be specified whenever there might be confusion. Figure (2.1) a person walks toward the front of a train at 5 km/h. The train is moving 80 km/h with respect to the ground, so the walking person's speed, relative to the ground, is 85 km / h.

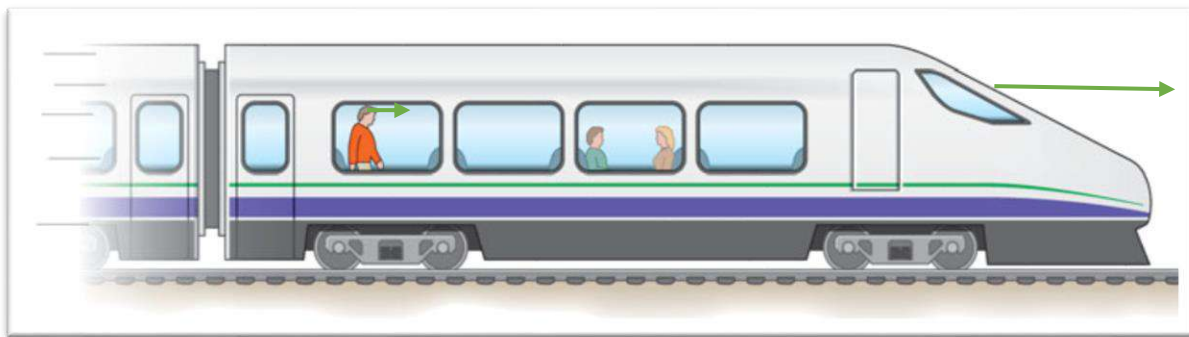
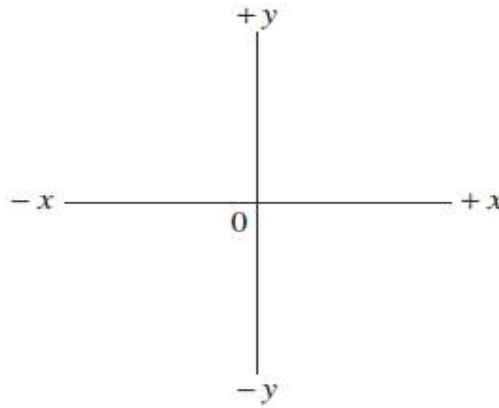


Figure (2.1)

When specifying the motion of an object, it is important to specify not only the speed but also the direction of motion. Often we can specify a direction by using north, east, south, and west, and by “up” and “down.” In physics, we often draw a set of **coordinate axes**, as shown in Fig. (2.2), to represent a frame of reference. We can always place the origin 0, and the directions of the x and y axes, as we like for convenience. The x and y -axes are always perpendicular to each other. The **origin** is where Objects positioned to the right of the origin of coordinates (0) on the x -axis have an x coordinate which we usually choose to be positive; then points to the left of 0 have a negative x coordinate. The position along the y -axis is usually considered positive when above 0, and negative when below 0, although the reverse convention can be used if convenient. Any point on the plane can be specified by giving its x and y coordinates. In three dimensions, a z -axis perpendicular to the x and y -axes is added. For one-dimensional motion, we often choose the x -axis as the line along which the motion takes place. Then the **position** of an object at any moment is given by its x coordinate. If the motion is vertical, as for a dropped object, we usually use the y -axis. $x = 0$, $y = 0$. Figure (2.2) Standard set of xy coordinate axes, sometimes called “rectangular coordinates.”

Fig 2.2

We need to make a distinction between the ***distances*** an object has traveled and its **displacement**, which is defined as the *change in position* of the object. That is, *displacement is how far the object is from its starting point*. To see the distinction between total distance and displacement, imagine a person walking 70 m to the east and then turning around and walking back (west) a distance of 30 m (see Fig. 2.3). The total *distance* traveled is 100 m, but the *displacement* is only 40 m since the person is now only 40 m from the starting point. Displacement is a quantity that has both magnitude and direction. Such quantities are called **vectors**, and are represented by arrows in diagrams. For example, in Fig. 2.3, the blue arrow represents the displacement whose magnitude is 40 m and whose direction is to the right (east).

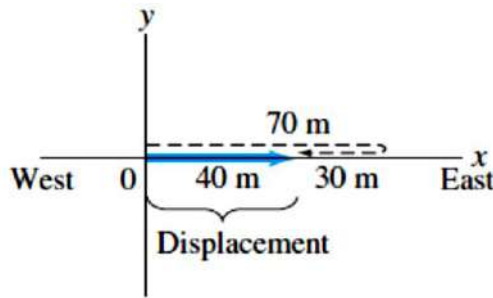


Figure (2.3) a person walks 70 m east, then 30 m west. The total distance traveled is 100 m (path is shown dashed in black); but the displacement, shown as a solid blue arrow, is 40 m to the east.

Vectors, which point in one direction, will be positive (typically to the right along the x -axis). Vectors that point in the opposite direction will have a negative sign in front of their magnitude. Consider the motion of an object over a particular time interval. Suppose that at some initial time, call it t_1 , the object is on the x -axis at the position x_1 in the coordinate system shown in Fig. 2.4. At some later time t_2 , suppose the object has moved to position x_2 . The displacement of our object is $x_2 - x_1$, and is represented by the arrow pointing to the right in Fig. 2.4. It is convenient to write $\Delta x = x_2 - x_1$, where the symbol Δ (Greek letter delta) means, “change in.” Then means “the change in x ,” or “change in position,” which is the displacement. The **change in** any quantity means *the final value of that quantity, minus the initial value*. Suppose $x_1 = 10.0$ m and $x_2 = 30.0$ m as in Fig. 2.4. Then $\Delta x = x_2 - x_1 = 30.0$ m - 10.0 m = 20.0 m, so the displacement is 20.0 m in the positive direction, Fig. 2.4. Now consider an object moving to the left as shown in Fig. 2.5. Here the object, a person, starts at $x_1 = 30.0$ m and walks to the left to the point $x_2 = 10.0$ m. In this case, her displacement is $\Delta x = x_2 - x_1 = 10.0$ m - 30.0 m = -20.0 m, moreover, the blue arrow representing the vector displacement points to the left. For one-dimensional motion along the x -axis, a vector pointing to the right is positive, whereas a vector pointing to the left has a negative sign.

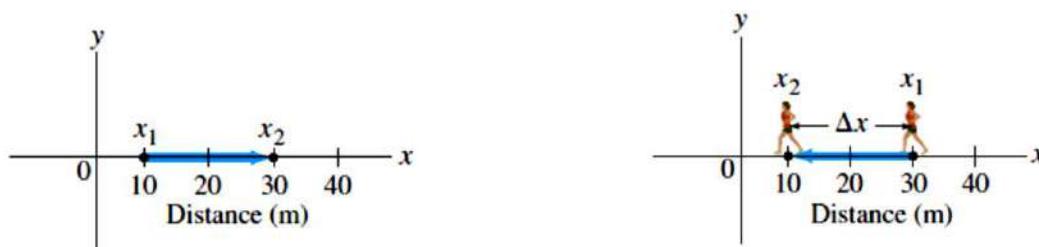


Figure (2.4) the arrow represents the displacement Distances are in meters $x_2 - x_1$.

Figure (2.5) for the displacement $\Delta x = x_2 - x_1 = 10.0 \text{ m} - 30.0 \text{ m}$ the displacement vector points left.

Exercise: An ant starts at $x = 20 \text{ cm}$ on a piece of graph paper and walks along the x -axis to it then turns around and walks back to determine (a) the ant's displacement and (b) the total distance traveled.

2.2 Average Velocity

An important aspect of the motion of a moving object is how *fast* it is moving its **speed or velocity**. The term “**speed**” refers to how far an object travels in a given time interval, regardless of direction. If a car travels 240 kilometers (km) in 3 hours (h), we say its average speed was In general, the **average speed** of an object is defined as *the total distance traveled along its path divided by the time it takes to travel this distance* as shown in equation (2.1):

$$\text{average speed} = \frac{\text{distance traveled}}{\text{time elapsed}}. \quad \dots\dots\dots (2.1)$$

The terms “**velocity**” and “**speed**” are often used interchangeably in ordinary language. However, in physics, we make a distinction between the two. **Speed** is simply a positive number, with units. **Velocity**, on the other hand, is used to signify both the *magnitude* (numerical value) of how fast an object is moving and the *direction* in which it is moving. **Velocity** is therefore a **vector**. There is a second difference between speed and velocity:

namely, the **average velocity** is defined in terms of **displacement**, rather than total distance traveled:

$$\text{average velocity} = \frac{\text{displacement}}{\text{time elapsed}} = \frac{\text{final position} - \text{initial position}}{\text{time elapsed}}.$$

Average speed and average velocity have the same magnitude when the motion is all in one direction. In other cases, they may differ: recall the walk we described earlier, in Fig. 2.2, where a person walked 70 m east and then 30 m west.

The total distance traveled was 70 m + 30 m = 100 m, but the displacement was 40 m. suppose this walk took 70 s to complete. Then the average speed was:

$$\frac{\text{distance}}{\text{time elapsed}} = \frac{100 \text{ m}}{70 \text{ s}} = 1.4 \text{ m/s}.$$

The magnitude of the average velocity, on the other hand, was:

$$\frac{\text{displacement}}{\text{time elapsed}} = \frac{40 \text{ m}}{70 \text{ s}} = 0.57 \text{ m/s}.$$

To discuss one-dimensional motion of an object in general, suppose that at some moment in time, call it t_1 the object is on the x axis at position x_1 in a coordinate system, and at some later time t_2 , suppose it is at position x_2 . The **elapsed time** (= change in time) is $\Delta t = t_2 - t_1$; during this time interval the displacement of our object is $\Delta x = x_2 - x_1$. Then the **average velocity**, defined as *the displacement divided by the elapsed time*, can be written:

$$\bar{v} = \frac{x_2 - x_1}{t_2 - t_1} = \frac{\Delta x}{\Delta t}, \quad \dots\dots\dots (2.2)$$

Where v stands for velocity and the bar (-) over the v is a standard symbol meaning “average.” For one-dimensional motion in the usual case of the $+x$ axis to the right, note that if x_2 is less

than x_1 , the object is moving to the left, and then $\Delta x = x_2 - x_1$ is less than zero. The sign of the displacement, and thus of the average velocity, indicates the direction: the average velocity is positive for an object moving to the right along the x -axis and negative when the object moves to the left. The direction of the average velocity is always the same as the direction of the displacement. It is always important to choose (and state) the *elapsed time*, or **time interval**, $t_2 - t_1$ the time that passes during our chosen period of observation.

Example2.1: Runner's average velocity. The position of a runner as a function of time is plotted as moving along the x -axis of a coordinate system. During a 3.00-s time interval, the runner's position changes from $x_1 = 30.5$ m, as shown in Fig. 2.6. What is the runner's average velocity?

Solution The displacement is:

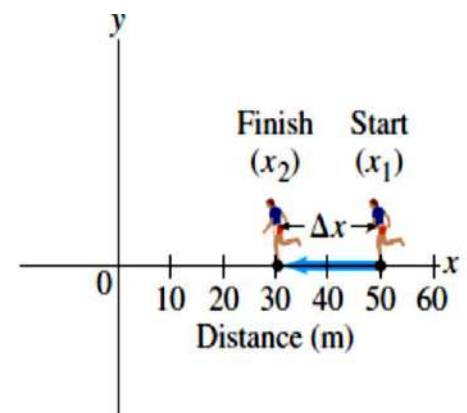
$$\begin{aligned}\Delta x &= x_2 - x_1 \\ &= 30.5 \text{ m} - 50.0 \text{ m} = -19.5 \text{ m}.\end{aligned}$$

The elapsed time, or time interval, is given as $\Delta t = 3.00$ s. The average velocity is

$$\bar{v} = \frac{\Delta x}{\Delta t} = \frac{-19.5 \text{ m}}{3.00 \text{ s}} = -6.50 \text{ m/s}.$$

The displacement and average velocity are negative, which tells us that the runner is moving to the left along the x -axis, as indicated by the arrow in Fig.

2–6. The runner's average velocity is 6.50 m/s to the left.



Example2.2: Distance a cyclist travels. How far can a cyclist travel in 2.5 h along a straight road if her average velocity is 18 km/h?

Answer: $\bar{v} = \Delta x / \Delta t$, we multiply both sides by Δt and obtain

$$\Delta x = \bar{v} \Delta t = (18 \text{ km/h})(2.5 \text{ h}) = 45 \text{ km}.$$

Example2.3: Car changes speed. A car travels at a constant for 100 km/h. It then speeds up to and is driven another 100 km. What is the car's average speed for the 200-km trip?

Solution Average velocity is:

$$\bar{v} = \frac{\Delta x}{\Delta t} = \frac{100 \text{ km} + 100 \text{ km}}{2.0 \text{ h} + 1.0 \text{ h}} = 67 \text{ km/h}.$$

2.3 Instantaneous Velocity

If you drive a car along a straight road for 150 km in 2.0 h, the magnitude of your average velocity is it is unlikely, though, that you were moving at precisely at every instant. To describe this situation we need the concept of *instantaneous velocity*, which is the velocity at any instant of time. (Its magnitude is the number, with units, indicated by a speedometer, Fig. (2.7). shows Car speedometer in white mi / h, and in orange km / h.



More precisely, the **instantaneous velocity** at any moment is defined as *the average velocity over an infinitesimally short time interval*. That is, Eq. 2–2 is to be evaluated in the limit of becoming extremely small, approaching zero. We can write the definition of instantaneous velocity, for one-dimensional motion as [instantaneous velocity]

$$v = \lim_{\Delta t \rightarrow 0} \frac{\Delta x}{\Delta t} \dots\dots\dots (2.3)$$

The notation **lim** _{$\Delta t \rightarrow 0$} means the ratio $\Delta x / \Delta t$ is to be evaluated in the limit of Δt approaching zero. For instantaneous velocity, we use the symbol **v** whereas for average velocity we use $\bar{v}$ with a bar above. In the rest of this book, when we use the term “velocity” it will refer to instantaneous velocity. When we want to speak of the average velocity, we will make this clear by including the word “average.” Note that the *instantaneous speed* always equals the magnitude of the instantaneous velocity. Why? Because distance traveled and the magnitude of the displacement become the same when they become infinitesimally small.

$\bar{v} = \Delta x / \Delta t = 15 \text{ km} / 0.50 \text{ h} = 30 \text{ km/h}$. If an object moves at a uniform (that is, constant) velocity during a particular time interval, then its instantaneous velocity at any instant is the same as its average velocity (see Fig. 2.8a). However, in many situations this is not the case. For example, a car may start from rest, speed up to 50 km/h, remain at that velocity for a time, then slow down to 20 km/h in a traffic jam, and finally stop at its destination after traveling a total of 15 km/h in 30 min. This trip is plotted on the graph of Fig. 2.8b. Also shown on the graph is the average velocity (dashed line), which is:

$$\bar{v} = \Delta x / \Delta t = 15 \text{ km} / 0.50 \text{ h} = 30 \text{ km/h}.$$

Graphs are often useful for analysis of motion.

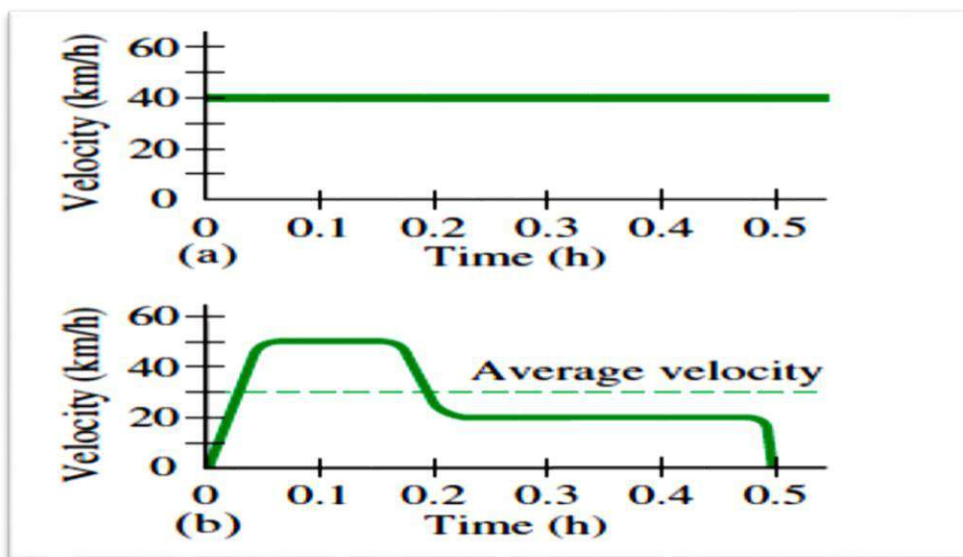


Figure (2.8 a, b)

2.4 Acceleration: An object whose velocity is changing is said to be accelerating. For instance, a car whose velocity increases in magnitude from zero to 80 km/h is accelerating.

Acceleration specifies how rapidly the velocity of an object is changing. Average acceleration is defined as the change in velocity divided by the time taken to make this change:

$$\text{average acceleration} = \frac{\text{change of velocity}}{\text{time elapsed}}.$$

In symbols, the **average acceleration**, $\bar{a}$ over a time interval $\Delta t = t_2 - t_1$, during which the velocity changes by $\Delta v = v_2 - v_1$ is defined as

$$\bar{a} = \frac{v_2 - v_1}{t_2 - t_1} = \frac{\Delta v}{\Delta t}. \quad [\text{average acceleration}] \quad (2-4)$$

We saw that velocity is a vector (it has magnitude and direction), so acceleration is a vector too. However, for one-dimensional motion, we need only use a plus or minus sign to indicate acceleration direction relative to a chosen coordinate axis.

(Usually, right is + left is−.) The **instantaneous acceleration**, a , can be defined in analogy to instantaneous velocity as the average acceleration over an infinitesimally short time interval at a given instant:

$$a = \lim_{\Delta t \rightarrow 0} \frac{\Delta v}{\Delta t}. \quad [\text{instantaneous acceleration}] \quad (2.5)$$

Here Δv is the very small change in velocity during the very short time interval Δt .

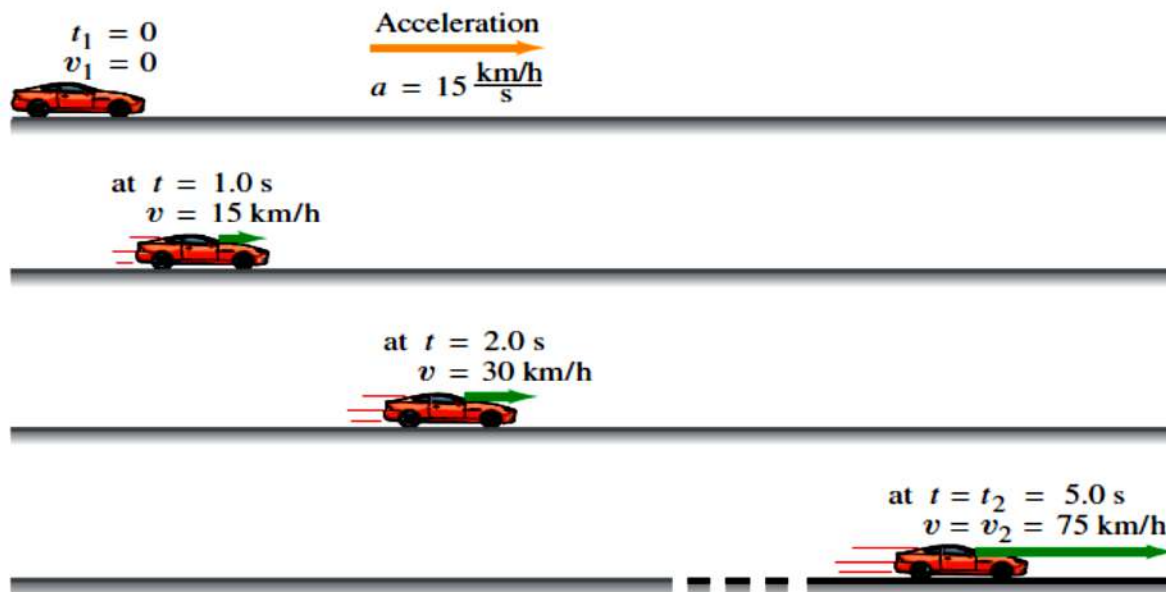
EXAMPLE 2-4 **Average acceleration.** A car accelerates on a straight road from rest to 75 km/h in 5.0 s, Fig. 2-10. What is the magnitude of its average acceleration?

APPROACH Average acceleration is the change in velocity divided by the elapsed time, 5.0 s. The car starts from rest, so $v_1 = 0$. The final velocity is $v_2 = 75$ km/h.

SOLUTION From Eq. 2-4, the average acceleration is

$$\bar{a} = \frac{v_2 - v_1}{t_2 - t_1} = \frac{75 \text{ km/h} - 0 \text{ km/h}}{5.0 \text{ s}} = 15 \frac{\text{km/h}}{\text{s}}.$$

This is read as “fifteen kilometers per hour per second” and means that, on average, the velocity changed by 15 km/h during each second. That is, assuming the acceleration was constant, during the first second the car’s velocity increased from zero to 15 km/h. During the next second its velocity increased by another 15 km/h, reaching a velocity of 30 km/h at $t = 2.0$ s, and so on. See Fig. 2-10.



Our result in Example 2–4 contains two different time units: hours and seconds. We usually prefer to use only seconds. To do so we can change km/h to m/s (see Section 1–6, and Example 1–5):

$$75 \text{ km/h} = \left(75 \frac{\cancel{\text{km}}}{\cancel{\text{h}}}\right) \left(\frac{1000 \text{ m}}{1 \cancel{\text{km}}}\right) \left(\frac{1 \cancel{\text{h}}}{3600 \text{ s}}\right) = 21 \text{ m/s}.$$

Then

$$\bar{a} = \frac{21 \text{ m/s} - 0.0 \text{ m/s}}{5.0 \text{ s}} = 4.2 \frac{\text{m/s}}{\text{s}} = 4.2 \frac{\text{m}}{\text{s}^2}.$$

We almost always write the units for acceleration as m/s^2 (meters per second squared) instead of m/s/s . This is possible because:

$$\frac{\text{m/s}}{\text{s}} = \frac{\text{m}}{\text{s} \cdot \text{s}} = \frac{\text{m}}{\text{s}^2}.$$

Note that *acceleration tells us how quickly the velocity changes*, whereas *velocity tells us how quickly the position changes*.

Deceleration

When an object is slowing down, we can say it is **decelerating**. However, be careful: deceleration does *not* mean that the acceleration is necessarily negative. The velocity of an object moving to the right along the positive x -axis is positive; if the object is slowing down (as in Fig. 2–10), the acceleration *is* negative. However, the same car moving to the left (decreasing x), and slowing down, has positive acceleration that points to the right, as shown in Fig. 2–11. We have a deceleration whenever the magnitude of the velocity is decreasing; thus the *velocity and acceleration point in opposite directions* when there is deceleration.

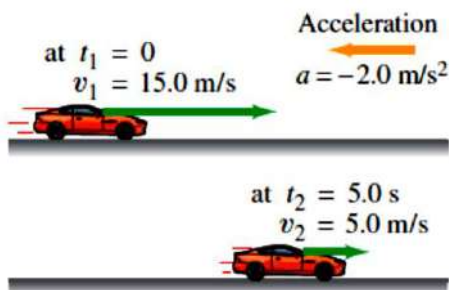


Fig 2.10

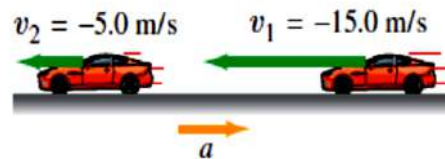


Fig 2.11

Exercise: A car moves along the x -axis. What is the sign of the car's acceleration if it is moving in the positive x direction with (a) increasing speed or (b) decreasing speed? What is the sign of the acceleration if the car moves in the negative x direction with (c) increasing speed or (d) decreasing speed?

2.5 Motion at Constant Acceleration

We now examine motion in a straight line when the magnitude of the acceleration is constant. In this case, the instantaneous and average accelerations are equal. We use the definitions of average velocity and acceleration to derive a set of valuable equations that relate x , v , a , and t , when a is constant, allowing us to determine any one of these variables if we know the others. We now have four equations relating **position, velocity, acceleration, and time**, when the acceleration a is constant. We collect these *kinematic equations for constant acceleration*, here in one place for future reference.

$v = v_0 + at$	2.6 a	$[a = \text{constant}]$
$x = x_0 + v_0t + \frac{1}{2}at^2$	2.6 b	$[a = \text{constant}]$
$v^2 = v_0^2 + 2a(x - x_0)$	2.6 c	$[a = \text{constant}]$
$\bar{v} = \frac{v + v_0}{2}$	2.6 d	$[a = \text{constant}]$

Example 2.5: Runway design. You are designing an airport for small planes. One kind of airplane that might use this airfield must reach a speed before takeoff of at least 27.8 m/s (100 km/h), and can accelerate at 2.00 m/s².

(a) If the runway is 150 m long, can this airplane reach the required speed for takeoff?

(b) If not, what minimum length must the runway have?

Approach Assuming the plane's acceleration is constant, we use the kinematic Equations for constant acceleration. In (a), we want to find v and what we are given is shown in the Table in the margin.

(a) Of the above four equations:

$$\begin{aligned} v^2 &= v_0^2 + 2a(x - x_0) \\ &= 0 + 2(2.00 \text{ m/s}^2)(150 \text{ m}) = 600 \text{ m}^2/\text{s}^2 \\ v &= \sqrt{600 \text{ m}^2/\text{s}^2} = 24.5 \text{ m/s.} \end{aligned}$$

This runway length is *not* sufficient, because the minimum speed is not reached.

(b) Now we want to find the minimum runway length, $x - x_0$, for a plane to reach $v = 27.8 \text{ m/s}$, given $a = 2.00 \text{ m/s}^2$. We again use Eq. 2-11c, but rewritten as

$$(x - x_0) = \frac{v^2 - v_0^2}{2a} = \frac{(27.8 \text{ m/s})^2 - 0}{2(2.00 \text{ m/s}^2)} = 193 \text{ m.}$$

A 200-m runway is more appropriate for this plane.

NOTE We did this Example as if the plane were a particle, so we round off our answer to 200 m.

Exercise: A car starts from rest and accelerates at a constant 10 m/s^2 during a $\frac{1}{4}$ mile (402 m) race. How fast is the car going at the finish line? (a) 8040 m/s (b) 90 m/s (c) 81m/s (d) 804 m/s.

2.6 Freely Falling Objects

One of the most common examples of uniformly accelerated motion is that of an object allowed to fall freely near the Earth's surface. That a falling object is accelerating may not be obvious at first. In addition, beware of thinking, as was widely believed before the time of Galileo, that heavier objects fall faster than lighter objects and that the speed of fall is proportional to how heavy the object is. *The speed of a falling object is not proportional to its mass.* Galileo made use of his new technique of imagining what would happen in idealized (simplified) cases. For free fall, he postulated that *all objects would fall with the same constant acceleration in the absence of air or other resistance.* He showed that this postulate predicts that for an object falling from rest, the distance traveled will be proportional to the square of the time; that is $d \propto t^2$. Galileo was the first to derive this mathematical relation. Galileo claimed that *all* objects, light or heavy, fall with the *same* acceleration, at least in the absence of air. Galileo was sure that air acts as a resistance to very light objects that have a large surface area. Nevertheless, in many ordinary circumstances this air resistance is negligible. In a chamber from which the air has been removed, even light objects as a feather or a horizontally held piece of paper will fall with the same acceleration as any other object (see Fig. 2.12).

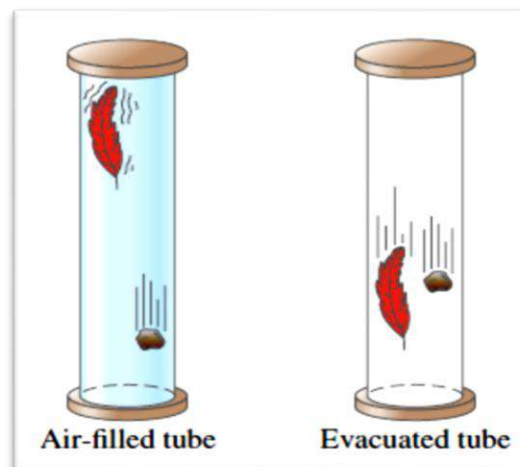


Figure (2.12)

Galileo's specific contribution to our understanding of the motion of falling objects can be summarized as follows: **at a given location on the Earth and in the absence of air resistance, all objects fall with the same constant acceleration.**

We call this acceleration the **acceleration due to gravity** at the surface of the Earth, and we give it the symbol **g** . Its magnitude is approximately

$$g = 9.80 \text{ m/s}^2 \quad [\text{acceleration due to a gravity at surface of earth}]$$

EXAMPLE **Falling from a tower.** Suppose that a ball is dropped ($v_0 = 0$) from a tower. How far will it have fallen after a time $t_1 = 1.00 \text{ s}$, $t_2 = 2.00 \text{ s}$, and $t_3 = 3.00 \text{ s}$? Ignore air resistance.

APPROACH Let us take y as positive downward, so the acceleration is $a = g = +9.80 \text{ m/s}^2$. We set $v_0 = 0$ and $y_0 = 0$. We want to find the position y of the ball after three different time intervals. Equation 2-11b, with x replaced by y , relates the given quantities (t , a , and v_0) to the unknown y .

SOLUTION We set $t = t_1 = 1.00 \text{ s}$ in Eq. 2-11b:

$$\begin{aligned} y_1 &= v_0 t_1 + \frac{1}{2} a t_1^2 \\ &= 0 + \frac{1}{2} a t_1^2 = \frac{1}{2} (9.80 \text{ m/s}^2) (1.00 \text{ s})^2 = 4.90 \text{ m}. \end{aligned}$$

The ball has fallen a distance of 4.90 m during the time interval $t = 0$ to $t_1 = 1.00 \text{ s}$. Similarly, after 2.00 s ($= t_2$), the ball's position is

$$y_2 = \frac{1}{2} a t_2^2 = \frac{1}{2} (9.80 \text{ m/s}^2) (2.00 \text{ s})^2 = 19.6 \text{ m}.$$

Finally, after 3.00 s ($= t_3$), the ball's position is (see Fig. 2-22)

$$y_3 = \frac{1}{2} a t_3^2 = \frac{1}{2} (9.80 \text{ m/s}^2) (3.00 \text{ s})^2 = 44.1 \text{ m}.$$

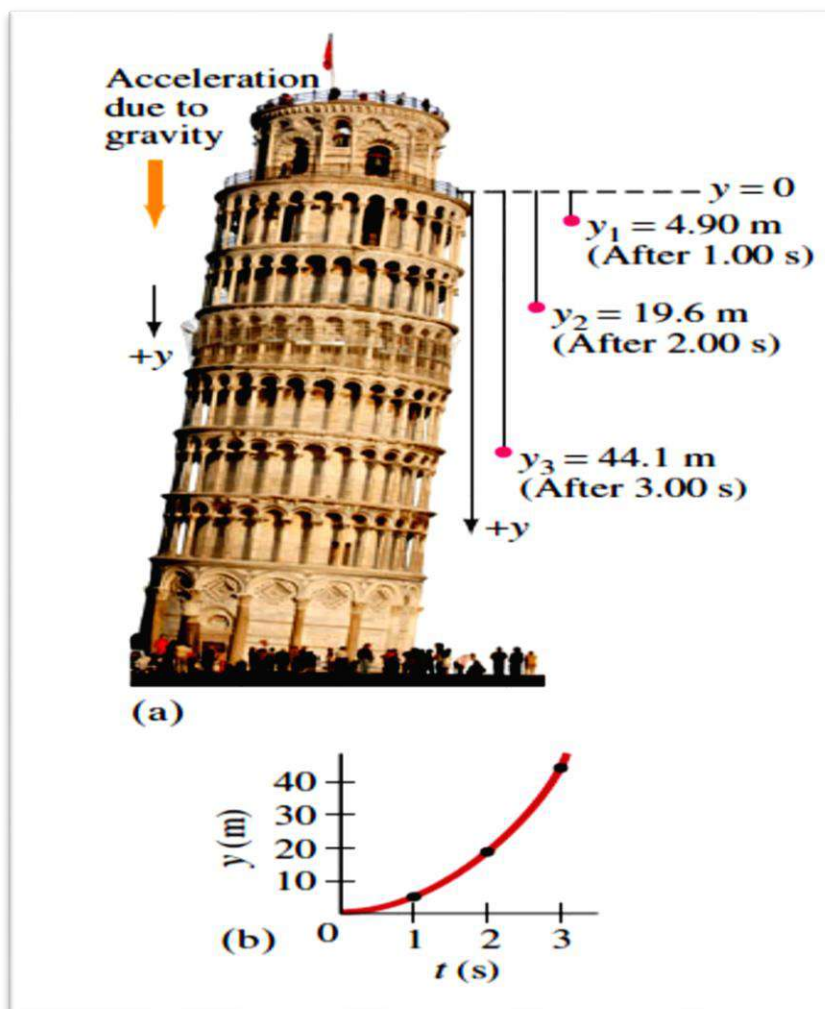


Figure (2.13)

3. Dynamics: Dynamics is one of the central branches of physics. The ideas of dynamics deal with motion and factors that affect motion. Dynamics are any external forces that increase or decrease an item's velocity or affect its acceleration. The four main forces that affect an object's velocity and acceleration are mass, force, momentum, and energy. In the context of dynamics, force is a push or pull influence on an object. The mass of an object is its size or the amount of matter it has. The more mass an object has the less effect external forces will have on its motion. Momentum is the potential movement that an object can undergo. The momentum of an object is affected by its velocity and its mass. The energy of an object is the level of its capacity to move or act on another object. This potential action is referred to as "work."

3.1 Force and Motion: A force, in the simplest sense, is a push or a pull. Its source may be gravitational, electrical, magnetic, or simply muscular effort. When more than a single force acts on an object, we consider the **net force**. For example, if you and a friend pull in the same direction with equal forces on an object, the forces combine to produce a net force twice as great as your single force. If you both pull with equal forces in *opposite* directions, the net force is zero. The equal but oppositely directed forces cancel each other. One of the forces can be considered the negative of the other, and they add algebraically to zero, with a resulting net force of zero. Figure 2.14 shows how forces combine to produce a net force on a box of cookies. A pair of 5-newton forces in the same direction produce a net force of 10 newton's (**the newton, N, is the scientific unit of force**). If the 5-newton forces are in opposite directions, the net force is zero. If 10 newtons of force are exerted to the right and 5 newtons to the left, the net force is 5 newtons to the right. The forces are shown by arrows. When the length and direction of such arrows are drawn to scale, we refer to the arrow as a **vector**.

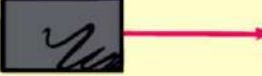
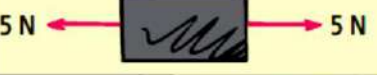
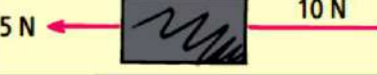
Applied forces	Net force
	
	
	

Figure (2.14). Net force (a force of 5 N is about 1.1 lb).

Any quantity that requires both magnitude and direction for a complete description is a **vector quantity** (Figure 2.15). Examples of **vector quantities include force, velocity, and acceleration**. By contrast, a quantity that can be described by **magnitude only, not involving direction, is called a scalar quantity. Mass, volume, and speed are scalar quantities**. Adding vectors that act along parallel directions is simple enough: If they act in the same direction, they add; if they act in opposite directions, they subtract. The sum of two or more vectors is called their **resultant**. To find the resultant of two vectors that do not act in exactly the same or opposite direction, we construct a parallelogram in which the two vectors are adjacent sides the diagonal of the parallelogram shows the resultant. In Figure 2.16, the parallelograms are rectangles.



Figure 2.15 This vector, scaled so that 1 cm equals 20 N, represents a force of 60 N to the right.



Figure 2.16 The pair of vectors at right angles to each other make two sides of a rectangle, the diagonal of which is their resultant.

SUMMARY OF TERMS

Force: In the simplest sense, a push or a pull.

Net force: The vector sum of forces that act on an object.

Vector: An arrow drawn to scale used to represent a vector quantity.

Vector quantity: A quantity that has both magnitude and direction, such as force.

Scalar quantity: A quantity that has magnitude but not direction, such as mass and volume.

Resultant: The net result of a combination of two or more vectors.

3.2 Mass: What is Mass?

Mass is a measure of the amount of matter in an object. It is a fundamental property of all objects and is often confused with weight, but the two are distinct. Weight is the force exerted on an object due to gravity, whereas mass remains constant regardless of an object's location

in the universe. Mass is typically measured in kilograms (kg) or grams (g), and it is a scalar quantity, meaning it has magnitude but no direction. The Importance of mass plays a crucial role in the laws of physics, and understanding it is fundamental to grasping many other physical concepts.

Newton's Second Law of Motion: This law states that the force acting on an object is directly proportional to its mass and the acceleration it experiences. In mathematical terms, $F = ma$, where F represents force, m represents mass, and a represents acceleration. This law helps us understand how objects move when subjected to forces.

Gravitational Attraction: Mass is essential in understanding how objects are attracted to each other due to gravity. The greater the mass of an object, the stronger its gravitational pull. This is why, for example, Earth's mass keeps us anchored to its surface.

Conservation of Mass: In many physical processes, mass is conserved. This means that the total mass of a closed system remains the same before and after a physical or chemical change. This concept is fundamental in chemistry and many other areas of science and engineering.

Measuring Mass: There are various methods for measuring mass, and the choice of method depends on the scale and precision of the measurement. Here are some common methods:

Balance Scales: These traditional devices compare the mass of an object to a set of known masses. They work on the principle of gravitational balance.

Spring Scales: Spring scales use Hooke's law to measure force, which can be used to calculate mass. These are commonly used for measuring the mass of lighter objects.

Electronic Scales: These modern scales use strain gauges or load cells to measure the force exerted by an object due to gravity. They can provide highly accurate measurements.

Gravitational Methods: In some cases, scientists use gravitational methods to measure the mass of celestial objects, like stars or planets, by studying their effects on nearby objects.

Summery

Mass is a fundamental property of matter that plays a central role in physics and our understanding of the natural world. It is distinct from weight and is critical for comprehending concepts such as force, gravity, conservation of mass, and even the relationship between mass and energy. As we continue to explore the universe and push the boundaries of our knowledge, our understanding of mass will remain at the core of our scientific endeavors. Mass in motion is a fundamental concept in physics that plays a crucial role in our understanding of how objects move and interact with one another. This concept is intimately connected with Newton's second law of motion and helps us explain the behavior of everything from a rolling ball to the orbits of planets.

Newton's Second Law of Motion

Newton's second law of motion, often expressed as $F = ma$, is one of the cornerstones of classical mechanics. It relates the force applied to an object (F), the mass of the object (m), and the resulting acceleration (a). In simple terms, it tells us that the force required to accelerate an object is directly proportional to the mass of the object and the desired acceleration. If the mass of an object is large, it will require a substantial force to produce a given acceleration. If the mass is small, the same force will result in a much larger acceleration. In essence, mass resists changes in motion. The more massive an object is, the more difficult it is to accelerate or decelerate.

Implications of Mass in Motion

Understanding mass in motion has several important implications:

Inertia: Mass in motion is closely linked to the concept of inertia. Inertia is the property of an object to resist changes in its state of motion. The greater the mass, the more inertia an object has, making it less responsive to changes in motion. For example, a heavy truck requires more force to stop or start than a small car.

Momentum: Momentum is another important concept related to mass in motion.

Momentum (p) is the product of an object's mass and its velocity ($p = mv$). It quantifies the motion of an object and is a conserved quantity in the absence of external forces. This means that in a closed system, the total momentum remains constant, demonstrating the importance of mass in motion. Figure (2.17) shows that the boulder, unfortunately, has considerable momentum.

Figure (2.17)



3.3 Collisions and Impulse: When objects collide, the relationship between mass and motion becomes evident. The change in momentum of an object during a collision (impulse) depends on both the force applied and the duration of the interaction. In these situations, the masses of the objects involved play a critical role in determining the outcome of the collision.

3.4 Projectile Motion: When an object is launched into the air, its mass influences how it follows a parabolic trajectory. Heavier objects, given the same initial velocity, will not deviate as much from their trajectory due to their greater mass. Mass in motion is an essential concept in physics, forming the basis of our understanding of how objects move and interact in the world. The relationship between mass, force, and acceleration, as described by Newton's second law, is fundamental to classical mechanics and has widespread applications in fields ranging from engineering to astrophysics. Whether we're examining the dynamics of a car crash, the orbit of a satellite, or the flight of a projectile, an understanding of mass in motion is essential for making sense of the physical world.

Example: Car Crash

Imagine two cars, Car A and Car B, both traveling at the same speed along a straight road. Car A is a compact car with a lower mass, while Car B is a larger, heavier SUV with a higher mass. Now, let us consider what happens when these two cars collide. Impact and Change in Motion: Car A and Car B collide head-on. Both cars experience a sudden change in motion due to the collision.

Inertia and Mass: Because Car B has a higher mass than Car A, it has greater inertia. Inertia is the tendency of an object to resist changes in its state of motion. When the collision occurs, Car B's greater mass makes it less responsive to changes in motion compared to Car A.

Example: Car Crash

Imagine two cars, Car A and Car B, both traveling at the same speed along a straight road. Car A is a compact car with a lower mass, while Car B is a larger, heavier SUV with a higher mass. Now, let us consider what happens when these two cars collide.

Impact and Change in Motion: Car A and Car B collide head-on. Both cars experience a sudden change in motion due to the collision. **Inertia and Mass:** Because Car B has a higher mass than Car A, it has greater inertia. Inertia is the tendency of an object to resist changes in its state of motion. When the collision occurs, Car B's greater mass makes it less responsive to changes in motion compared to Car A.

Force and Deceleration: The force of the collision is equal and opposite for both cars, according to Newton's third law of motion. However, Car A experiences a more significant deceleration (negative acceleration) due to its lower mass. This means that Car A comes to a stop relatively quickly, while Car B, with its higher mass, decelerates more slowly. **Damage and Consequences:** Due to its lower mass and the higher deceleration, Car A may suffer significant damage and potentially injure its occupants. Car B, with its greater mass and slower deceleration, may sustain less damage and provide better protection to its occupants. This example illustrates how mass in motion influences the outcome of a car crash. The mass of the vehicles affects their ability to resist changes in motion (inertia), the force they experience during the collision, and the resulting deceleration. It demonstrates the practical application of the principles of mass, motion, and force, as described by Newton's laws of motion.

Example 1: Calculating Force and Acceleration

Suppose you have a 500 kg car moving at a constant velocity of 20 m/s, and you want to calculate the force needed to accelerate it to a new velocity of 30 m/s over a distance of 100 meters. What will be the car's acceleration?

Solution:

First, calculate the change in velocity (Δv):

$$\Delta v = \text{final velocity} - \text{initial velocity}, \Delta v = 30 \text{ m/s} - 20 \text{ m/s} = 10 \text{ m/s}$$

Now, calculate the acceleration (a) using Newton's second law ($F = ma$):

$$F = m * a$$

Rearrange the formula to solve for acceleration: $a = F / m$

We need to find the force (F) required to accelerate the car. To do that, we can use the formula for force:

$$F = (\text{mass}) * (\text{change in velocity}) / (\text{distance})$$

$$F = (500 \text{ kg}) * (10 \text{ m/s}) / (100 \text{ m})$$

$$F = 50 \text{ N}$$

Now, calculate acceleration:

$$a = F / m$$

$$a = 50 \text{ N} / 500 \text{ kg}$$

$$a = 0.1 \text{ m/s}^2$$

Therefore, the car's acceleration is 0.1 m/s^2 .

Example 2: Conservation of Momentum

Let us consider a collision between two objects. A 2 kg ball is moving to the right at 4 m/s and collides with a stationary 3 kg ball. After the collision, the 2 kg ball moves to the left at 1 m/s. determine the velocity of the 3 kg ball after the collision.

Solution:

First, calculate the initial momentum of each ball:

Initial momentum of the 2 kg ball (before the collision) = (mass) * (initial velocity)

Momentum of the 2 kg ball = (2 kg) * (4 m/s) = 8 kg·m/s

The 3 kg ball is initially stationary, so its initial momentum is 0.

According to the law of conservation of momentum, the total momentum before the collision should equal the total momentum after the collision.

Total initial momentum = Total final momentum

8 kg·m/s (2 kg ball) + 0 (3 kg ball) = (2 kg ball) * (final velocity) + (3 kg ball) * (final velocity)

Simplify the equation:

8 kg·m/s = 5 kg * (final velocity)

Solve for the final velocity of the 3 kg ball:

final velocity = 8 kg·m/s / 5 kg

final velocity = 1.6 m/s

So, after the collision, the 3 kg ball moves to the right at 1.6 m/s.

These examples demonstrate how the principles of mass, motion, force, and momentum can be applied to solve real-world physics problems. These principles are essential for understanding the behavior of objects in motion and their interactions.



Ministry of Higher Education and Scientific Research
Middle Technical University
Electrical Engineering Technical College
Department of Cybersecurity Technology Engineering



General Physics

Lecture Three

Mass, Weight and Density

First Stage

By

Lecturer: Dr. Thamir. Al-hamadany

4. Mass, Weight and Density.

4.1 Mass: Mass is that property of an object that specifies how much resistance an object exhibits to changes in its velocity, the SI unit of mass is the kilogram. The greater the mass of an object, the less that object accelerates under the action of a given applied force. To describe mass quantitatively, we begin by experimentally comparing the accelerations a given force produces on different objects. Suppose a force acting on an object of mass m_1 produces an acceleration a_1 , and the *same force* acting on an object of mass m_2 produces an acceleration a_2 . The ratio of the two masses is defined as the *inverse* ratio of the magnitudes of the accelerations produced by the force:

$$\frac{m_1}{m_2} \equiv \frac{a_2}{a_1}$$

For example, if a given force acting on a 3-kg object produces an acceleration of 4 m/s², the same force applied to a 6-kg object produces an acceleration of 2 m/s². If one object has a known mass, the mass of the other object can be obtained from acceleration measurements. Mass is an inherent property of an object and is independent of the objects surroundings and of the method used to measure it. In addition, mass is a scalar quantity and thus obeys the rules of ordinary arithmetic. That is, several masses can be combined in simple numerical fashion. For example, if you combine a 3-kg mass with a 5-kg mass, the total mass is 8 kg. We can verify this result experimentally by comparing the accelerations that a known force gives to several objects separately with the acceleration that the same force gives to the same objects combined as one unit. Mass should not be confused with weight. Mass and weight are two different quantities. The weight of an object is equal to the magnitude of the

On the gravitational force exerted on the object and varies with location. (For example) a person who weighs 180 lb on the Earth weighs only about 30 lb on the Moon. On the other hand, the mass of an object is the same everywhere: an object having a mass of 2 kg on the Earth also has a mass of 2 kg on the Moon.

4.2 Weight: We defined the *weight* of a body as the attractive gravitational force exerted on it by the earth. We can now broaden our definition:

The weight of a body is the total gravitational force exerted on the body by all other bodies in the universe. When the body is near the surface of the earth, we can neglect all other gravitational forces and consider the weight as just the earth's gravitational attraction. At the surface of the *moon*, we consider a body's weight to be the gravitational attraction of the moon, and so on. If we again model the earth as a spherically symmetric body with radius R_E and mass m_E , the weight W of a small body of mass m at the earth's surface (a distance R_E from its center) is:

$$w = F_g = \frac{Gm_E m}{R_E^2} \quad (\text{weight of a body of mass } m \text{ at the earth's surface})$$

However, we also know that the weight w of a body is the force that causes the acceleration g of free fall, so by Newton's second law:

$$w = mg$$

Equating this with Eq.above and dividing by m , we find:

$$g = \frac{Gm_E}{R_E^2} \quad (\text{acceleration due to gravity at the earth's surface})$$

The acceleration due to gravity g is independent of the mass m of the body because m doesn't appear in this equation. We already knew that, but we can now see how it follows from the law of gravitation. At a point above, the earth's surface a distance r from the center of the earth (a distance $r - R_E$ above the surface), the weight of a body is replaced by r :

$$w = F_g = \frac{Gm_E m}{r^2} \dots\dots\dots$$

The weight of a body decreases inversely with the square of its distance from the earth's center. The weight varies with height above the earth for an astronaut who weighs 700 N at the earth's surface. The *apparent* weight of a body on earth differs slightly from the earth's gravitational force because the earth rotates and is therefore not precisely an inertial frame of reference assumed that the earth is an inertial system.

Example 1:

A robotic lander with an earth weight of 3430 N is sent to Mars, which has radius $R_M = 3.40 \times 10^6$ m and mass $m_M = 6.42 \times 10^{23}$ kg. Find the weight F_g of the lander on the Martian surface and the acceleration there due to gravity, g_M .

EXECUTE: The lander's earth weight is $w = mg$, so

$$m = \frac{w}{g} = \frac{3430 \text{ N}}{9.80 \text{ m/s}^2} = 350 \text{ kg}$$

The mass is the same no matter where the lander is. From Eq. (3), the lander's weight on Mars is

$$\begin{aligned} F_g &= \frac{Gm_M m}{R_M^2} \\ &= \frac{(6.67 \times 10^{-11} \text{ N} \cdot \text{m}^2/\text{kg}^2)(6.42 \times 10^{23} \text{ kg})(350 \text{ kg})}{(3.40 \times 10^6 \text{ m})^2} \\ &= 1.30 \times 10^3 \text{ N} \end{aligned}$$

The acceleration due to gravity on Mars is

$$g_M = \frac{F_g}{m} = \frac{1.30 \times 10^3 \text{ N}}{350 \text{ kg}} = 3.7 \text{ m/s}^2$$

4.3 Density: An important property of any material is its **density**, defined as its mass per unit volume. A homogeneous material such as ice or iron has the same density throughout.

We use ρ (the Greek letter rho) for density. If a mass, m of homogeneous material has volume V , the density is $\rho = \frac{m}{V}$ (definition of density) (1)

Two objects made of the same material have the same density even though they may have different masses and different volumes. That is because the *ratio* of mass to volume is the same for both objects (Fig. 1).

1 Two objects with different masses and different volumes but the same density.

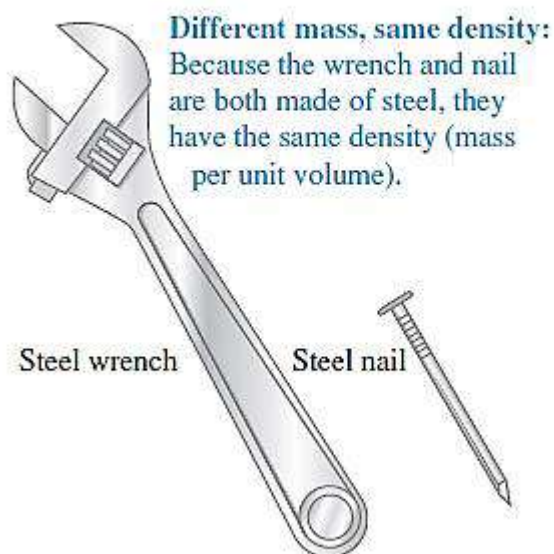


Figure. 1

Table 1 Densities of Some Common Substances

Material	Density (kg/m ³)*	Material	Density (kg/m ³)*
Air (1 atm, 20°C)	1.20	Iron, steel	7.8×10^3
Ethanol	0.81×10^3	Brass	8.6×10^3
Benzene	0.90×10^3	Copper	8.9×10^3
Ice	0.92×10^3	Silver	10.5×10^3
Water	1.00×10^3	Lead	11.3×10^3
Seawater	1.03×10^3	Mercury	13.6×10^3
Blood	1.06×10^3	Gold	19.3×10^3
Glycerine	1.26×10^3	Platinum	21.4×10^3
Concrete	2×10^3	White dwarf star	10^{10}
Aluminum	2.7×10^3	Neutron star	10^{18}

*To obtain the densities in grams per cubic centimeter, simply divide by 10^3 .

The SI unit of density is the kilogram per cubic meter (1kg/ m^3). The cgs unit, the gram per cubic centimeter (1 g/cm^3) is also widely used:

$$1\text{ g/cm}^3 = 1000\text{ kg/m}^3$$

The densities of some common substances at ordinary temperatures are given in Table 1. Note the wide range of magnitudes. The densest material found on earth is the metal osmium ($\rho = 22,500\text{ kg/m}^3$), but its density pales by comparison to the densities of exotic astronomical objects such as white dwarf stars and neutron stars.

The **specific gravity** of a material is the ratio of its density to the density of water at 4.0°C, 1000 kg/m³; it is a pure number without units. For example, the specific gravity of aluminum is 2.7. “Specific gravity” is a poor term, since it has nothing to do with gravity; “relative density” would have been better. The density of some materials varies from point to point within the material. One example is the material of the human body, which includes low-density fat (about 940 kg/ m³) and high-density bone (from 1700 to 2500 kg / m³). Two others are the earth’s atmosphere (which is less dense at high altitudes) and oceans (which are denser at greater depths). For these materials, Eq. (1) describes the **average density**. In general, the density of a material depends on environmental factors such as **temperature and pressure**.

Example 2: Find the mass and weight of the air at 20°C in a living room with a 4.0 m × 5.0 m floor and a ceiling 3.0 m high, and the mass and weight of an equal volume of water.

Solution: We assume that the air density is the same throughout the room. (Air is less dense at high elevations than near sea level, but the density varies negligibly over the room’s 3.0 m height.

EXECUTE: We have $V = (4.0 \text{ m})(5.0 \text{ m})(3.0 \text{ m}) = 60 \text{ m}^3$, so from Eq. (1),

$$m_{\text{air}} = \rho_{\text{air}} V = (1.20 \text{ kg/m}^3)(60 \text{ m}^3) = 72 \text{ kg}$$

$$w_{\text{air}} = m_{\text{air}} g = (72 \text{ kg})(9.8 \text{ m/s}^2) = 700 \text{ N} = 160 \text{ lb}$$

The mass and weight of an equal volume of water are

$$m_{\text{water}} = \rho_{\text{water}} V = (1000 \text{ kg/m}^3)(60 \text{ m}^3) = 6.0 \times 10^4 \text{ kg}$$

$$\begin{aligned} w_{\text{water}} &= m_{\text{water}} g = (6.0 \times 10^4 \text{ kg})(9.8 \text{ m/s}^2) \\ &= 5.9 \times 10^5 \text{ N} = 1.3 \times 10^5 \text{ lb} = 66 \text{ tons} \end{aligned}$$

EVALUATE: A roomful of air weighs about the same as an average adult. Water is nearly a thousand times denser than air, so its mass and weight are larger by the same factor. The weight of a roomful of water would collapse the floor of an ordinary house.



Ministry of Higher Education and Scientific Research
Middle Technical University
Electrical Engineering Technical College
Department of Cybersecurity Technology Engineering



General Physics

Lecture Four

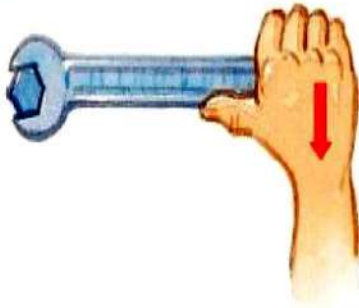
Turning Effect of Force

First Stage

By

Lecturer: Dr. Thamir. Al-hamadany

5. Turning Effect of Force



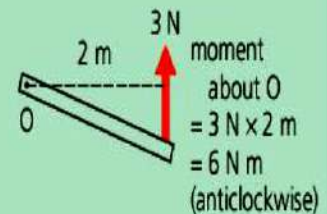
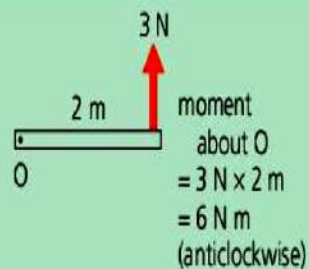
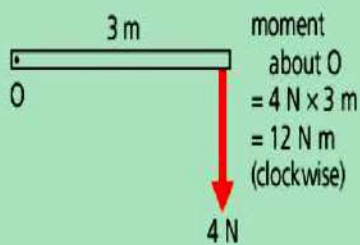
▲ A large force at the end of a long spanner gives a large turning effect.

Moment of a force

It is difficult to tighten a nut with your fingers. But with a spanner, you can produce a larger turning effect. The turning effect is even greater if you increase the force or use a longer spanner. The turning effect of a force is called a **moment**. It is calculated like this:

$$\text{moment of a force about a point} = \text{force} \times \text{perpendicular distance from the point}$$

Below, there are some examples of forces and their moments. Moments are described as **clockwise** or **anticlockwise**, depending on their direction. The moment of a force is also called a **torque**.



Unit of force

Force is measured in newtons (N)

**The principle of moments**

In diagram A below, the bar is in a state of balance, or **equilibrium**. Note that the anticlockwise moment about O is equal to the clockwise moment. One turning effect balances the other. In diagram B, there are more forces acting but, once again, the bar is in equilibrium. This time, the *total* clockwise moment about O is equal to the anticlockwise moment.

Taking moments

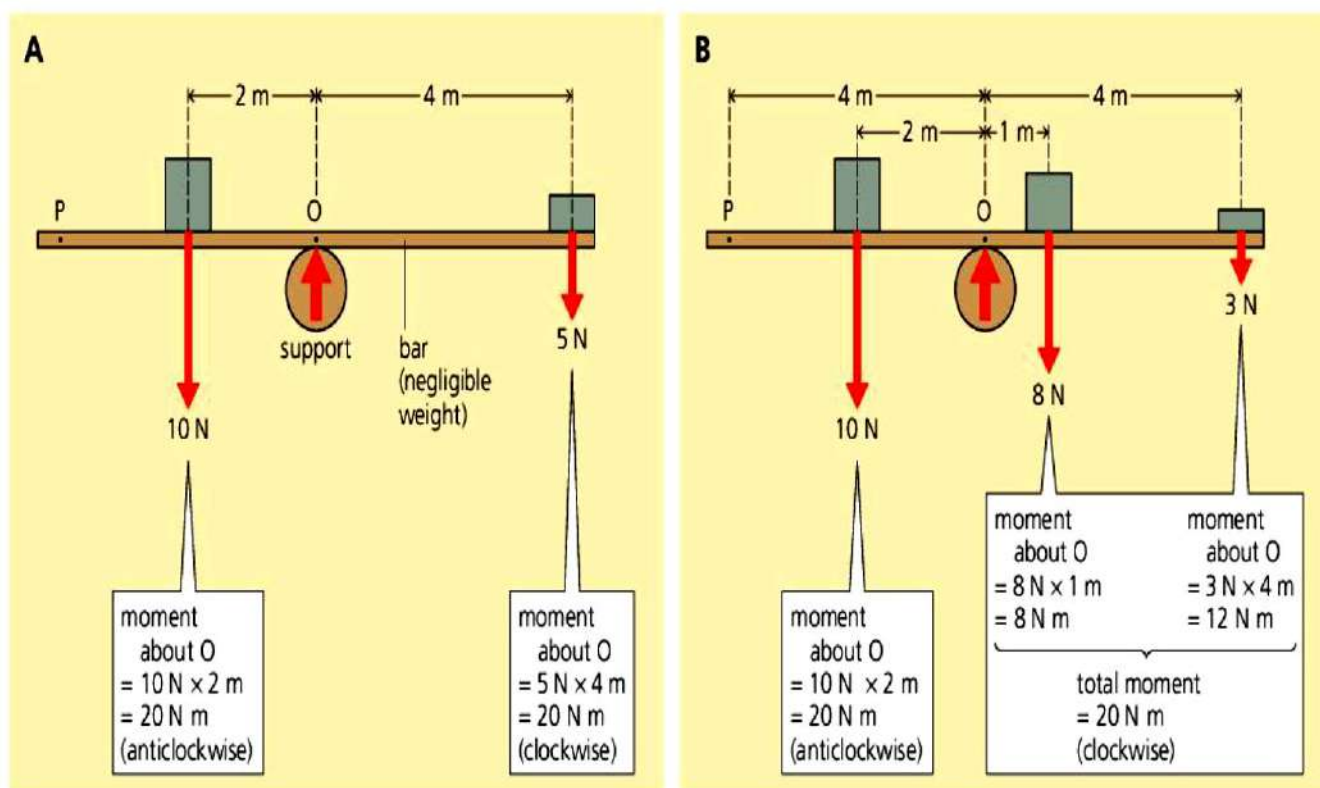
Calculating the moments about a point is called *taking moments* about the point.



These examples illustrate the **principle of moments**.

If an object is in equilibrium:

the sum of the clockwise moments about any point is equal to the sum of the anticlockwise moments about that point.





Ministry of Higher Education and Scientific Research
Middle Technical University
Electrical Engineering Technical College
Department of Cybersecurity Technology Engineering



General Physics

Lecture Five

First Stage

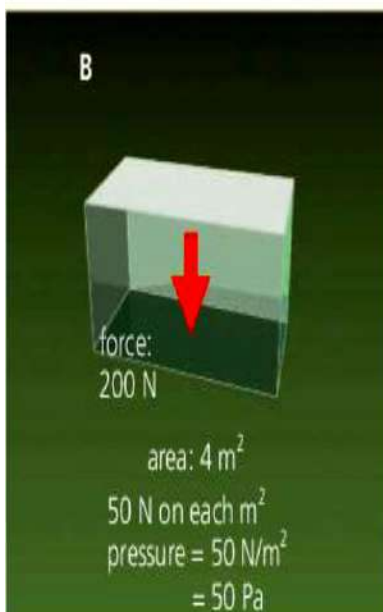
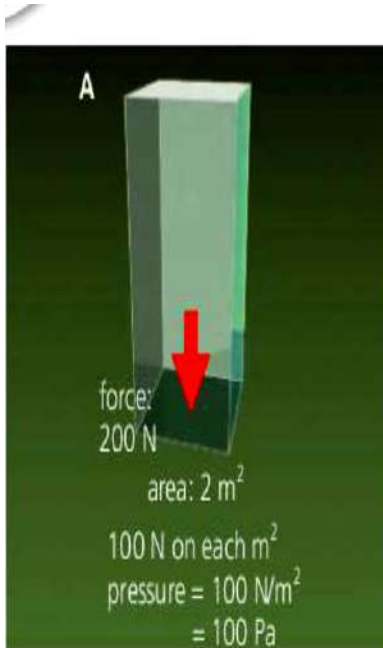
Pressure, Energy, Work and Power

By

Lecturer: Dr. Thamir. Al-hamadany

6. Pressure, Energy, Work and Power

6.1 Pressure



Blocks A and B on the left are resting on soft ground. Both weigh the same and exert the same force on the ground. But the force from block B is spread over a larger area, so the force *on each square metre* is reduced. The **pressure** under block B is less than that under block A.

For a force acting at right angles to a surface, the pressure is calculated like this:

$$\text{pressure} = \frac{\text{force}}{\text{area}}$$

$$\text{In symbols: } p = \frac{F}{A}$$

If force is measured in newtons (N) and area in square metres (m²), pressure is measured in newton/square metre (N/m²). 1 N/m² is called 1 **pascal** (Pa):

If a 100N force is spread over an area of 1 m², the pressure is 100 Pa.

If a 100N force is spread over an area of 2 m², the pressure is 50 Pa.

If a 100N force is spread over an area of 0.2 m², the pressure is 500 Pa.

If a 200N force is spread over an area of 0.2 m², the pressure is 1000 Pa.

For most pressure measurements, the pascal is a very small unit. In practical situations, it is often more convenient to use the **kilopascal (kPa)**.

$$1\text{kPa} = 1000\text{ Pa}$$

Increasing the pressure by reducing the area

The studs on a football boot have only a small area of contact with the ground. The pressure under the studs is high enough for them to sink into the ground, which gives extra grip.



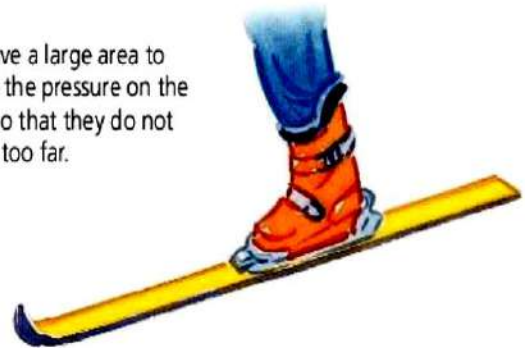
The area under the edge of a knife's blade is extremely small. Beneath it, the pressure is high enough for the blade to push easily through the material.



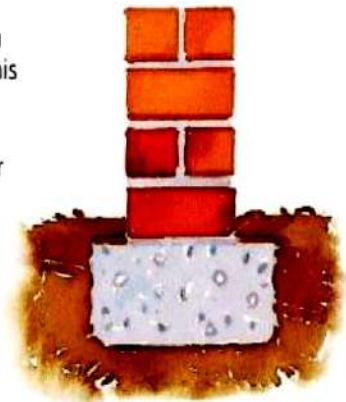
Under the tiny area of the point of a drawing pin, the pressure is far too high for the wood to withstand.

**Reducing the pressure by increasing the area**

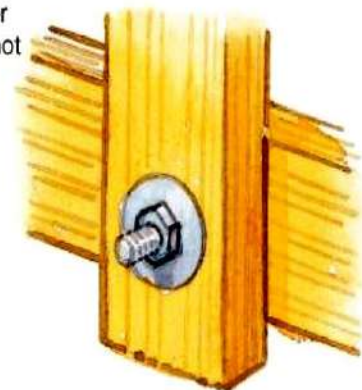
Skis have a large area to reduce the pressure on the snow so that they do not sink in too far.



Wall foundations have a large horizontal area. This reduces the pressure underneath so that the walls do not sink further into the ground.



A load-spreading washer ensures that the nut is not pulled into the wood when tightened up.



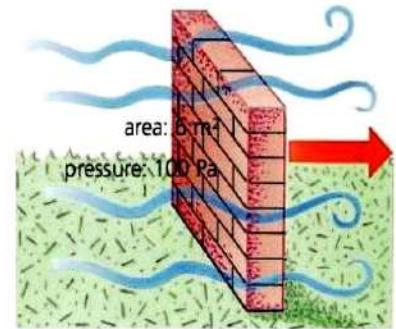
Pressure problems

Example 1 The wind pressure on the wall on the right is 100 Pa. If the wall has an area of 6 m², what is the force on it?

To solve this problem, you need to rearrange the pressure equation:

$$\begin{aligned}\text{force} &= \text{pressure} \times \text{area} \\ &= 100 \text{ Pa} \times 6 \text{ m}^2 = 600 \text{ N}\end{aligned}$$

So the force on the wall is 600 N.



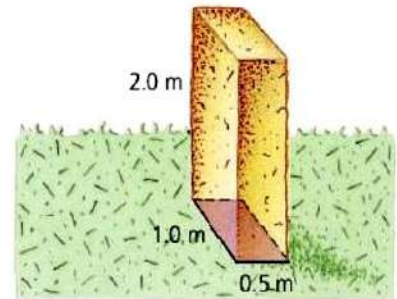
Example 2 A concrete block has a mass of 2600 kg. If the block measures 0.5 m by 1.0 m by 2.0 m, what is the maximum pressure it can exert when resting on the ground? ($g = 10 \text{ N/kg}$)

As g is 10 N/kg, the 2600 kg block has a weight of 26 000 N, so the force on the ground is also 26 000 N.

To exert *maximum* pressure, the block must be resting on the side with the *smallest* area. This is the side measuring 1.0 m $\times$ 0.5 m, as shown on the right. Its area = 1.0 m $\times$ 0.5 m = 0.5 m². So:

$$\text{pressure} = \frac{\text{force}}{\text{area}} = \frac{26\,000 \text{ N}}{0.5 \text{ m}^2} = 52\,000 \text{ Pa}$$

So the maximum pressure is 52 000 Pa, or 52 kPa.



Q

Assume that $g = 10 \text{ N/kg}$, and that all forces are acting at right angles to any area mentioned.

1 A force of 200 N acts on an area of 4 m².

- What pressure is produced?
- What would the pressure be if the same force acted on half the area?

2 What force is produced if:

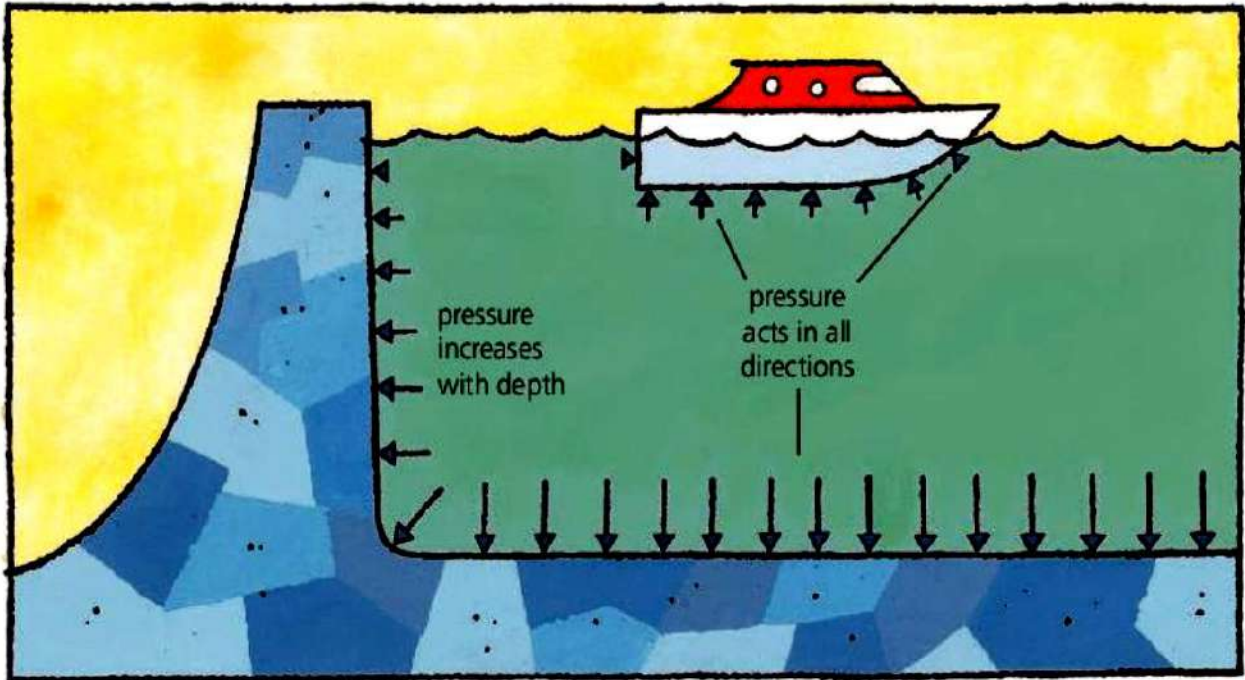
- A pressure of 1000 Pa acts on an area of 0.2 m²?
- A pressure of 2 kPa acts on an area of 0.2 m²?

3 Explain why a tractor's big tyres stop it sinking too far into soft soil.

4 A rectangular block of mass 30 kg measures 0.1 m by 0.4 m by 1.5 m.

- Calculate the weight of the block.
- Draw diagrams to show how the block must rest on the ground to exert *i* *maximum* pressure *ii* *minimum* pressure.
- Calculate the maximum and minimum pressures in part b.

1.1.6 Pressure in liquids



A liquid is held in its container by its weight. This causes pressure on the container, and pressure on any object in the liquid.

The following properties apply to any stationary liquid in an open container. The experiments on the left demonstrate three of them.

Pressure acts in all directions The liquid pushes on every surface in contact with it, no matter which way the surface is facing. For example, the deep-sea vessel below has to withstand the crushing effect of sea water pushing in on it from all sides, not just downwards.

Pressure increases with depth The deeper into a liquid you go, the greater the weight of liquid above and the higher the pressure. Dams are made thicker at the bottom to withstand the higher pressure there.

Pressure depends on the density of the liquid The more dense the liquid, the higher the pressure at any particular depth.

Pressure doesn't depend on the shape of the container Whatever the shape or width, the pressure at any particular depth is the same.

Useful connections

For calculations like those below, you need to know the connections between these:

volume (in m^3) density (in kg/m^3) mass (in kg) weight (in N) g (10 N/kg)

For example, you might know the volume and density of a liquid, but need to find its weight. For this, the equations required are:

$$\text{density} = \frac{\text{mass}}{\text{volume}} \quad \text{weight} = \text{mass} \times g$$

From these equations, it follows that:

$$\text{mass} = \text{density} \times \text{volume} \quad \text{weight} = \text{density} \times \text{volume} \times g$$

Pressure and weight essentials

For a force acting at right angles to a surface:

$$\text{pressure} = \frac{\text{force}}{\text{area}}$$

If force is in newtons (N) and area in square metres (m^2), then pressure is in pascals (Pa).

**Calculating the pressure in a liquid**

The container on the right has a base area A . It is filled to a depth h with a liquid of density ρ (Greek letter 'rho'). To calculate the pressure on the base due to the liquid, you first need to know the weight of the liquid on it:

$$\text{volume of liquid} = \text{base area} \times \text{depth} = Ah$$

$$\text{mass of liquid} = \text{density} \times \text{volume} = \rho Ah$$

$$\text{weight of liquid} = \text{mass} \times g = \rho g Ah \quad (g = 10 \text{ N/kg})$$

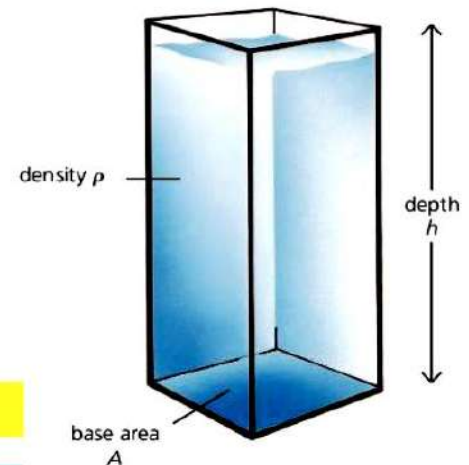
So: force on base = $\rho g Ah$

This force is acting on an area A .

So: pressure = $\frac{\text{force}}{\text{area}} = \frac{\rho g Ah}{A} = \rho gh$

At a depth h in a liquid of density ρ :

$$\text{pressure} = \rho gh$$



Example If the density of water is $1000 \text{ kg}/\text{m}^3$, what is the pressure due to the water at the bottom of a swimming pool 2 m deep?

$$\text{pressure} = \rho gh = 1000 \text{ kg}/\text{m}^3 \times 10 \text{ N/kg} \times 2 \text{ m} = 20\,000 \text{ Pa}$$

$g = 10 \text{ N/kg}$; density of water = $1000 \text{ kg}/\text{m}^3$; density of paraffin = $800 \text{ kg}/\text{m}^3$

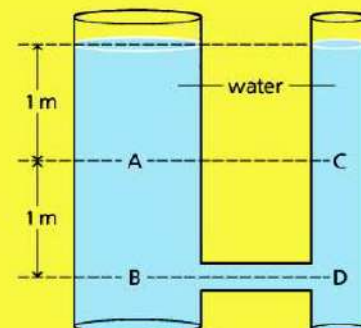
1 In the diagram on the right:

- How does the pressure at A compare with the pressure at B?
- How does the pressure at B compare with the pressure at D?
- How does the pressure at A compare with the pressure at C?
- If the water in the system were replaced with paraffin, how would this affect the pressure at B?

2 A rectangular storage tank 4 m long by 3 m wide is filled with paraffin to a depth of 2 m. Calculate:

- the volume of paraffin
- the mass of paraffin
- the weight of paraffin
- the pressure at the bottom of the tank due to the paraffin

3 In the diagram on the right, calculate the pressure at B due to the water.



2.6 Energy: *Energy* is one of the most important concepts in the world of science. In everyday use, energy is associated with the fuel needed for transportation and heating, with electricity for lights and appliances, and with the foods we consume. These associations, however, don't tell us what energy *is*, only what it *does*, and that producing it requires fuel. Energy is present in the Universe in a variety of forms, including mechanical, chemical, electromagnetic, and nuclear energy. Even the inert mass of everyday matter contains a very large amount of energy. Although energy can be transformed from one kind to another, all observations and experiments to date suggest that the total amount of energy in the Universe never changes. There are two types of energy ***mechanical energy***, which is the sum of ***kinetic energy***, the energy associated with motion, and ***potential energy*** the energy associated with relative position.

1.2.6 Kinetic energy (K_E)

What property of a system does $\frac{1}{2} mv_i^2$ describe? A massive, fast-moving vehicle can do damage to objects around it, and a baseball hit at high speed can rise high into the air. That is, an object with this property can produce a change in itself or the world around it. This property, the ability of an object to produce a change in itself or the world around it, is called **energy**. The fast-moving vehicle and the baseball possess energy that is associated with their motion. This energy resulting from motion is called **kinetic energy** and is represented by the symbol ***KE***.

Kinetic Energy $KE = \frac{1}{2}mv^2$

The kinetic energy of an object is equal to $\frac{1}{2}$ times the mass of the object multiplied by the speed of the object squared.

SI unit: joule (J) = kg. m²/s²

2.1.6 Potential Energy (P_E)

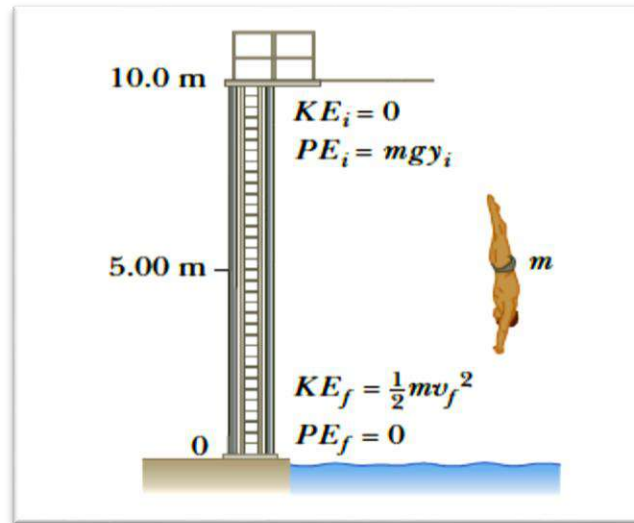
An object with kinetic energy (energy of motion) can do work on another object, just as a moving hammer can drive a nail into a wall. A brick on a high shelf can also do work: it can fall off the shelf, accelerate downward, and hit a nail squarely, driving it into the floorboards. The brick is said to have **potential energy** associated with it, because from its location on the shelf it can potentially do work. Potential energy is a property of a **system**, rather than of a single object, because it is due to the relative positions of interacting objects in the system, such as the position of the diver relative to the Earth. In this topic, we define a system as a collection of objects interacting via forces or other processes that are internal to the system. It turns out that potential energy is another way of looking at the work done by conservative forces. Potential energy, like work and kinetic energy, is a scalar quantity. The gravitational potential energy of a system consisting of Earth and an object of mass m near Earth's surface is given by:

$$PE = mgy$$

Where g is the acceleration of gravity and y is the vertical position of the mass relative the surface of Earth (or some other reference point).

SI unit: joule (J)

PROBLEM A diver of mass m drops from a board 10.0 m above the water's surface, as in Figure. Neglect air resistance. **(a)** Use conservation of mechanical energy to find his speed 5.00 m above the water's surface. **(b)** Find his speed as he hits the water.



SOLUTION

(a) Find the diver's speed halfway down, at $y = 5.00$ m.

Step 4: Write the energy conservation equation and supply the proper terms:

$$KE_i + PE_i = KE_f + PE_f$$

$$\frac{1}{2}mv_i^2 + mgy_i = \frac{1}{2}mv_f^2 + mgy_f$$

Step 5: Substitute $v_i = 0$, cancel the mass m , and solve for v_f :

$$0 + gy_i = \frac{1}{2}v_f^2 + gy_f$$

$$v_f = \sqrt{2g(y_i - y_f)} = \sqrt{2(9.80 \text{ m/s}^2)(10.0 \text{ m} - 5.00 \text{ m})}$$

$$v_f = 9.90 \text{ m/s}$$

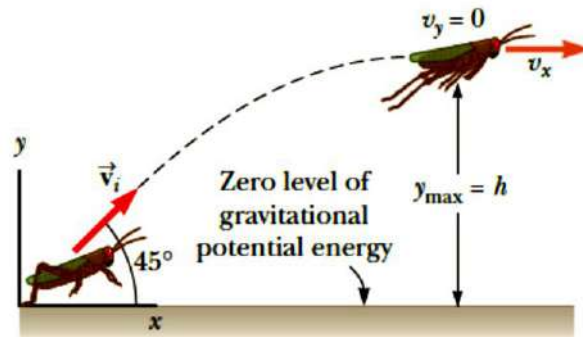
(b) Find the diver's speed at the water's surface, $y = 0$.

Use the same procedure as in part (a), taking $y_f = 0$:

$$0 + mgy_i = \frac{1}{2}mv_f^2 + 0$$

$$v_f = \sqrt{2gy_i} = \sqrt{2(9.80 \text{ m/s}^2)(10.0 \text{ m})} = 14.0 \text{ m/s}$$

Problem: A powerful grasshopper launches itself at an angle of 45° above the horizontal and rises to a maximum height of 1.00 m during the leap. (See the Figure) With what speed v_i did it leave the ground? Neglect air resistance.



SOLUTION

Use energy conservation:

$$\frac{1}{2}mv_i^2 + mgy_i = \frac{1}{2}mv_f^2 + mgy_f$$

Substitute $y_i = 0$, $v_f = v_x$ and $y_f = h$:

$$\frac{1}{2}mv_i^2 = \frac{1}{2}mv_x^2 + mgh$$

Multiply each side by $2/m$, obtaining one equation and two unknowns:

$$(1) \quad v_i^2 = v_x^2 + 2gh$$

Eliminate v_x by substituting $v_x = v_i \cos 45^\circ$ into Equation (1), solving for v_i , and substituting known values:

$$v_i^2 = (v_i \cos 45^\circ)^2 + 2gh = \frac{1}{2}v_i^2 + 2gh$$

$$v_i = 2\sqrt{gh} = 2\sqrt{(9.80 \text{ m/s}^2)(1.00 \text{ m})} = 6.26 \text{ m/s}$$

3.6 Work

Work has a different meaning in physics than it does in everyday usage. In the physics definition, a physics textbook author does very little work typing away at a computer. A mason, by contrast, may do a lot of work laying concrete blocks. In physics, work is done only if an object is moved through some displacement while a force is applied to it. If either the force or displacement is doubled, the work is doubled. Double them both, and the work is quadrupled. Doing work involves applying a force to an object while moving it a given distance. The definition for work W might be taken as:

$$W = Fd \dots\dots\dots$$

Where F is the magnitude of the force acting on the object and d is the magnitude of the object's displacement. That definition, however, gives only the magnitude of work done on an object when the force is constant and parallel to the displacement, which must be along a line. A more sophisticated definition is required. Figure shows a block undergoing a displacement $\Delta \vec{x}$ along a straight line while acted on by a constant force $\vec{F}$ in the same direction. We have the following definition:

The work W done on an object by a constant force $\vec{F}$ during a linear displacement along the x - axis is:

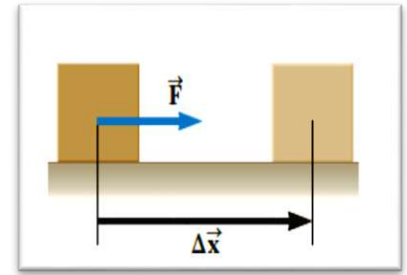
$$W = F_x \Delta x$$

where F_x is the x - component of the force $\vec{F}$ and $\Delta x = x_f - x_i$ is the object's displacement.

SI unit: joule (J) = newton. meter (N. m) = kg. m²/s²

Work is a Scalar Quantity Work is a simple number a scalar, not a vector so there is no direction associated with it. Energy and energy transfer are also scalars.

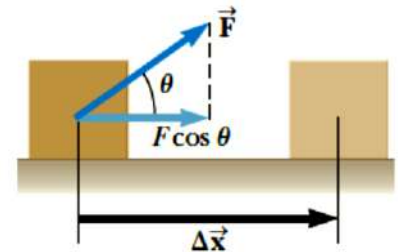
A constant force $\vec{F}$ in the same direction as the displacement, $\Delta\vec{x}$, does work $F \Delta x$.



Note that in one dimension, $\Delta x = x_f - x_i$ is a vector quantity not a magnitude as might be inferred from definitions of a vector and its magnitude. Therefore, Δx can be either positive or negative. Work as defined in Equation is rigorous for displacements of any object along the x – axis while a constant force acts on it and, therefore, is suitable for many one – dimensional problems. Work is a positive number if F_x and Δx are either positive or both negative, the work increases the mechanical energy of the object. If F_x is positive and Δx is negative, or vice versa, then the work is negative, and the object loses mechanical energy. The definition in Equation of works even when the constant force $\vec{F}$ is not parallel to the x - axis. Work is only done by the part of the force acting parallel to the object’s direction of motion. Work is a scalar quantity a number rather than a vector and consequently is easier to handle. No direction is associated with it. Further, work doesn’t depend explicitly on time, which can be an advantage in problems involving only velocities and positions. Because the units of work are those of force and distance, the SI unit is the **newton - meter** (N. m). Another name for the newton - meter is the **joule (J)** (rhymes with “pool”). The U.S. customary unit of work is the **foot - pound**, because distances are measured in feet and forces in pounds in that system. A useful alternate definition relates the work done on an object to the angle the displacement makes with respect to the force. This definition exploits the triangle shown in Figure below. The components of the vector $\vec{F}$ can be written as $F_x = F \cos \theta$ and $F_y = F \sin \theta$. However,

only the x - component, which is parallel to the direction of motion, makes a nonzero contribution to the work done on the object.

Figure shows a constant force $\vec{F}$ exerted at an angle θ with respect to the displacement, $\Delta\vec{x}$, does work $(F \cos \theta) \Delta x$.



Work by a constant force at an angle to the displacement the work W done on an object by a constant force $\vec{F}$ during a linear displacement $\Delta\vec{x}$ is:

$$W = (F \cos \theta) d$$

where d is the magnitude of the displacement and θ is the angle between the vectors $\vec{F}$ and $\Delta\vec{x}$. **SI unit: joule (J).**

Note: No work is done on an object when it is moved horizontally because the applied force $\vec{F}$ is perpendicular to the displacement.

For example, when a student lifts a box as in Figure below, the work he does on the box is positive because the force he exerts on the box is upward, in the same direction as the displacement. In lowering the box slowly back down, however, the student still exerts an upward force on the box, but the motion of the box is downwards. Because the vectors $\vec{F}$ and $\Delta\vec{x}$ are now in opposite directions, the angle between them is 180° , and $\cos 180^\circ = -1$ and the work done by the student is negative. In general, when the part of $\vec{F}$ parallel to $\Delta\vec{x}$ points in the same direction as $\Delta\vec{x}$, the work is positive; otherwise, it's negative.

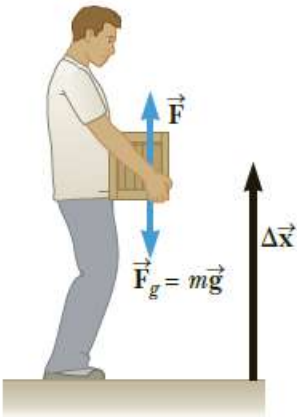
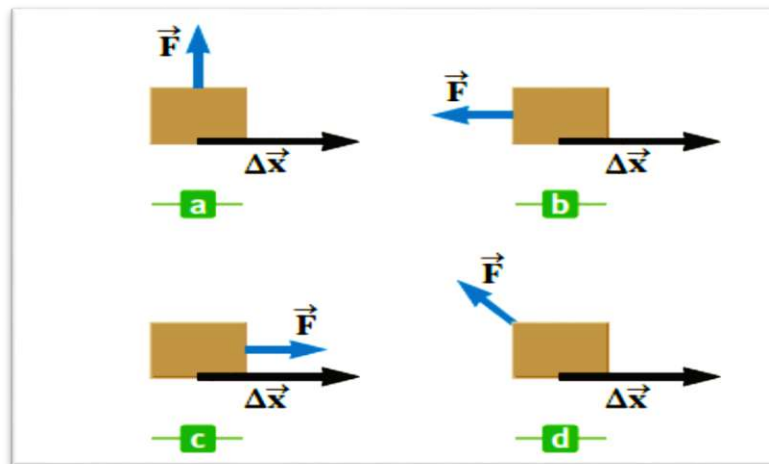


Figure shows the student does positive work when he lifts the box from the floor, because the applied force $\vec{F}$ is in the same direction as the displacement. When he lowers the box to the floor, he does negative work.

In Figure (a)–(d), a block moves to the right in the positive x – direction through the displacement $\Delta\vec{x}$ while under the influence of a force with the same magnitude $\vec{F}$. Which of the following is the correct order of the amount of work done by the force $\vec{F}$, from most positive to most negative? (a) d, c, a, b (b) c, a, b, d (c) c, a, d, b.



4.6 Power

Power, the rate at which energy is transferred, is important in the design and use of practical devices, such as electrical appliances and engines of all kinds. The concept of power, however, is essential whenever a transfer of any kind of energy takes place. The issue is particularly interesting for living creatures because the maximum work per second, or power output, of an animal varies greatly with output duration. **Power** is defined as the rate of energy transfer with time: average power If an external force does work W on an object in the time interval Δt , then the **average power** delivered to the object is the work done divided by the time interval, or :

$$\bar{P} = \frac{W}{\Delta t} \dots\dots\dots$$

SI unit: watt (W = J/s)

It's sometimes useful to rewrite Equation above by substituting $W = F \Delta x$ and noticing that $\Delta x / \Delta t$ is the average velocity of the object during the time Δt :

$$\bar{P} = \frac{W}{\Delta t} = \frac{F \Delta x}{\Delta t} = F \bar{v}$$

According to Equation above, average power is a constant force times the average velocity.

The force F is the component of force in the direction of the average velocity. A more general definition, called the **instantaneous power**, can be written down with a little calculus and has the same form as Equation: $P = Fv$ ◀ **Instantaneous power** .In

Equation above both the force F and the velocity v must be parallel, but can change with time. The SI unit of power is the joule per second (J/s), also called the

Watt, named after James Watt:

$$1 \text{ W} = 1 \text{ J/s} = 1 \text{ kg} \cdot \text{m}^2/\text{s}^3$$

The unit of power in the U.S. customary system is the horsepower (hp), where:

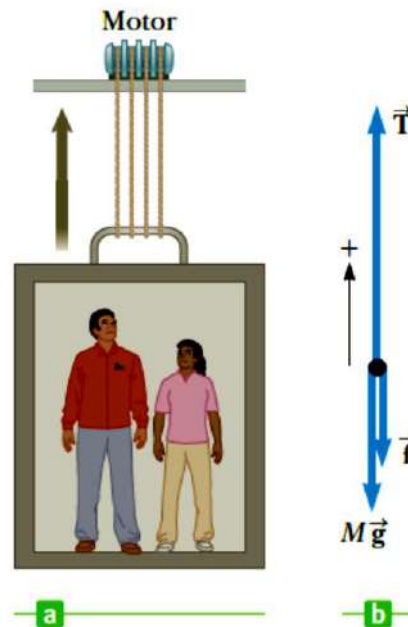
$$1 \text{ hp} \equiv 550 \frac{\text{ft} \cdot \text{lb}}{\text{s}} = 746 \text{ W}$$

fyi Do not confuse the non-italic symbol for watts, W, with the italic symbol *W* for work.

A watt is a unit, the same as joules per second. Work is a concept, carrying units of joules.

Watt, who needed a large power unit to rate the power output of his new invention, the steam engine, first defined the horsepower. The watt is commonly used in electrical applications, but it can be used in other scientific areas as well. For example, European sports car engines are rated in kilowatts. In electric power generation, it is customary to use the kilowatt-hour as a measure of energy. One kilowatt-hour (kWh) is the energy transferred in 1 h at the constant rate of $1 \text{ kW} = 1\,000 \text{ J/s}$. Therefore, $1 \text{ kWh} = (10^3 \text{ W})(3\,600 \text{ s}) = 3.60 \times 10^6 \text{ J}$. It is important to realize that a kilowatt-hour is a unit of energy, *not* power. When you pay your electric bill, you are buying energy, and that is why your bill lists a charge for electricity of about 10 cents/kWh. The amount of electricity used by an appliance can be calculated by multiplying its power rating (usually expressed in watts and valid only for normal household electrical circuits) by the length of time, the appliance is operated. For example, an electric bulb rated at 100 W ($=0.100 \text{ kW}$) “consumes” $3.6 \times 10^5 \text{ J}$ of energy in 1 h.

Problem: a 1.00×10^3 - kg elevator car carries a maximum load of $8.00 \times 10^2 \text{ kg}$. A constant frictional force of $4.00 \times 10^3 \text{ N}$ retards its motion upward, as in Figure below. What minimum power, in kilowatts and in horsepower, must the motor deliver to lift the fully loaded elevator car at a constant speed of 3.00 m/s ?

**SOLUTION**

Apply Newton's second law to the elevator car:

The velocity is constant, so the acceleration is zero. The forces acting on the elevator car are the force of tension in the cable, $\vec{T}$, the friction $\vec{f}$, and gravity $M\vec{g}$, where M is the mass of the elevator car.

Write the equation in terms of its components:

Solve this equation for the tension T and evaluate it:

Substitute this value of T for F in the power equation:

$$\sum \vec{F} = m\vec{a}$$

$$\vec{T} + \vec{f} + M\vec{g} = 0$$

$$T - f - Mg = 0$$

$$\begin{aligned} T &= f + Mg \\ &= 4.00 \times 10^3 \text{ N} + (1.80 \times 10^3 \text{ kg})(9.80 \text{ m/s}^2) \\ T &= 2.16 \times 10^4 \text{ N} \end{aligned}$$

$$P = Fv = (2.16 \times 10^4 \text{ N})(3.00 \text{ m/s}) = 64.8 \text{ kW}$$

$$P = 64.8 \text{ kW} = 86.9 \text{ hp}$$

Problem: Killer whales are known to reach 32 ft in length and have a mass of over 8 000 kg. They are also very quick, able to accelerate up to 30 mi/h in a matter of seconds. Disregarding the considerable drag force of water, calculate the average power a killer whale named Shamu with mass 8.00×10^3 kg would need to generate to reach a speed of 12.0 m/s in 6.00 s.

SOLUTION

Calculate the change in Shamu's kinetic energy. By the work-energy theorem, this equals the minimum work Shamu must do:

$$\begin{aligned}\Delta KE &= \frac{1}{2}mv_f^2 - \frac{1}{2}mv_i^2 \\ &= \frac{1}{2} \cdot 8.00 \times 10^3 \text{ kg} \cdot (12.0 \text{ m/s})^2 - 0 \\ &= 5.76 \times 10^5 \text{ J}\end{aligned}$$

Divide by the elapsed time (Eq. 5.23), noting that $W = \Delta KE$:

$$\bar{P} = \frac{W}{\Delta t} = \frac{5.76 \times 10^5 \text{ J}}{6.00 \text{ s}} = 9.60 \times 10^4 \text{ W}$$



*Ministry of Higher Education and Scientific Research
Middle Technical University
Electrical Engineering Technical College
Department of Cybersecurity Technology Engineering*



General Physics

Lecture six

First Stage

Motion along a straight line

By

Lecturer: Dr. Thamir. Al-hamadany

7. Motion along a straight line

What distance must an airliner travel down a runway before reaching takeoff speed? When you throw a baseball straight up in the air, how high does it go? When a glass slips from your hand, how much time do you have to catch it before it hits the floor? These are the kinds of questions you will learn to answer in this subject. We are beginning our study of physics with *mechanics*, the study of the relationships among force, matter, and motion. The simplest kind of motion: a body moving along a straight line. To describe this motion, we introduce the physical quantities *velocity* and *acceleration*. In physics, these quantities have definitions that are more precise and slightly different from the ones used in everyday language. Both velocity and acceleration are *vectors*: this means that they have both magnitude and direction. Our concern is a motion along a straight line only, so we will not need the full mathematics of vectors just yet. But using vectors will be essential when we consider motion in two or three dimensions. In physics, these quantities have definitions that are more precise and slightly different from the ones used in everyday language. **Both velocity and acceleration are vectors: this means that they have both magnitude and direction.** We will develop simple equations to describe straight-line motion in the important special case when the acceleration is constant. An example is the motion of a freely falling body. We will also consider situations in which the acceleration varies during the motion.

Displacement, Time, and Average Velocity

Key Ideas The **position** x of a particle on an x -axis locates the particle with respect to the origin, or zero point, of the axis. The position is either positive or negative, according to which side of the origin the particle is on, or zero if the particle is at the origin. The positive direction on an axis is the direction of increasing positive numbers; the opposite direction is

the negative direction on the axis. The displacement Δx of a particle is the change in its position:

$$\Delta x = x_2 - x_1$$

Displacement is a vector quantity. It is positive if the particle has moved in the positive direction of the x -axis and negative if the particle has moved in the negative direction.

When a particle has moved from position x_1 to position x_2 during a time interval $\Delta t = t_2 - t_1$, its average velocity during that interval is:

$$v_{\text{avg}} = \frac{\Delta x}{\Delta t} = \frac{x_2 - x_1}{t_2 - t_1}.$$

The algebraic sign of v_{avg} indicates the direction of motion (v_{avg} is a vector quantity).

Average velocity does not depend on the actual distance a particle moves, but instead depends on its original and final positions. On a graph of x versus t , the average velocity for a time interval Δt is the slope of the straight line connecting the points on the curve that represent the two ends of the interval. The average speed s_{avg} of a particle during a time interval Δt depends on the total distance the particle moves in that time interval:

$$s_{\text{avg}} = \frac{\text{total distance}}{\Delta t}.$$

Motion

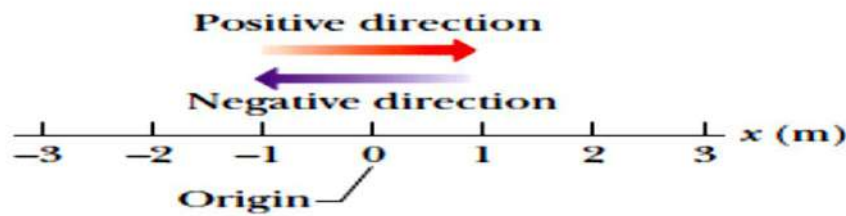
The world and everything in it, moves. Even seemingly, stationary things, such as a roadway, move with Earth's rotation, Earth's orbit around the Sun, and the Sun's orbit around the center of the Milky Way galaxy, and that galaxy's migration relative to other galaxies. The classification and comparison of motions (called **kinematics**) is often challenging. What exactly do you measure, and how do you compare? Before we attempt an answer, we shall examine some general properties of motion that is restricted in three ways.

1. The motion is along a straight line only. The line may be vertical, horizontal, or slanted, but it must be straight.
2. Forces (pushes and pulls) cause motion only the motion itself and changes in the motion. Does the moving object speed up, slow down, stop, or reverse direction? If the motion does change, how is time involved in the change?
3. The moving object is either a **particle** (by which we mean a point-like object such as an electron) or an object that moves like a particle (such that every portion moves in the same direction and at the same rate). A stiff pig slipping down a straight playground slide might be considered moving like a particle; however, a tumbling tumbleweed would not.

Position and Displacement

To locate an object means to find its position relative to some reference point, often the **origin** (or zero point) of an axis such as the x -axis in Fig below. The **positive direction** of the axis is in the direction of increasing numbers (coordinates), which is to the right in Fig. The opposite is the **negative direction**. For example, a particle might be located at $x = 5$ m, which means it is 5 m in the positive direction from the origin. If it were at $x = -5$ m, it would be just as far from the origin but in the opposite direction. On the axis, a coordinate of -5 m is less than a coordinate of -1 m, and both coordinates are less than a coordinate of 5 m. A plus sign for a coordinate need not be shown, but a minus sign must always be shown. A change from position x_1 to position x_2 is called a **displacement** Δx , where:

$$\Delta x = x_2 - x_1.$$



(The symbol Δ , the Greek uppercase delta, represents a change in a quantity, and it means the final value of that quantity minus the initial value.) When numbers are inserted for the position values x_1 and x_2 in Eq. above, a displacement in the positive direction (to the right in Fig. above) always comes out positive, and a displacement in the opposite direction (left in the figure) always comes out negative. For example, if the particle moves from $x_1 = 5$ m to $x_2 = 12$ m, then the displacement is $\Delta x = (12 \text{ m}) - (5 \text{ m}) = 7 \text{ m}$. The positive result indicates that the motion is in the positive direction. If, instead, the particle moves from $x_1 = 5$ m to $x_2 = 1$ m, then $\Delta x = (1 \text{ m}) - (5 \text{ m}) = -4 \text{ m}$. The negative result indicates that the motion is in the negative direction. The actual number of meters covered for a trip is irrelevant; displacement involves only the original and final positions. **For example**, if the particle moves from $x = 5$ m out to $x = 200$ m and then back to $x = 5$ m, the displacement from start to finish is $\Delta x = (5 \text{ m}) - (5 \text{ m}) = 0$.

Signs. A plus sign for a displacement need not be shown, but a **minus sign** must always be shown. If we ignore the sign (and thus the direction) of a displacement, we are left with the **magnitude** (or absolute value) of the displacement. For example, a displacement of $\Delta x = -4$ m has a magnitude of 4 m.

Displacement is an example of a **vector quantity**, which is a quantity that has both a direction and a magnitude. Displacement has two features: (1) its *magnitude* is the distance (such as the number of meters) between the original and final positions.

(2) Its **direction**, from an original position to a final position, can be represented by a plus sign or a minus sign if the motion is along a single axis.

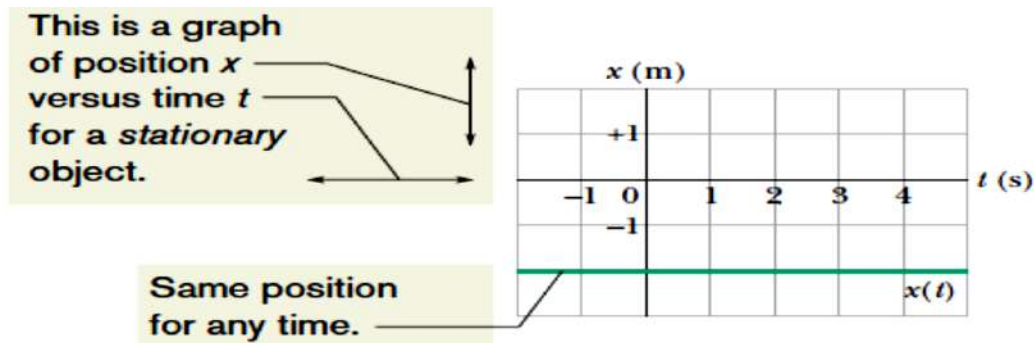


Checkpoint 1

Here are three pairs of initial and final positions, respectively, along an x axis. Which pairs give a negative displacement: (a) -3 m, $+5$ m; (b) -3 m, -7 m; (c) 7 m, -3 m?

Average Velocity and Average Speed

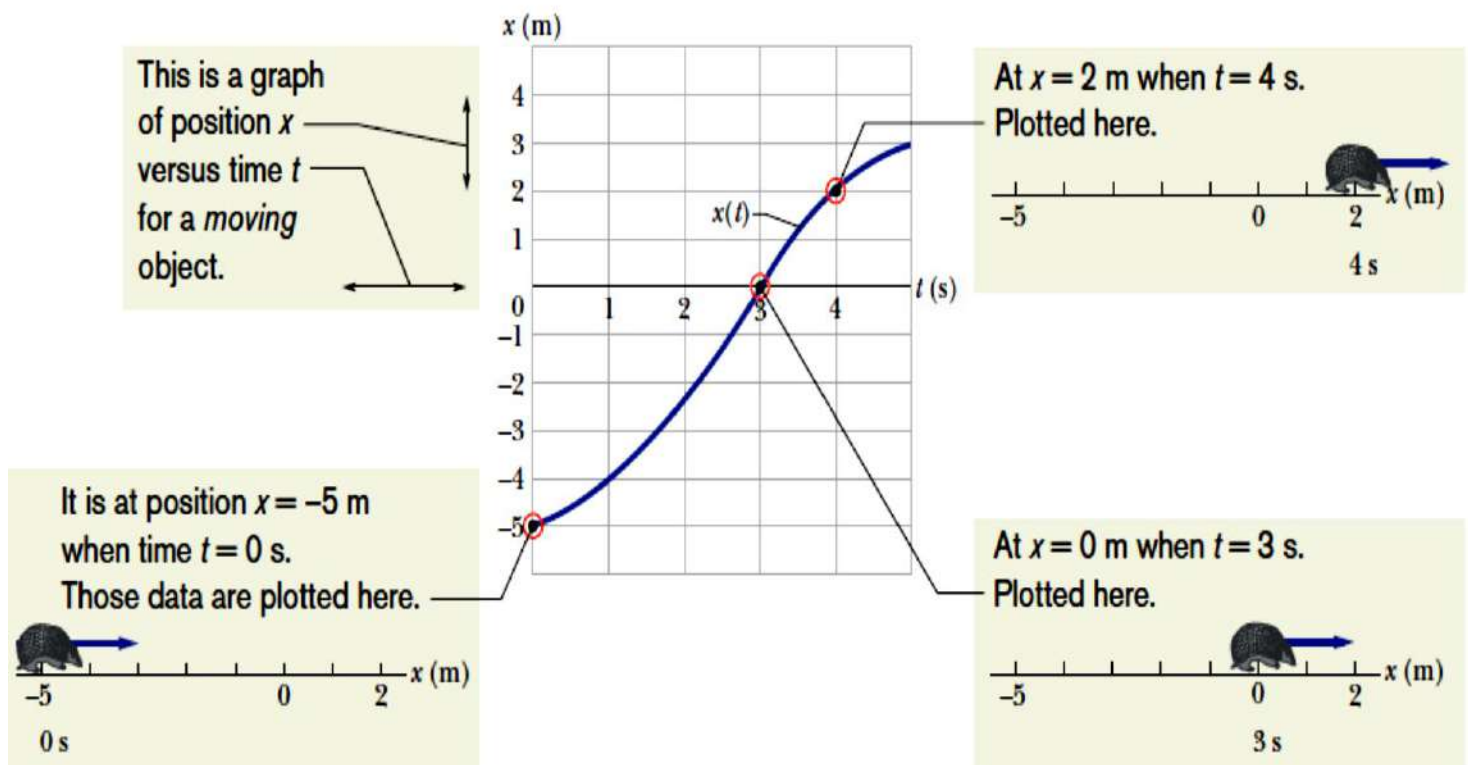
A compact way to describe position is with a graph of position x plotted as a function of time t —a graph of $x(t)$. (The notation $x(t)$ represents a function x of t , not the product x times t .) As a simple example, Fig. below shows the position function $x(t)$ for a stationary armadillo (which we treat as a particle) over a 7 s time interval.



The graph of $x(t)$ for an armadillo that is stationary at $x = -2$ m. The value of x is -2 m for all times t .

The animal's position stays at $x = -2$ m.

Figure below is more interesting, because it involves motion. The armadillo is apparently first noticed at $t = 0$ when it is at the position $x = -5$ m.



It moves toward $x = 0$, passes through that point at $t = 3$ s, and then moves on to increasingly larger positive values of x . Figure above also depicts the straight-line motion of the armadillo (at three times) and is something like what you would see. The graph in Fig. Is more abstract, but it reveals how fast the armadillo moves. Actually, several quantities are associated with the phrase “how fast.” One of them is the **average velocity** v_{avg} , which is the ratio of the displacement Δx that occurs during a particular time interval Δt to that interval:

$$v_{\text{avg}} = \frac{\Delta x}{\Delta t} = \frac{x_2 - x_1}{t_2 - t_1}.$$

The notation means that the position is x_1 at time t_1 and then x_2 at time t_2 . A common unit for v_{avg} is the meter per second (m/s). You may see other units in the problems, but they are always in the form of length/time.

Graphs. On a graph of x versus t , v_{avg} is the **slope** of the straight line that connects two particular points on the $x(t)$ curve: one is the point that corresponds to x_2 and t_2 , and the other is the point that corresponds to x_1 and t_1 . Like displacement, v_{avg} has both magnitude and direction (it is another vector quantity). Its magnitude is the magnitude of the line's slope. A positive v_{avg} (and slope) tells us that the line slants upward to the right; a negative v_{avg} (and slope) tells us that the line slants downward to the right. The average velocity v_{avg} always has the same sign as the displacement Δx because Δt in Eq. above is always positive. Figure below shows how to find v_{avg} in Fig. above for the time interval $t = 1 \text{ s}$ to $t = 4 \text{ s}$. We draw the straight line that connects the point on the position curve at the beginning of the interval and the point on the curve at the end of the interval. Then we find the slope $\Delta x / \Delta t$ of the straight line. For the given time interval, the average velocity is:

$$v_{\text{avg}} = \frac{6 \text{ m}}{3 \text{ s}} = 2 \text{ m/s}.$$

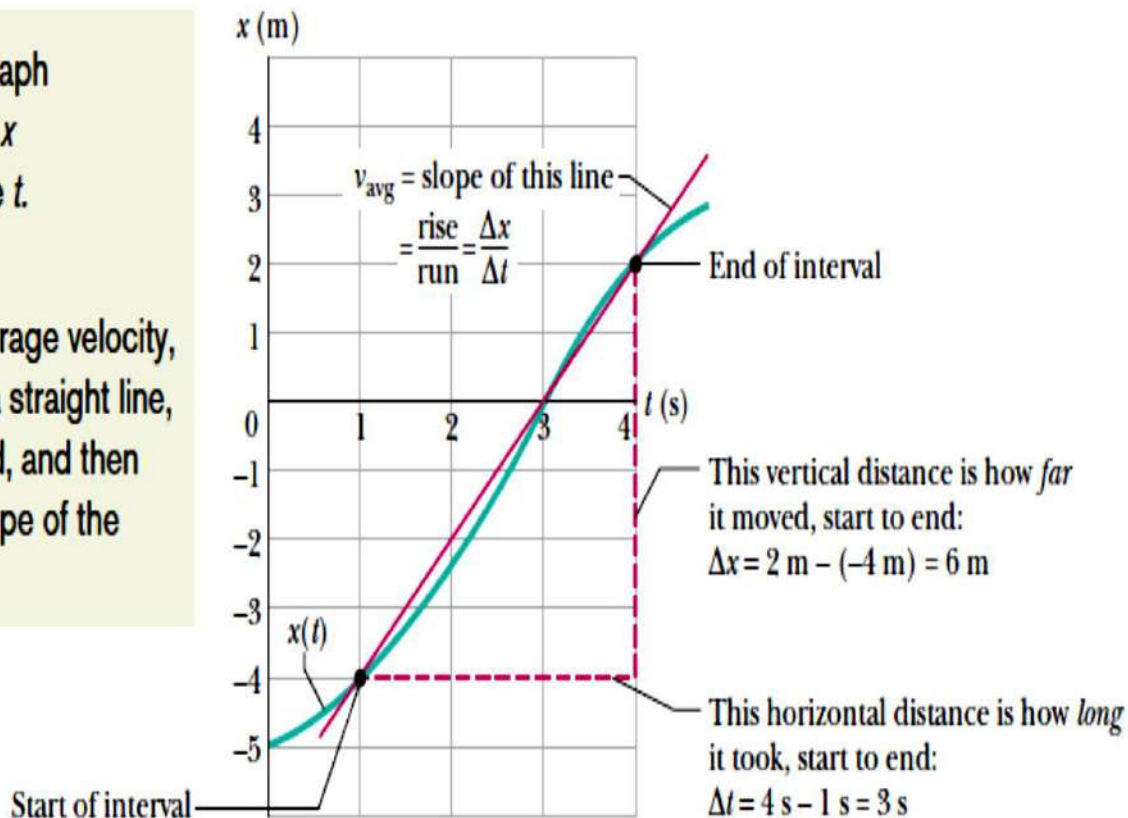
Average speed s_{avg} is a different way of describing “how fast” a particle moves. Whereas the average velocity involves the particle's displacement Δx , the average speed involves the total distance covered (for example, the number of meters moved), independent of direction; that is,

$$s_{\text{avg}} = \frac{\text{total distance}}{\Delta t}.$$

Because average speed does *not* include direction, it lacks any algebraic sign. Sometimes s_{avg} is the same (except for the absence of a sign) as v_{avg} . However, the two can be quite different.

This is a graph of position x versus time t .

To find average velocity, first draw a straight line, start to end, and then find the slope of the line.



Example: You drive a beat-up pickup truck along a straight road for 8.4 km at 70 km/h, at which point the truck runs out of gasoline and stops. Over the next 30 min, you walk another 2.0 km farther along the road to a gasoline station.

(a) What is your overall displacement from the beginning of your drive to your arrival at the station?

Assume, for convenience, that you move in the positive direction of an x -axis, from a first position of $x_1 = 0$ to a second position of x_2 at the station. That second position must be at

$x_2 = 8.4 \text{ km} + 2.0 \text{ km} = 10.4 \text{ km}$. Then your displacement Δx along the x -axis is the second position minus the first position. **Calculation:** From Eq., we have

$$\Delta x = x_2 - x_1 = 10.4 \text{ km} - 0 = 10.4 \text{ km. (Answer)}$$

Thus, your overall displacement is 10.4 km in the positive direction of the x -axis.

(b) What is the time interval Δt from the beginning of your drive to your arrival at the station?

Calculation: Here we find:

$$\begin{aligned} v_{\text{avg}} &= \frac{\Delta x}{\Delta t} = \frac{10.4 \text{ km}}{0.62 \text{ h}} \\ &= 16.8 \text{ km/h} \approx 17 \text{ km/h.} \end{aligned}$$

To find v_{avg} graphically, first we graph the function $x(t)$ as shown in Fig. below, where the beginning and arrival points on the graph are the origin and the point labeled as “Station.” Your average velocity is the slope of the straight line connecting those points; that is, v_{avg} is the ratio of the *rise* ($\Delta x = 10.4 \text{ km}$) to the *run* ($\Delta t = 0.62 \text{ h}$), which gives us $v_{\text{avg}} 16.8 \text{ km/h}$.

(d) Suppose that to pump the gasoline, pay for it, and walk back to the truck takes you another 45 min. What is your average speed from the beginning of your drive to your return to the truck with the gasoline?

Your average speed is the ratio of the total distance you move to the total time interval you take to make that move.

Calculation: The total distance is $8.4 \text{ km} + 2.0 \text{ km} + 2.0 \text{ km} = 12.4 \text{ km}$. The total time interval is $0.12 \text{ h} + 0.50 \text{ h} + 0.75 \text{ h} = 1.37 \text{ h}$. Thus:

$$s_{\text{avg}} = \frac{12.4 \text{ km}}{1.37 \text{ h}} = 9.1 \text{ km/h.}$$

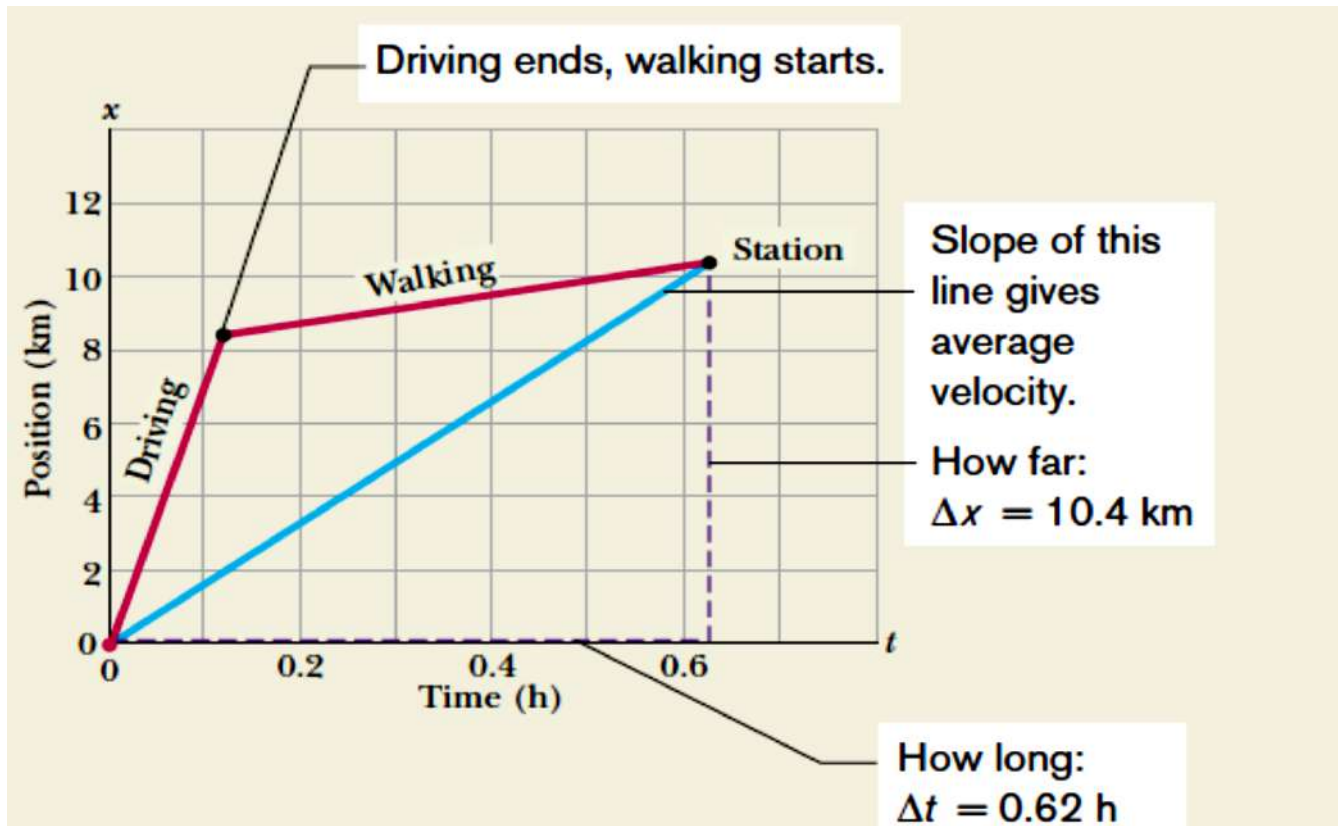


Figure above shows the lines marked “Driving” and “Walking” are the position–time plots for the driving and walking stages. (The plot for the walking stage assumes a constant rate of walking.) The slope of the straight line joining the origin and the point labeled “Station” is the average velocity for the trip, from the beginning to the station.



Ministry of Higher Education and Scientific Research
Middle Technical University
Electrical Engineering Technical College
Department of Cybersecurity Technology Engineering



General Physics

Lecture Seven

First Stage

Motion in Two or Three Dimension

By

Lecturer: Dr. Thamir. Al-hamadany

8. Motion in Two or Three Dimension

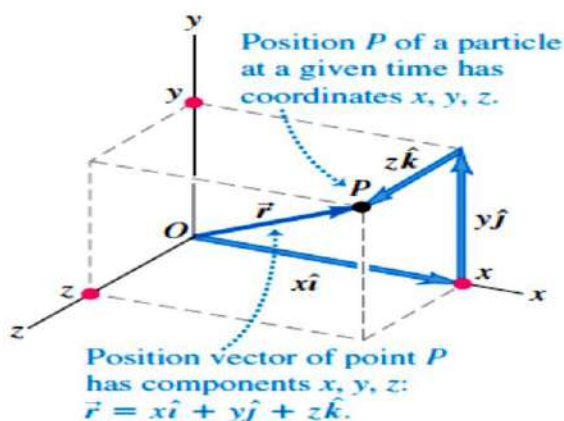
What determines where a batted baseball lands? How do you describe the motion of a roller coaster car along a curved track or the flight of a circling hawk? Which hits the ground first: a baseball that you simply drop or one that you throw horizontally? We cannot answer these kinds of questions using techniques that were relevant when particles moved only along a straight line. Instead, we need to extend our descriptions of motion to two- and three-dimensional situations.

Position and Velocity Vectors

To describe the *motion* of a particle in space, we must first be able to describe the particle's *position*. Consider a particle that is at a point P at a certain instant. The **position vector** $\vec{r}$ of the particle at this instant is a vector that goes from the origin of the coordinate system to the point P (Fig. 1). The Cartesian coordinates x , y , and z of point P are the x -, y -, and z -components of vector. Using unit vectors we can write:

$$\vec{r} = x\hat{i} + y\hat{j} + z\hat{k} \quad (\text{position vector})$$

1 The position vector $\vec{r}$ from the origin to point P has components x , y , and z . The path that the particle follows through space is in general a curve (Fig. 2).



During a time interval Δt the particle moves from P_1 , where its position vector is $\vec{r}_1$, to P_2 , where its position vector is $\vec{r}_2$. The change in position (the displacement) during this interval is $\Delta\vec{r} = \vec{r}_2 - \vec{r}_1 = (x_2 - x_1)\hat{i} + (y_2 - y_1)\hat{j} + (z_2 - z_1)\hat{k}$. We define the **average velocity** $\vec{v}_{av}$ during this interval in the same way we did in for straight-line motion, as the displacement divided by the time interval:

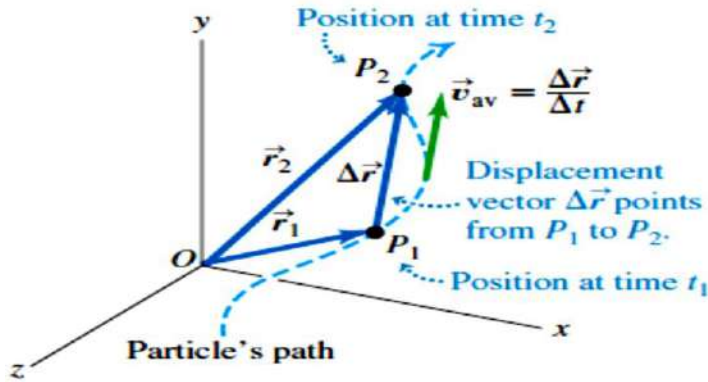
$$\vec{v}_{av} = \frac{\vec{r}_2 - \vec{r}_1}{t_2 - t_1} = \frac{\Delta\vec{r}}{\Delta t} \quad (\text{average velocity vector}) \quad (2)$$

Dividing a vector by a scalar is really a special case of *multiplying* a vector by a scalar, the average velocity $\vec{v}_{av}$ is equal to the displacement vector $\Delta\vec{r}$ multiplied by $1/\Delta t$, the reciprocal of the time interval. Note that the x -component of Eq. (2) is $v_{av-x} = (x_2 - x_1)/(t_2 - t_1) = \Delta x/\Delta t$. This is just the expression for average x -velocity that we found in for one-dimensional motion.

We now define **instantaneous velocity**: It is the limit of the average velocity as the time interval approaches zero, and it equals the instantaneous rate of change of position with time. The key difference is that position $\vec{r}$ and instantaneous velocity $\vec{v}$ are now both vectors:

$$\vec{v} = \lim_{\Delta t \rightarrow 0} \frac{\Delta\vec{r}}{\Delta t} = \frac{d\vec{r}}{dt} \quad (\text{instantaneous velocity vector}) \quad (3)$$

2 The average velocity $\vec{v}_{av}$ between points P_1 and P_2 has the same direction as the displacement $\Delta\vec{r}$.



The *magnitude* of the vector $\vec{v}$ at any instant is the *speed* v of the particle at that instant. The *direction* of $\vec{v}$ at any instant is the same as the direction in which the particle is moving at that instant.

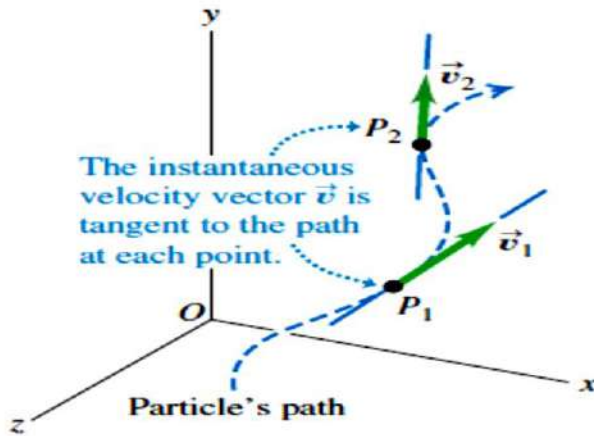
Note that as $\Delta t \rightarrow 0$, points P_1 and P_2 in Fig. 2 move closer and closer together. In this limit, the vector $\Delta\vec{r}$ becomes tangent to the path. The direction of $\Delta\vec{r}$ in this limit is also the direction of the instantaneous velocity $\vec{v}$. This leads to an important conclusion: *At every point along the path, the instantaneous velocity vector is tangent to the path at that point* (Fig. 3).

It's often easiest to calculate the instantaneous velocity vector using components. During any displacement $\Delta\vec{r}$, the changes Δx , Δy , and Δz in the three coordinates of the particle are the *components* of $\Delta\vec{r}$. It follows that the components v_x , v_y , and v_z of the instantaneous velocity $\vec{v}$ are simply the time derivatives of the coordinates x , y , and z . That is,

$$v_x = \frac{dx}{dt} \quad v_y = \frac{dy}{dt} \quad v_z = \frac{dz}{dt} \quad \text{(components of instantaneous velocity)} \quad (4)$$

The x -component of $\vec{v}$ is $v_x = dx/dt$, is the same as the expression for instantaneous velocity for straight-line motion. Hence Eq. (4) is a direct extension of the idea of instantaneous velocity to motion in three dimensions.

3 The vectors $\vec{v}_1$ and $\vec{v}_2$ are the instantaneous velocities at the points P_1 and P_2 shown in Fig. 2.



We can also get Eq. (4) by taking the derivative of Eq. (1). The unit vectors $\hat{i}$, $\hat{j}$, and $\hat{k}$ are constant in magnitude and direction, so their derivatives are zero, and we find

$$\vec{v} = \frac{d\vec{r}}{dt} = \frac{dx}{dt}\hat{i} + \frac{dy}{dt}\hat{j} + \frac{dz}{dt}\hat{k} \quad (5)$$

This shows again that the components of $\vec{v}$ are dx/dt , dy/dt , and dz/dt .

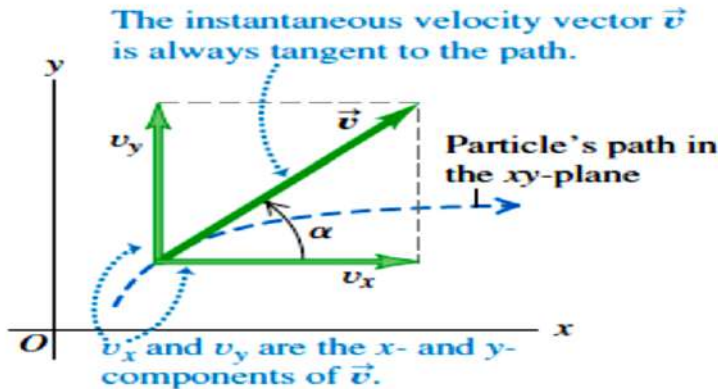
The magnitude of the instantaneous velocity vector $\vec{v}$ —that is, the speed—is given in terms of the components v_x , v_y , and v_z by the Pythagorean relation:

$$|\vec{v}| = v = \sqrt{v_x^2 + v_y^2 + v_z^2} \quad (6)$$

Figure 4 shows the situation when the particle moves in the xy -plane. In this case, z and v_z are zero. Then the speed (the magnitude of $\vec{v}$) is

$$v = \sqrt{v_x^2 + v_y^2}$$

4 The two velocity components for motion in the xy -plane.



and the direction of the instantaneous velocity $\vec{v}$ is given by the angle α (the Greek letter alpha) in the figure. We see that

$$\tan \alpha = \frac{v_y}{v_x} \quad (7)$$

(We always use Greek letters for angles. We use α for the direction of the instantaneous velocity vector to avoid confusion with the direction θ of the *position* vector of the particle.)

The instantaneous velocity vector is usually more interesting and useful than the average velocity vector. From now on, when we use the word “velocity,” we will always mean the instantaneous velocity vector $\vec{v}$ (rather than the average velocity vector). Usually, we won’t even bother to call $\vec{v}$ a vector; it’s up to you to remember that velocity is a vector quantity with both magnitude and direction.

A robotic vehicle, or rover, is exploring the surface of Mars. The stationary Mars lander is the origin of coordinates, and the surrounding Martian surface lies in the xy -plane. The rover, which we represent as a point, has x - and y -coordinates that vary with time:

$$x = 2.0 \text{ m} - (0.25 \text{ m/s}^2)t^2$$

$$y = (1.0 \text{ m/s})t + (0.025 \text{ m/s}^3)t^3$$

(a) Find the rover's coordinates and distance from the lander at $t = 2.0 \text{ s}$. (b) Find the rover's displacement and average velocity vectors for the interval $t = 0.0 \text{ s}$ to $t = 2.0 \text{ s}$. (c) Find a general expression for the rover's instantaneous velocity vector $\vec{v}$. Express $\vec{v}$ at $t = 2.0 \text{ s}$ in component form and in terms of magnitude and direction.

SOLUTION

IDENTIFY and SET UP: This problem involves motion in two dimensions, so we must use the vector equations obtained in this section. Figure 5 shows the rover's path (dashed line). We'll use Eq. (1) for position $\vec{r}$, the expression $\Delta\vec{r} = \vec{r}_2 - \vec{r}_1$ for displacement, Eq. (2) for average velocity, and Eqs. (5), (6), and (7) for instantaneous velocity and its magnitude and direction. The target variables are stated in the problem.

EXECUTE: (a) At $t = 2.0 \text{ s}$ the rover's coordinates are

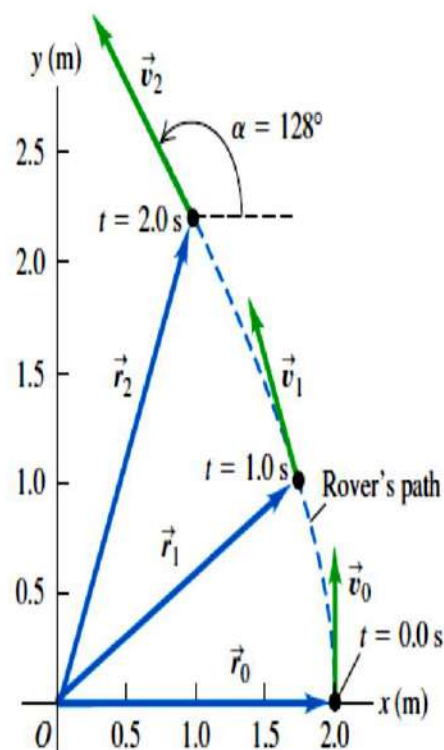
$$x = 2.0 \text{ m} - (0.25 \text{ m/s}^2)(2.0 \text{ s})^2 = 1.0 \text{ m}$$

$$y = (1.0 \text{ m/s})(2.0 \text{ s}) + (0.025 \text{ m/s}^3)(2.0 \text{ s})^3 = 2.2 \text{ m}$$

The rover's distance from the origin at this time is

$$r = \sqrt{x^2 + y^2} = \sqrt{(1.0 \text{ m})^2 + (2.2 \text{ m})^2} = 2.4 \text{ m}$$

5 At $t = 0.0 \text{ s}$ the rover has position vector $\vec{r}_0$ and instantaneous velocity vector $\vec{v}_0$. Likewise, $\vec{r}_1$ and $\vec{v}_1$ are the vectors at $t = 1.0 \text{ s}$; $\vec{r}_2$ and $\vec{v}_2$ are the vectors at $t = 2.0 \text{ s}$.



Continued

(b) To find the displacement and average velocity over the given time interval, we first express the position vector $\vec{r}$ as a function of time t . From Eq. (1) this is

$$\begin{aligned}\vec{r} &= x\hat{i} + y\hat{j} \\ &= [2.0 \text{ m} - (0.25 \text{ m/s}^2)t^2]\hat{i} \\ &\quad + [(1.0 \text{ m/s})t + (0.025 \text{ m/s}^3)t^3]\hat{j}\end{aligned}$$

At $t = 0.0 \text{ s}$ the position vector $\vec{r}_0$ is

$$\vec{r}_0 = (2.0 \text{ m})\hat{i} + (0.0 \text{ m})\hat{j}$$

From part (a), the position vector $\vec{r}_2$ at $t = 2.0 \text{ s}$ is

$$\vec{r}_2 = (1.0 \text{ m})\hat{i} + (2.2 \text{ m})\hat{j}$$

The displacement from $t = 0.0 \text{ s}$ to $t = 2.0 \text{ s}$ is therefore

$$\begin{aligned}\Delta\vec{r} &= \vec{r}_2 - \vec{r}_0 = (1.0 \text{ m})\hat{i} + (2.2 \text{ m})\hat{j} - (2.0 \text{ m})\hat{i} \\ &= (-1.0 \text{ m})\hat{i} + (2.2 \text{ m})\hat{j}\end{aligned}$$

During this interval the rover moves 1.0 m in the negative x -direction and 2.2 m in the positive y -direction. From Eq. (2), the average velocity over this interval is the displacement divided by the elapsed time:

$$\begin{aligned}\vec{v}_{\text{av}} &= \frac{\Delta\vec{r}}{\Delta t} = \frac{(-1.0 \text{ m})\hat{i} + (2.2 \text{ m})\hat{j}}{2.0 \text{ s} - 0.0 \text{ s}} \\ &= (-0.50 \text{ m/s})\hat{i} + (1.1 \text{ m/s})\hat{j}\end{aligned}$$

The components of this average velocity are $v_{\text{av-}x} = -0.50 \text{ m/s}$ and $v_{\text{av-}y} = 1.1 \text{ m/s}$

(c) From Eq. (4) the components of *instantaneous* velocity are the time derivatives of the coordinates:

$$v_x = \frac{dx}{dt} = (-0.25 \text{ m/s}^2)(2t)$$

$$v_y = \frac{dy}{dt} = 1.0 \text{ m/s} + (0.025 \text{ m/s}^3)(3t^2)$$

Hence the instantaneous velocity vector is

$$\begin{aligned}\vec{v} &= v_x \hat{i} + v_y \hat{j} = (-0.50 \text{ m/s}^2)t \hat{i} \\ &\quad + [1.0 \text{ m/s} + (0.075 \text{ m/s}^3)t^2] \hat{j}\end{aligned}$$

At $t = 2.0 \text{ s}$ the velocity vector $\vec{v}_2$ has components

$$v_{2x} = (-0.50 \text{ m/s}^2)(2.0 \text{ s}) = -1.0 \text{ m/s}$$

$$v_{2y} = 1.0 \text{ m/s} + (0.075 \text{ m/s}^3)(2.0 \text{ s})^2 = 1.3 \text{ m/s}$$

The magnitude of the instantaneous velocity (that is, the speed) at $t = 2.0 \text{ s}$ is

$$\begin{aligned}v_2 &= \sqrt{v_{2x}^2 + v_{2y}^2} = \sqrt{(-1.0 \text{ m/s})^2 + (1.3 \text{ m/s})^2} \\ &= 1.6 \text{ m/s}\end{aligned}$$

Figure 5 shows the direction of the velocity vector $\vec{v}_2$, which is at an angle α between 90° and 180° with respect to the positive x -axis. From Eq. (7) we have

$$\arctan \frac{v_y}{v_x} = \arctan \frac{1.3 \text{ m/s}}{-1.0 \text{ m/s}} = -52^\circ$$

This is off by 180° ; the correct value of the angle is $\alpha = 180^\circ - 52^\circ = 128^\circ$, or 38° west of north.

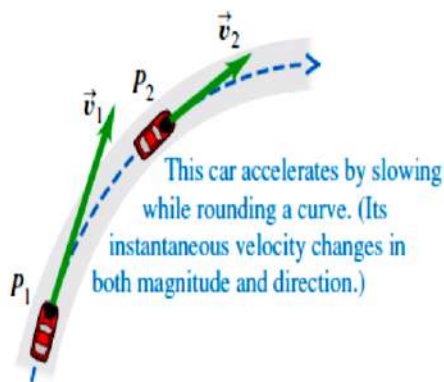
2 The Acceleration Vector

Now let's consider the *acceleration* of a particle moving in space. Just as for motion in a straight line, acceleration describes how the velocity of the particle changes. But since we now treat velocity as a vector, acceleration will describe changes in the velocity magnitude (that is, the speed) *and* changes in the direction of velocity (that is, the direction in which the particle is moving).

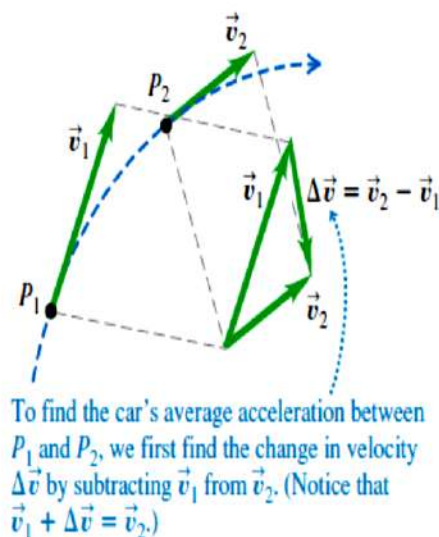
In Fig. 6a, a car (treated as a particle) is moving along a curved road. The vectors $\vec{v}_1$ and $\vec{v}_2$ represent the car's instantaneous velocities at time t_1 , when the car

6 (a) A car moving along a curved road from P_1 to P_2 . (b) How to obtain the change in velocity $\Delta\vec{v} = \vec{v}_2 - \vec{v}_1$ by vector subtraction. (c) The vector $\vec{a}_{av} = \Delta\vec{v}/\Delta t$ represents the average acceleration between P_1 and P_2 .

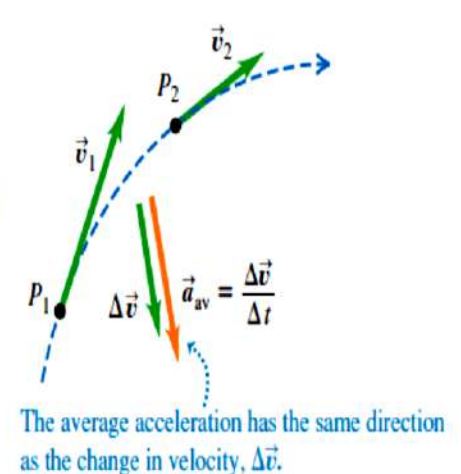
(a)



(b)



(c)



is at point P_1 , and at time t_2 , when the car is at point P_2 . The two velocities may differ in both magnitude and direction. During the time interval from t_1 to t_2 , the *vector change in velocity* is $\vec{v}_2 - \vec{v}_1 = \Delta\vec{v}$, so $\vec{v}_2 = \vec{v}_1 + \Delta\vec{v}$ (Fig. 6b). We define the **average acceleration** $\vec{a}_{av}$ of the car during this time interval as the velocity change divided by the time interval $t_2 - t_1 = \Delta t$:

$$\vec{a}_{av} = \frac{\vec{v}_2 - \vec{v}_1}{t_2 - t_1} = \frac{\Delta\vec{v}}{\Delta t} \quad (\text{average acceleration vector}) \quad (8)$$

Average acceleration is a *vector* quantity in the same direction as the vector $\Delta\vec{v}$ (Fig. 6c). The x -component of Eq. (8) is $a_{av-x} = (v_{2x} - v_{1x})/(t_2 - t_1) = \Delta v_x/\Delta t$, which is just the equation for the average acceleration in straight-line motion.

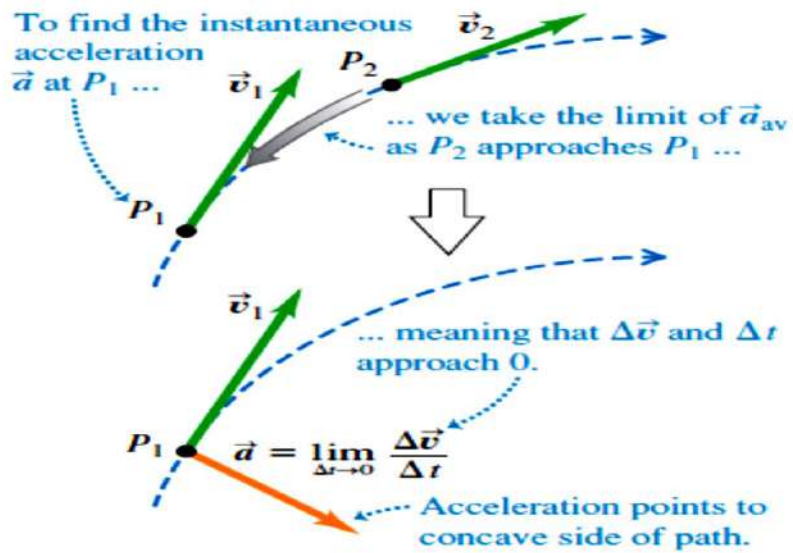
We define the **instantaneous acceleration** $\vec{a}$ (a *vector* quantity) at point P_1 as the limit of the average acceleration vector when point P_2 approaches point P_1 , so $\Delta\vec{v}$ and Δt both approach zero (Fig. 7). The instantaneous acceleration is also equal to the instantaneous rate of change of velocity with time:

$$\vec{a} = \lim_{\Delta t \rightarrow 0} \frac{\Delta\vec{v}}{\Delta t} = \frac{d\vec{v}}{dt} \quad (\text{instantaneous acceleration vector}) \quad (9)$$

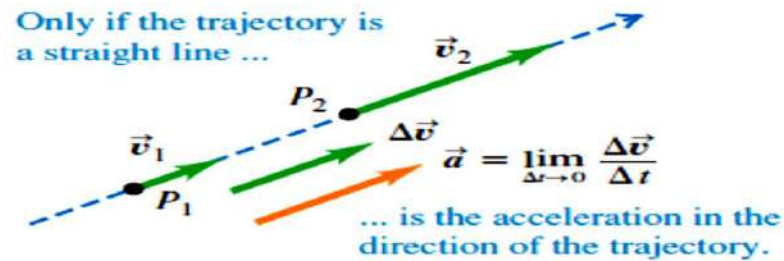
The velocity vector $\vec{v}$, as we have seen, is tangent to the path of the particle. The instantaneous acceleration vector $\vec{a}$, however, does *not* have to be tangent to the path. Figure 7a shows that if the path is curved, $\vec{a}$ points toward the concave side of the path—that is, toward the inside of any turn that the particle is making. The acceleration is tangent to the path only if the particle moves in a straight line (Fig. 7b).

7 (a) Instantaneous acceleration $\vec{a}$ at point P_1 in Fig. 6. **(b)** Instantaneous acceleration for motion along a straight line.

(a) Acceleration: curved trajectory



(b) Acceleration: straight-line trajectory



CAUTION Any particle following a curved path is accelerating. When a particle is moving in a curved path, it always has nonzero acceleration, even when it moves with constant speed. This conclusion may seem contrary to your intuition, but it's really just contrary to the everyday use of the word "acceleration" to mean that speed is increasing. The more precise definition given in Eq. (9) shows that there is a nonzero acceleration whenever the velocity vector changes in any way, whether there is a change of speed, direction, or both. **I**

To convince yourself that a particle has a nonzero acceleration when moving on a curved path with constant speed, think of your sensations when you ride in a car. When the car accelerates, you tend to move inside the car in a

direction *opposite* to the car's acceleration. Thus you tend to slide toward the back of the car when it accelerates forward (speeds up) and toward the front of the car when it accelerates backward (slows down). If the car makes a turn on a level road, you tend to slide toward the outside of the turn; hence the car has an acceleration toward the inside of the turn.

We will usually be interested in the instantaneous acceleration, not the average acceleration. From now on, we will use the term "acceleration" to mean the instantaneous acceleration vector $\vec{a}$.

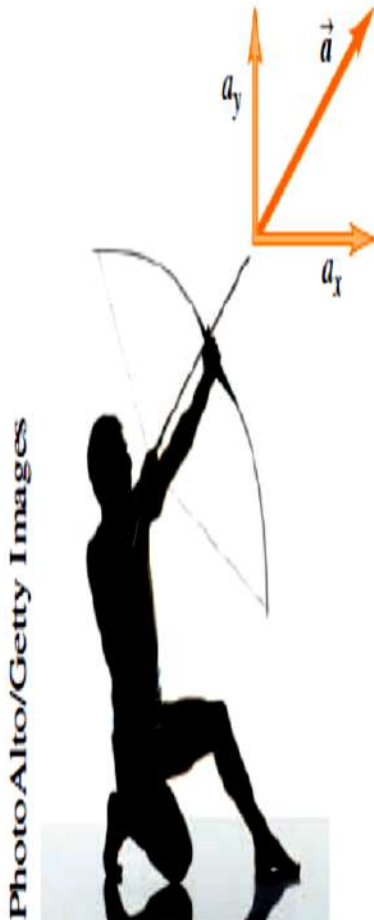
Each component of the acceleration vector is the derivative of the corresponding component of velocity:

$$a_x = \frac{dv_x}{dt} \quad a_y = \frac{dv_y}{dt} \quad a_z = \frac{dv_z}{dt} \quad (\text{components of instantaneous acceleration}) \quad (10)$$

In terms of unit vectors,

$$\vec{a} = \frac{dv_x}{dt} \hat{i} + \frac{dv_y}{dt} \hat{j} + \frac{dv_z}{dt} \hat{k} \quad (11)$$

8 When the arrow is released, its acceleration vector has both a horizontal component (a_x) and a vertical component (a_y).



The x -component of Eqs. (10) and (11), $a_x = dv_x/dt$, is the expression for instantaneous acceleration in one dimension. Figure 8 shows an example of an acceleration vector that has both x - and y -components.

Since each component of velocity is the derivative of the corresponding coordinate, we can express the components a_x , a_y , and a_z of the acceleration vector $\vec{a}$ as

$$a_x = \frac{d^2x}{dt^2} \quad a_y = \frac{d^2y}{dt^2} \quad a_z = \frac{d^2z}{dt^2} \quad (12)$$

The acceleration vector $\vec{a}$ itself is

$$\vec{a} = \frac{d^2x}{dt^2}\hat{i} + \frac{d^2y}{dt^2}\hat{j} + \frac{d^2z}{dt^2}\hat{k} \quad (13)$$

Let's return to the motions of the Mars rover in Example 1. (a) Find the components of the average acceleration for the interval $t = 0.0$ s to $t = 2.0$ s. (b) Find the instantaneous acceleration at $t = 2.0$ s.

SOLUTION

IDENTIFY and SET UP: In Example 1 we found the components of the rover's instantaneous velocity at any time t :

$$v_x = \frac{dx}{dt} = (-0.25 \text{ m/s}^2)(2t) = (-0.50 \text{ m/s}^2)t$$

$$v_y = \frac{dy}{dt} = 1.0 \text{ m/s} + (0.025 \text{ m/s}^3)(3t^2)$$

$$= 1.0 \text{ m/s} + (0.075 \text{ m/s}^3)t^2$$

We'll use the vector relationships among velocity, average acceleration, and instantaneous acceleration. In part (a) we determine the values of v_x and v_y at the beginning and end of the interval and

then use Eq. (8) to calculate the components of the average acceleration. In part (b) we obtain expressions for the instantaneous acceleration components at any time t by taking the time derivatives of the velocity components as in Eqs. (10).

EXECUTE: (a) In Example 1 we found that at $t = 0.0$ s the velocity components are

$$v_x = 0.0 \text{ m/s} \quad v_y = 1.0 \text{ m/s}$$

and that at $t = 2.00$ s the components are

$$v_x = -1.0 \text{ m/s} \quad v_y = 1.3 \text{ m/s}$$

Thus the components of average acceleration in the interval $t = 0.0$ s to $t = 2.0$ s are

$$a_{\text{av-}x} = \frac{\Delta v_x}{\Delta t} = \frac{-1.0 \text{ m/s} - 0.0 \text{ m/s}}{2.0 \text{ s} - 0.0 \text{ s}} = -0.50 \text{ m/s}^2$$

$$a_{\text{av-}y} = \frac{\Delta v_y}{\Delta t} = \frac{1.3 \text{ m/s} - 1.0 \text{ m/s}}{2.0 \text{ s} - 0.0 \text{ s}} = 0.15 \text{ m/s}^2$$

(b) Using Eqs. (10), we find

$$a_x = \frac{dv_x}{dt} = -0.50 \text{ m/s}^2 \quad a_y = \frac{dv_y}{dt} = (0.075 \text{ m/s}^3)(2t)$$

Hence the instantaneous acceleration vector $\vec{a}$ at time t is

$$\vec{a} = a_x \hat{i} + a_y \hat{j} = (-0.50 \text{ m/s}^2) \hat{i} + (0.15 \text{ m/s}^3)t \hat{j}$$

At $t = 2.0 \text{ s}$ the components of acceleration and the acceleration vector are

$$a_x = -0.50 \text{ m/s}^2 \quad a_y = (0.15 \text{ m/s}^3)(2.0 \text{ s}) = 0.30 \text{ m/s}^2$$

$$\vec{a} = (-0.50 \text{ m/s}^2) \hat{i} + (0.30 \text{ m/s}^2) \hat{j}$$

The magnitude of acceleration at this time is

$$a = \sqrt{a_x^2 + a_y^2}$$

$$= \sqrt{(-0.50 \text{ m/s}^2)^2 + (0.30 \text{ m/s}^2)^2} = 0.58 \text{ m/s}^2$$

A sketch of this vector (Fig. 9) shows that the direction angle β of $\vec{a}$ with respect to the positive x -axis is between 90° and 180° . From Eq. (7) we have

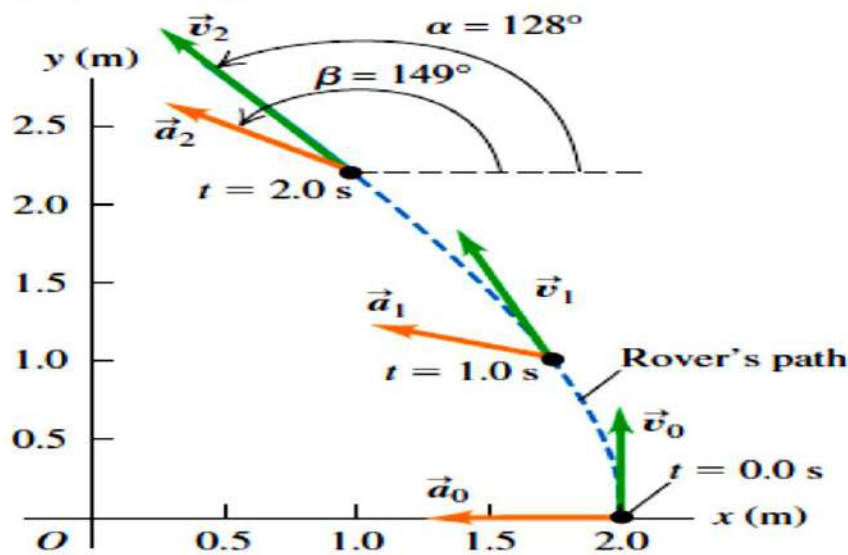
$$\arctan \frac{a_y}{a_x} = \arctan \frac{0.30 \text{ m/s}^2}{-0.50 \text{ m/s}^2} = -31^\circ$$

Hence $\beta = 180^\circ + (-31^\circ) = 149^\circ$

EVALUATE: Figure 9 shows the rover's path and the velocity and acceleration vectors at $t = 0.0 \text{ s}$, 1.0 s , and 2.0 s . (You should use

the results of part (b) to calculate the instantaneous acceleration at $t = 0.0$ s and $t = 1.0$ s for yourself.) Note that $\vec{v}$ and $\vec{a}$ are *not* in the same direction at any of these times. The velocity vector $\vec{v}$ is tangent to the path at each point (as is always the case), and the acceleration vector $\vec{a}$ points toward the concave side of the path.

9 The path of the robotic rover, showing the velocity and acceleration at $t = 0.0$ s ($\vec{v}_0$ and $\vec{a}_0$), $t = 1.0$ s ($\vec{v}_1$ and $\vec{a}_1$), and $t = 2.0$ s ($\vec{v}_2$ and $\vec{a}_2$).



3 Projectile Motion

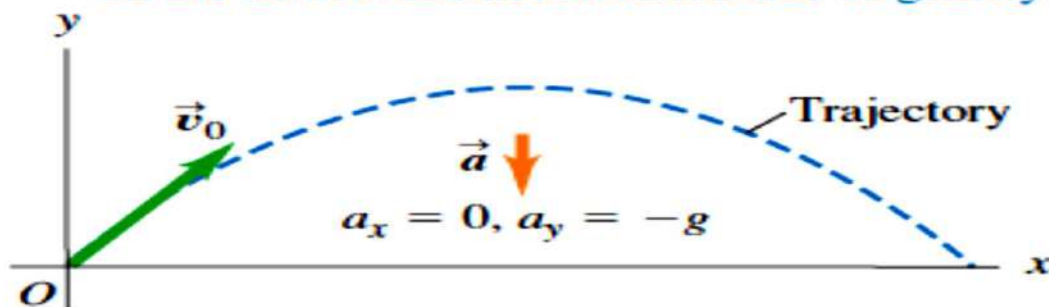
A **projectile** is any body that is given an initial velocity and then follows a path determined entirely by the effects of gravitational acceleration and air resistance. A batted baseball, a thrown football, a package dropped from an airplane, and a bullet shot from a rifle are all projectiles. The path followed by a projectile is called its **trajectory**.

To analyze this common type of motion, we'll start with an idealized model, representing the projectile as a particle with an acceleration (due to gravity) that is constant in both magnitude and direction. We'll neglect the effects of air resistance and the curvature and rotation of the earth. Like all models, this one has limitations. Curvature of the earth has to be considered in the flight of long-range missiles, and air resistance is of crucial importance to a sky diver. Nevertheless, we can learn a lot from analysis of this simple model. For the remainder of this chapter the phrase "projectile motion" will imply that we're ignoring air resistance.

Projectile motion is always confined to a vertical plane determined by the direction of the initial velocity (Fig. 15). This is because the acceleration due to

15 The trajectory of an idealized projectile.

- A projectile moves in a vertical plane that contains the initial velocity vector $\vec{v}_0$.
- Its trajectory depends only on $\vec{v}_0$ and on the downward acceleration due to gravity.



gravity is purely vertical; gravity can't accelerate the projectile sideways. Thus projectile motion is *two-dimensional*. We will call the plane of motion the *xy*-coordinate plane, with the *x*-axis horizontal and the *y*-axis vertically upward.

The key to analyzing projectile motion is that we can treat the *x*- and *y*-coordinates separately. The *x*-component of acceleration is zero, and the *y*-component is constant and equal to $-g$. (By definition, g is always positive; with our choice of coordinate directions, a_y is negative.) So *we can analyze projectile motion as a combination of horizontal motion with constant velocity and vertical motion with constant acceleration*. Figure 16 shows two projectiles with different *x*-motion but identical *y*-motion; one is dropped from rest and the other is projected horizontally, but both projectiles fall the same distance in the same time.

We can then express all the vector relationships for the projectile's position, velocity, and acceleration by separate equations for the horizontal and vertical components. The components of $\vec{a}$ are

$$a_x = 0 \quad a_y = -g \quad (\text{projectile motion, no air resistance}) \quad (14)$$

Since the *x*-acceleration and *y*-acceleration are both constant, we can use the relevant equations directly. For example, suppose that at time $t = 0$ our particle is at the point (x_0, y_0) and that at this time its velocity components have the initial values v_{0x} and v_{0y} . The components of acceleration are $a_x = 0$, $a_y = -g$. Considering the *x*-motion first, we find

$$v_x = v_{0x} \quad (15)$$

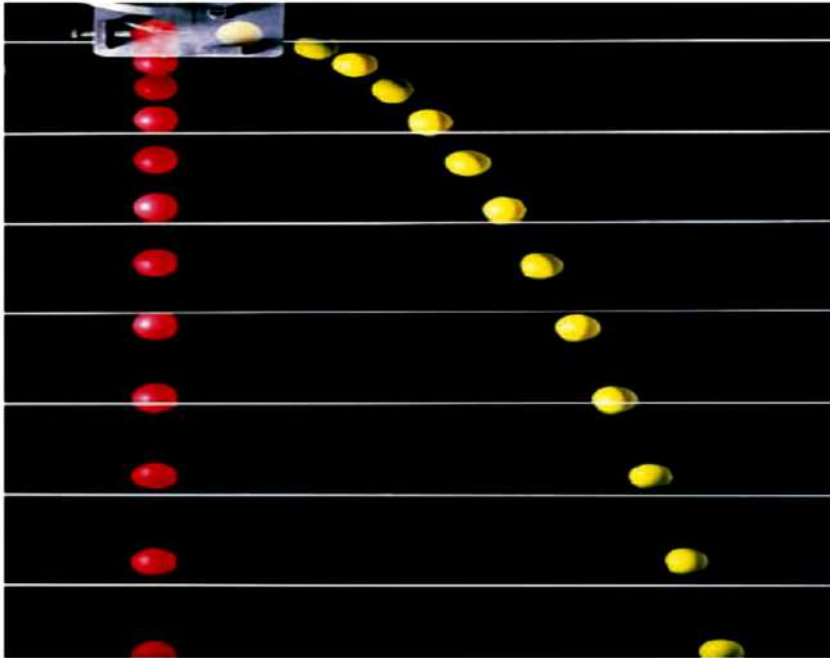
$$x = x_0 + v_{0x}t \quad (16)$$

For the *y*-motion we substitute *y* for *x*, v_y for v_x , v_{0y} for v_{0x} , and $a_y = -g$ for a_x :

$$v_y = v_{0y} - gt \quad (17)$$

$$y = y_0 + v_{0y}t - \frac{1}{2}gt^2 \quad (18)$$

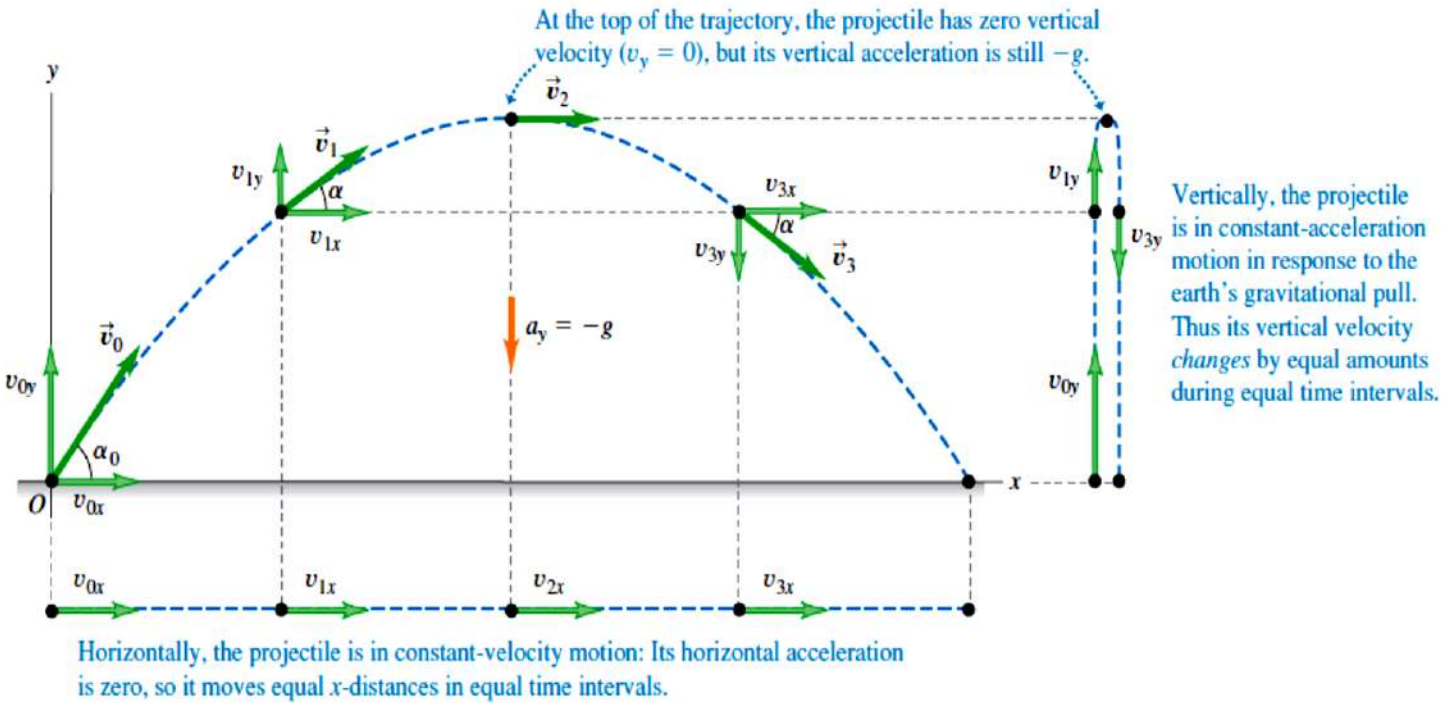
16 The red ball is dropped from rest, and the yellow ball is simultaneously projected horizontally; successive images in this stroboscopic photograph are separated by equal time intervals. At any given time, both balls have the same y -position, y -velocity, and y -acceleration, despite having different x -positions and x -velocities.



It's usually simplest to take the initial position (at $t = 0$) as the origin; then $x_0 = y_0 = 0$. This might be the position of a ball at the instant it leaves the thrower's hand or the position of a bullet at the instant it leaves the gun barrel.

Figure 17 shows the trajectory of a projectile that starts at (or passes through) the origin at time $t = 0$, along with its position, velocity, and velocity

17 If air resistance is negligible, the trajectory of a projectile is a combination of horizontal motion with constant velocity and vertical motion with constant acceleration.



components at equal time intervals. The x -component of acceleration is zero, so v_x is constant. The y -component of acceleration is constant and not zero, so v_y changes by equal amounts in equal times, just the same as if the projectile were launched vertically with the same initial y -velocity.

We can also represent the initial velocity $\vec{v}_0$ by its magnitude v_0 (the initial speed) and its angle α_0 with the positive x -axis (Fig. 18). In terms of these quantities, the components v_{0x} and v_{0y} of the initial velocity are

$$v_{0x} = v_0 \cos \alpha_0 \quad v_{0y} = v_0 \sin \alpha_0 \quad (19)$$

If we substitute these relationships in Eqs. (15) through (18) and set $x_0 = y_0 = 0$, we find

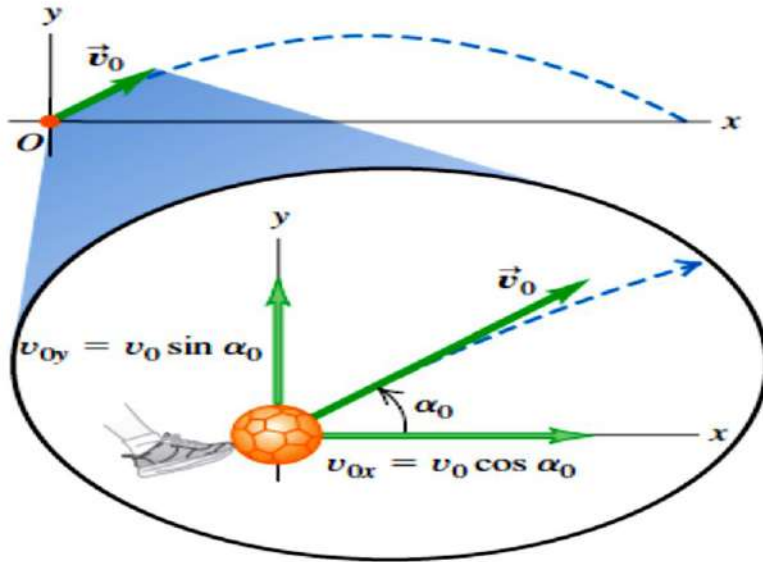
$$x = (v_0 \cos \alpha_0)t \quad (\text{projectile motion}) \quad (20)$$

$$y = (v_0 \sin \alpha_0)t - \frac{1}{2}gt^2 \quad (\text{projectile motion}) \quad (21)$$

$$v_x = v_0 \cos \alpha_0 \quad (\text{projectile motion}) \quad (22)$$

$$v_y = v_0 \sin \alpha_0 - gt \quad (\text{projectile motion}) \quad (23)$$

18 The initial velocity components v_{0x} and v_{0y} of a projectile (such as a kicked soccer ball) are related to the initial speed v_0 and initial angle α_0 .



These equations describe the position and velocity of the projectile in Fig. 17 at any time t .

We can get a lot of information from Eqs. (20) through (23). For example, at any time the distance r of the projectile from the origin (the magnitude of the position vector $\vec{r}$) is given by

$$r = \sqrt{x^2 + y^2} \quad (24)$$

The projectile's speed (the magnitude of its velocity) at any time is

$$v = \sqrt{v_x^2 + v_y^2} \quad (25)$$

The *direction* of the velocity, in terms of the angle α it makes with the positive x -direction (see Fig. 17), is given by

$$\tan \alpha = \frac{v_y}{v_x} \quad (26)$$

The velocity vector $\vec{v}$ is tangent to the trajectory at each point.

We can derive an equation for the trajectory's shape in terms of x and y by eliminating t . From Eqs. (20) and (21), which assume $x_0 = y_0 = 0$, we find $t = x/(v_0 \cos \alpha_0)$ and

$$y = (\tan \alpha_0)x - \frac{g}{2v_0^2 \cos^2 \alpha_0} x^2 \quad (27)$$

Don't worry about the details of this equation; the important point is its general form. Since v_0 , $\tan \alpha_0$, $\cos \alpha_0$, and g are constants, Eq. (27) has the form

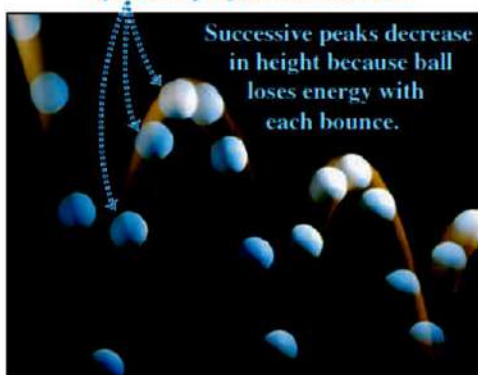
$$y = bx - cx^2$$

where b and c are constants. This is the equation of a *parabola*. In our simple model of projectile motion, the trajectory is always a parabola (Fig. 19).

When air resistance *isn't* always negligible and has to be included, calculating the trajectory becomes a lot more complicated; the effects of air resistance depend on velocity, so the acceleration is no longer constant. Figure 20 shows a

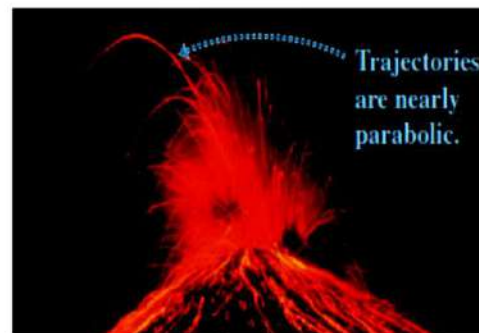
19 The nearly parabolic trajectories of (a) a bouncing ball and (b) blobs of molten rock ejected from a volcano.

(a) Successive images of ball are separated by equal time intervals.



Richard Megna/Fundamental Photographs

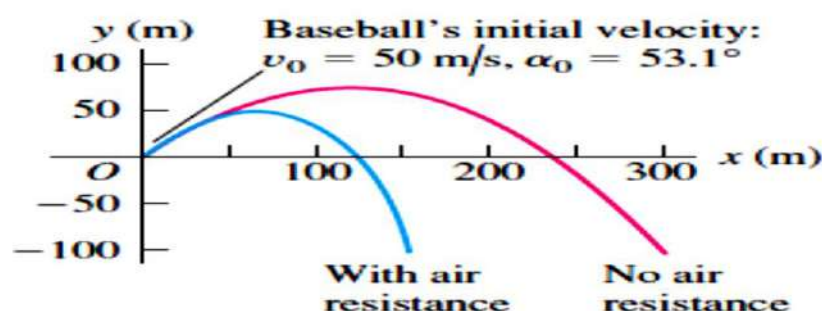
(b)



Fotolia

computer simulation of the trajectory of a baseball both without air resistance and with air resistance proportional to the square of the baseball's speed. We see that air resistance has a very large effect; the maximum height and range both decrease, and the trajectory is no longer a parabola. (If you look closely at Fig. 19b, you'll see that the trajectories of the volcanic blobs deviate in a similar way from a parabolic shape.)

20 Air resistance has a large cumulative effect on the motion of a baseball. In this simulation we allow the baseball to fall below the height from which it was thrown (for example, the baseball could have been thrown from a cliff).



Example 9

Different initial and final heights

You throw a ball from your window 8.0 m above the ground. When the ball leaves your hand, it is moving at 10.0 m/s at an angle of 20° below the horizontal. How far horizontally from your window will the ball hit the ground? Ignore air resistance.

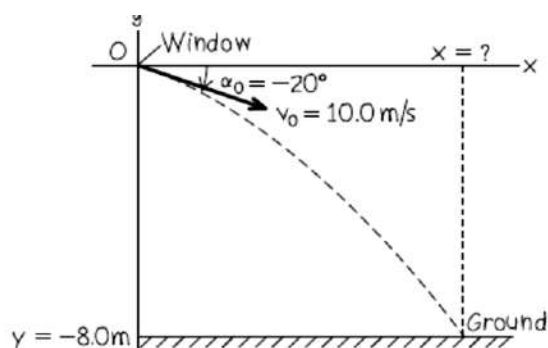


Fig. 25

SOLUTION

IDENTIFY and SET UP: As in Examples 7 and 8, we want to find the horizontal coordinate of a projectile when it is at a given y -value. The difference here is that this value of y is *not* the same as the initial value. We again choose the x -axis to be horizontal and the y -axis to be upward, and place the origin of coordinates at the point where the ball leaves your hand (Fig. 25). We have $v_0 = 10.0 \text{ m/s}$ and $\alpha_0 = -20^\circ$ (the angle is negative because the initial velocity is below the horizontal). Our target variable is the value of x when the ball reaches the ground at $y = -8.0 \text{ m}$. We'll use Eq. (21) to find the time t when this happens, then use Eq. (20) to find the value of x at this time.

EXECUTE: To determine t , we rewrite Eq. (21) in the standard form for a quadratic equation for t :

$$\frac{1}{2}gt^2 - (v_0 \sin \alpha_0)t + y = 0$$

Continued

The roots of this equation are

$$\begin{aligned} t &= \frac{v_0 \sin \alpha_0 \pm \sqrt{(-v_0 \sin \alpha_0)^2 - 4\left(\frac{1}{2}g\right)y}}{2\left(\frac{1}{2}g\right)} \\ &= \frac{v_0 \sin \alpha_0 \pm \sqrt{v_0^2 \sin^2 \alpha_0 - 2gy}}{g} \\ &= \frac{\left[(10.0 \text{ m/s}) \sin(-20^\circ) \pm \sqrt{(10.0 \text{ m/s})^2 \sin^2(-20^\circ) - 2(9.80 \text{ m/s}^2)(-8.0 \text{ m})} \right]}{9.80 \text{ m/s}^2} \\ &= -1.7 \text{ s} \quad \text{or} \quad 0.98 \text{ s} \end{aligned}$$

We discard the negative root, since it refers to a time before the ball left your hand. The positive root tells us that the ball reaches the ground at $t = 0.98$ s. From Eq. (20), the ball's x -coordinate at that time is

$$\begin{aligned}x &= (v_0 \cos \alpha_0)t = (10.0 \text{ m/s})[\cos(-20^\circ)](0.98 \text{ s}) \\&= 9.2 \text{ m}\end{aligned}$$

The ball hits the ground a horizontal distance of 9.2 m from your window.

EVALUATE: The root $t = -1.7$ s is an example of a “fictional” solution to a quadratic equation.



*Ministry of Higher Education and Scientific Research
Middle Technical University
Electrical Engineering Technical College
Department of Cybersecurity Technology Engineering*



General Physics

Lecture Eight

First Stage

Newton's Laws of Motion

By

Lecturer: Dr. Thamir. Al-hamadany

9. Newton's Laws of Motion

What *causes* bodies to move the way that they do? For example, how can a tugboat push a cruise ship that's much heavier than the tug? Why is it harder to control a car on wet ice than on dry concrete? The answers to these and similar questions take us into the subject of **dynamics**, the relationship of motion to the forces that cause it. In this subject, we will use two new concepts, *force* and *mass*, to analyze the principles of dynamics. These principles were clearly stated for the first time by Sir Isaac Newton (1642–1727); today we call them **Newton's laws of motion**. The **first law** states that when the net force on a body is zero, its motion does not change. The **second law** relates force to acceleration when the net force is *not* zero. The **third law** is a relationship between the forces that two interacting bodies exert on each other.

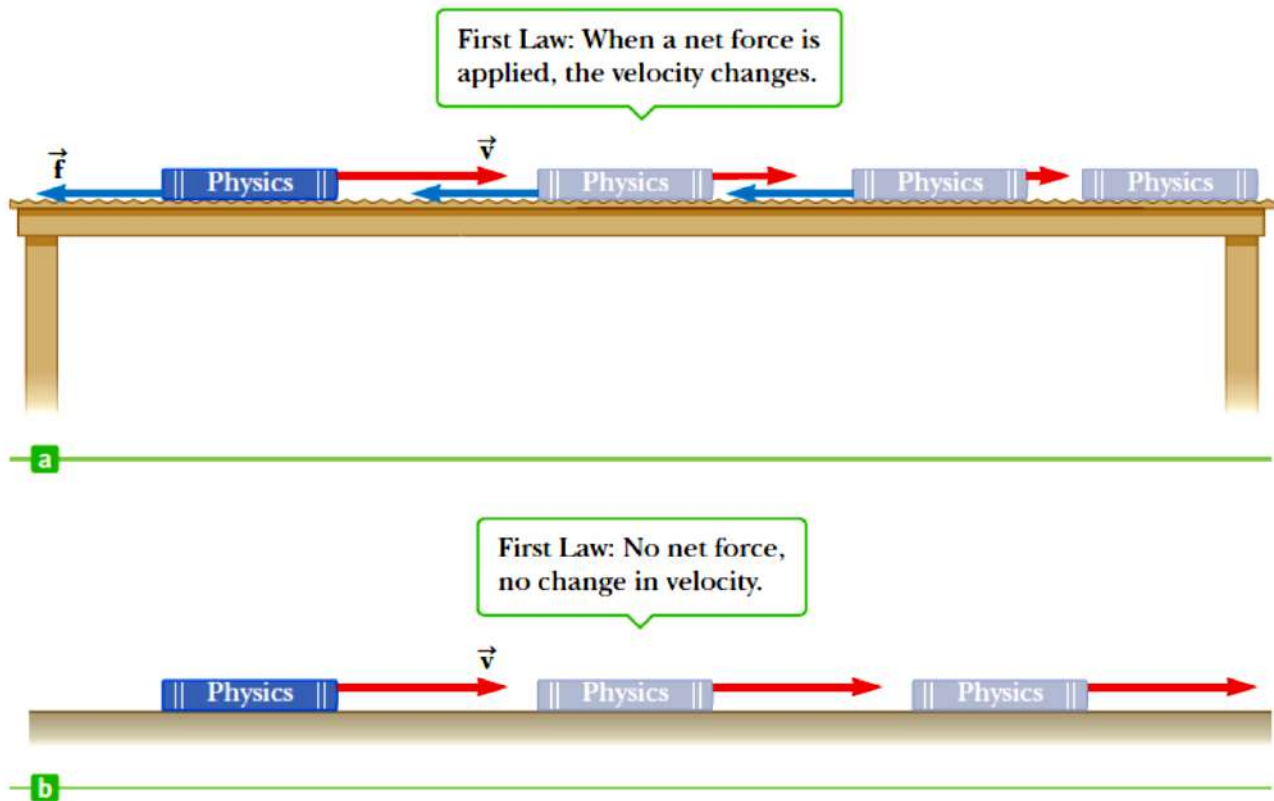
1.9 Newton's First Law

Consider a book lying on a table. Obviously, the book remains at rest if left alone. Now imagine pushing the book with a horizontal force great enough to overcome the force of friction between the book and the table, setting the book in motion. Because the magnitude of the applied force exceeds the magnitude of the friction force, the book accelerates. When the applied force is withdrawn (Fig. 4.3a), friction soon slows the book to a stop. Now imagine pushing the book across a smooth, waxed floor. The book again comes to rest once the force is no longer applied, but not as quickly as before. Finally, if the book is moving on a horizontal frictionless surface (Fig. 4.3b), it continues to move in a straight line with constant velocity until it hits a wall or some other obstruction. Before about 1600, scientists felt that the natural state of matter was the state of rest. Galileo, however, devised thought experiments such as an object.

moving on a frictionless surface, as just described and concluded that **it's not the nature of an object to stop once set in motion, but rather to continue in its original state of motion.**

This approach was later formalized as **Newton's first law of motion:**

An object moves with a velocity that is constant in magnitude and direction unless a nonzero net force acts on it.



The net force on an object is defined as the vector sum of all external forces exerted on the object. External forces come from the object's environment. If an object's velocity isn't changing in either magnitude or direction, then its acceleration and the net force acting on it must both be zero. Internal forces originate within the object itself and can't change the object's velocity. As a result, internal forces aren't included in Newton's first law. It's not

really possible to “pull yourself up by your own bootstraps.” A consequence of the first law is the feasibility of space travel. After just a few moments of powerful thrust, the spacecraft coasts for months or years, its velocity only slowly changing with time under the relatively faint influence of the distant Sun and planets.

Mass and Inertia Imagine hitting a golf ball off a tee with a driver. If you’re a good golfer, the ball will sail over two hundred yards down the fairway. Now imagine teeing up a bowling ball and striking it with the same club. Your club would probably break, you might sprain your wrist, and the bowling ball, at best, would fall off the tee, take half a roll, and come to rest. From this thought experiment, we conclude that although both balls resist changes in their state of motion, the bowling ball offers much more effective resistance. The tendency of an object to continue in its original state of motion is called **inertia**. Although inertia is the tendency of an object to continue its motion in the absence of a force, **mass** is a measure of the object’s resistance to changes in its motion due to a force. This kind of mass is often called *inertial mass* because it’s associated with inertia. The greater the mass of a body, the less it accelerates under the action of a given applied force. The SI unit of mass is the kilogram. Mass is a scalar quantity that obeys the rules of ordinary arithmetic. Inertia can be used to explain the operation of one type of seat belt mechanism. The purpose of the seat belt is to hold the passenger firmly in place relative to the car, to prevent serious injury in the event of an accident. Figure 4.5 illustrates how one type of shoulder harness operates. Under normal conditions, the ratchet turns freely to allow the harness to wind on or unwind from the pulley as the passenger moves. In an accident, the car undergoes a large acceleration and rapidly comes to rest. Because of its inertia, the large block under the seat continues to slide forward along the tracks. The pin connection between the block and the rod causes the rod to pivot

about its center and engage the ratchet wheel. At this point, the ratchet wheel locks in place and the harness no longer unwinds.

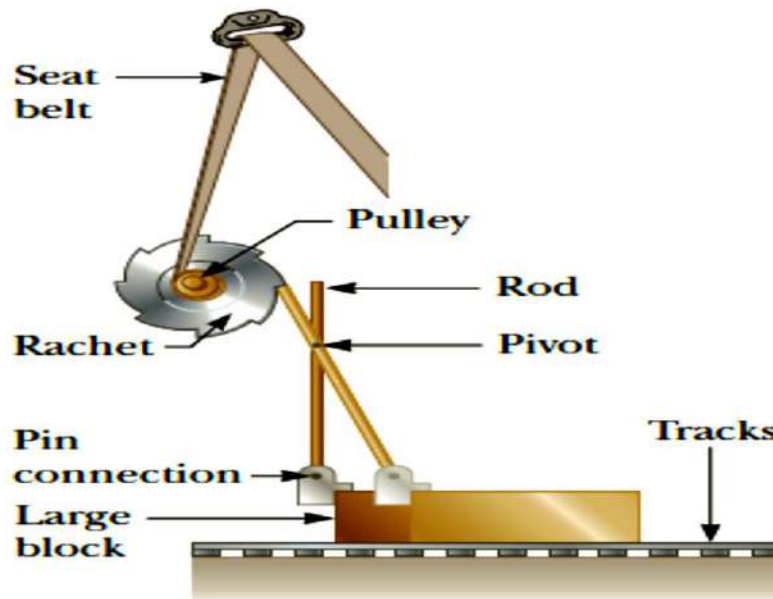


Figure 4.5 A mechanical arrangement for an automobile seat belt.

2.9 Newton's Second Law

Newton's first law explains what happens to an object that has no net force acting on it: The object either remains at rest or continues moving in a straight line with constant speed. Newton's second law answers the question of what happens to an object that *does* have a net force acting on it. Imagine pushing a block of ice across a frictionless horizontal surface. When you exert some horizontal force on the block, it moves with an acceleration of, say,

2 m/s². If you apply a force twice as large, the acceleration doubles to 4 m/s². Pushing three times as hard triples the acceleration, and so on. From such observations, we conclude that **the acceleration of an object is directly proportional to the net force acting on it**. Mass also affects acceleration. Suppose you stack identical blocks of ice on top of each other while pushing the stack with constant force. If the force applied to one block produces an acceleration of 2 m/s², then the acceleration drops to half that value, 1 m/s², when two blocks are pushed, to one-third the initial value when three blocks are pushed, and so on. We conclude that **the acceleration of an object is inversely proportional to its mass**. These observations are summarized in **Newton's second law**:

The acceleration $\vec{a}$ of an object is directly proportional to the net force acting on it and inversely proportional to its mass.

The constant of proportionality is equal to one, so in mathematical terms the preceding statement can be written:

$$\vec{a} = \frac{\sum \vec{F}}{m}$$

where $\vec{a}$ is the acceleration of the object, m is its mass, and $\sum \vec{F}$ is the vector sum of all forces acting on it. Multiplying through by m , we have

$$\sum \vec{F} = m\vec{a} \quad \text{.....4.1}$$

Physicists commonly refer to this equation as ' $F = ma$.' Figure 4.6 illustrates the relationship between the mass, acceleration, and the net force. The second law is a vector equation, equivalent to the following three component equations:

$$\sum F_x = ma_x \quad \sum F_y = ma_y \quad \sum F_z = ma_z \quad \text{[4.2]}$$

When there is no net force on an object, its acceleration is zero, which means the velocity is constant.



Figure 4.6 The second law of motion. For the block of mass m , the net force $\Sigma \vec{F}$ acting on the block equals the mass m times the acceleration vector $\vec{a}$.

Units of Force and Mass The SI unit of force is the **newton**. When 1 newton of force acts on an object that has a mass of 1 kg, it produces an acceleration of 1 m/s^2 in the object. From this definition and Newton's second law, we see that the newton can be expressed in terms of the fundamental units of mass, length, and time as

$$1 \text{ N} \equiv 1 \text{ kg} \cdot \text{m/s}^2 \quad [4.3]$$

In the U.S. customary system, the unit of force is the **pound**. The conversion from newtons to pounds is given by

$$1 \text{ N} = 0.225 \text{ lb} \quad [4.4]$$

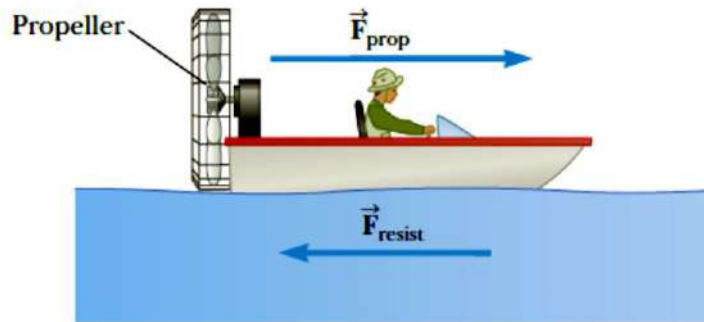
The units of mass, acceleration, and force in the SI and U.S. customary systems are summarized in Table 4.1.

Table 4.1 Units of Mass, Acceleration, and Force

System	Mass	Acceleration	Force
SI	kg	m/s^2	$\text{N} = \text{kg} \cdot \text{m/s}^2$
U.S. customary	slug	ft/s^2	$\text{lb} = \text{slug} \cdot \text{ft/s}^2$

Problem: An airboat with mass $3.50 \times 10^2 \text{ kg}$, including the passenger, has an engine that produces a net horizontal force of $7.70 \times 10^2 \text{ N}$, after accounting for forces of resistance (see Fig. 4.7). **(a)** Find the acceleration of the airboat. **(b)** Starting from rest, how long does it

take the airboat to reach a speed of 12.0 m/s? (c) After reaching that speed, the pilot turns off the engine and drifts to a stop over a distance of 50.0 m. Find the resistance force, assuming it is constant.



SOLUTION

(a) Find the acceleration of the airboat.

Apply Newton's second law and solve for the acceleration:

$$ma = F_{\text{net}} \quad \rightarrow \quad a = \frac{F_{\text{net}}}{m} = \frac{7.70 \times 10^2 \text{ N}}{3.50 \times 10^2 \text{ kg}} = 2.20 \text{ m/s}^2$$

(b) Find the time necessary to reach a speed of 12.0 m/s.

Apply the kinematics velocity equation:

$$v = at + v_0 = (2.20 \text{ m/s}^2)t = 12.0 \text{ m/s} \quad \rightarrow \quad t = 5.45 \text{ s}$$

(c) Find the resistance force after the engine is turned off.

Using kinematics, find the net acceleration due to resistance forces:

$$v^2 - v_0^2 = 2a \Delta x$$

$$0 - (12.0 \text{ m/s})^2 = 2a(50.0 \text{ m}) \quad \rightarrow \quad a = -1.44 \text{ m/s}^2$$

Substitute the acceleration into Newton's second law, finding the resistance force:

$$F_{\text{resist}} = ma = (3.50 \times 10^2 \text{ kg})(-1.44 \text{ m/s}^2) = -504 \text{ N}$$

3.9 Newton's third Law

Consider the task of driving a nail into a block of wood, for example, as illustrated in Figure 4.11a. To accelerate the nail and drive it into the block, the hammer must exert a net force on the nail. Newton recognized, however, that a single isolated force couldn't exist. Instead, **forces in nature always exist in pairs**. According to Newton, as the nail is driven into the block by the force exerted by the hammer, the hammer is slowed down and stopped by the force exerted by the nail. Newton described such paired forces with his **third law**:

If object 1 and object 2 interact, the force $\vec{F}_{12}$ exerted by object 1 on object 2 is equal in magnitude but opposite in direction to the force $\vec{F}_{21}$ exerted by object 2 on object 1.

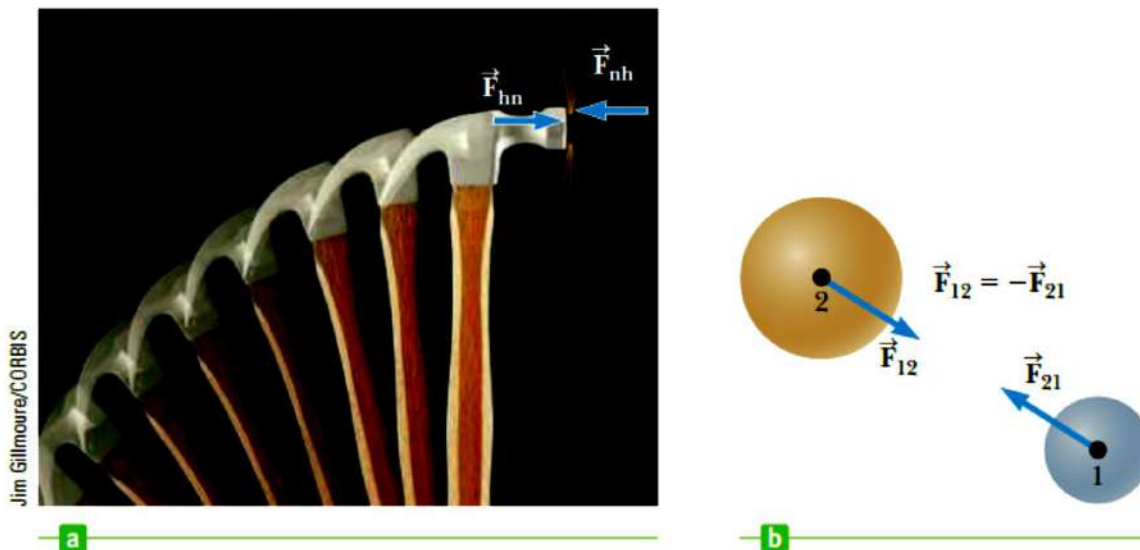


Figure 4.11 Newton's third law. (a) The force exerted by the hammer on the nail is equal in magnitude and opposite in direction to the force exerted by the nail on the hammer.

(b) The force $\vec{F}_{12}$ exerted by object 1 on object 2 is equal in magnitude and opposite in direction to the force $\vec{F}_{21}$ exerted by object 2 on object 1.

This law, which is illustrated in Figure 4.11b, states that **a single isolated force can't exist**. The force $\vec{F}_{12}$ exerted by object 1 on object 2 is sometimes called the *action force*, and the force $\vec{F}_{21}$ exerted by object 2 on object 1 is called the *reaction force*. In reality, either force can be labeled the action or reaction force. **The action force is equal in magnitude to the reaction force and opposite in direction. In all cases, the action and reaction forces act on different objects.** For example, the force acting on a freely falling projectile is the force of gravity exerted by Earth on the projectile, $\vec{F}_g$, and the magnitude of this force is its weight mg . The reaction to force $\vec{F}_g$ is the gravitational force exerted by the projectile on Earth, $\vec{F}_g' = -\vec{F}_g$. The reaction force $\vec{F}_g'$ must accelerate the Earth towards the projectile, just as the action force $\vec{F}_g$ accelerates the projectile towards Earth. Because Earth has such a large mass, however, its acceleration due to this reaction force is negligibly small.

PROBLEM A man of mass $M = 75.0$ kg and woman of mass $m = 55.0$ kg stand facing each other on an ice rink, both wearing ice skates. The woman pushes the man with a horizontal force of $F = 85.0$ N in the positive x -direction. Assume the ice is frictionless. (a) What is the man's acceleration? (b) What is the reaction force acting on the woman? (c) Calculate the woman's acceleration.

SOLUTION

(a) What is the man's acceleration?

Write the second law for the man:

$$Ma_M = F$$

Solve for the man's acceleration and substitute values:

$$a_M = \frac{F}{M} = \frac{85.0 \text{ N}}{75.0 \text{ kg}} = 1.13 \text{ m/s}^2$$

(b) What is the reaction force acting on the woman?

Apply Newton's third law of motion, finding that the reaction force R acting on the woman has the same magnitude and opposite direction:

$$R = -F = -85.0 \text{ N}$$

(Continued)

(c) Calculate the woman's acceleration.

Write Newton's second law for the woman:

$$ma_W = R = -F$$

Solve for the woman's acceleration and substitute values:

$$a_W = \frac{-F}{m} = \frac{-85.0 \text{ N}}{55.0 \text{ kg}} = -1.55 \text{ m/s}^2$$



*Ministry of Higher Education and Scientific Research
Middle Technical University
Electrical Engineering Technical College
Department of Cybersecurity Technology Engineering*



General Physics

Lecture Nine

First Stage

General Wave Properties

By

Lecturer: Dr. Thamir. Al-hamadany

9. General Wave Properties

Both particles and waves carry energy, but there is an important difference in how they do this. Think of a ball as a particle. If you toss the ball to a friend, the ball moves from you to your friend and carries energy. However, if you and your friend hold the ends of a rope and you give your end a quick shake, the rope remains in your hand. Even though no matter is transferred, the rope still carries energy through the wave that you created. A **wave** is a disturbance that carries energy through matter or space. You have learned how Newton's laws of motion and principles of conservation of energy govern the behavior of particles. These laws and principles also govern the motion of waves. There are many kinds of waves that transmit energy, including the waves you cannot see.

9.1 Mechanical Waves

Water waves, sound waves, and the waves that travel down a rope or spring are types of mechanical waves. Mechanical waves require a medium, such as water, air, ropes, or a spring. Because many other waves cannot be directly observed, mechanical waves can serve as models.

9.1 a Transverse waves the two disturbances shown in **Figure 14-5a** are called wave pulses. A **wave pulse** is a single bump or disturbance that travels through a medium. If the wave moves up and down at the same rate, a **periodic wave** is generated. Notice in Figure 14-5a that the rope is disturbed in the vertical direction, but the pulse travels horizontally. A wave with this type of motion is called a transverse wave. A **transverse wave** is one that vibrates perpendicular to the direction of the wave's motion.

9.1 b longitudinal waves in a coiled-spring toy, you can create a wave pulse in a different way. If you squeeze together several turns of the coiled-spring toy and then suddenly release

them, pulses of closely-spaced turns will move away in both directions, as shown in **Figure 14-5b**. This is called a **longitudinal wave**. The disturbance is in the same direction as, or parallel to, the direction of the wave's motion. Sound waves are longitudinal waves. Fluids usually transmit only longitudinal waves.

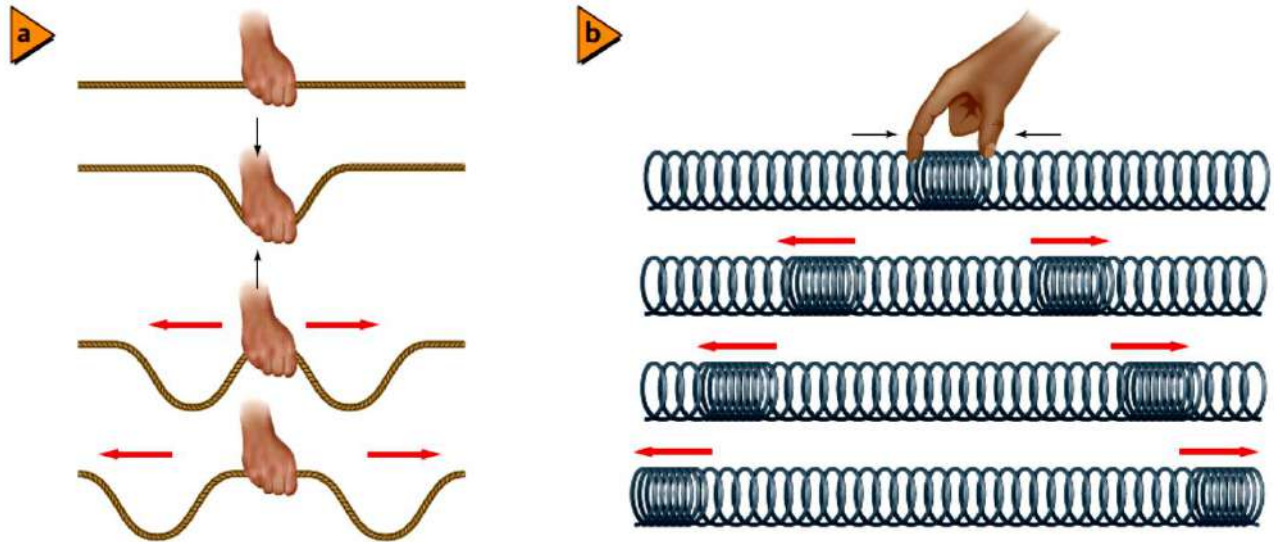


Figure 14-5 A quick shake of a rope sends out transverse wave pulses in both directions **(a)**. The squeeze and release of a coiled spring toy sends out longitudinal wave pulses in both directions **(b)**.

9.2 Surface waves: Waves that are deep in a lake or ocean are longitudinal; at the surface of the water, however, the particles move in a direction that is both parallel and perpendicular to the direction of wave motion, as shown in **Figure 14-6**. Each of the waves is a **surface wave**, which has characteristics of both transverse and longitudinal waves. The energy of water waves usually comes from distant storms, whose energy initially came from the heating of Earth by solar energy. This energy, in turn, was carried to Earth by transverse electromagnetic waves from the Sun.

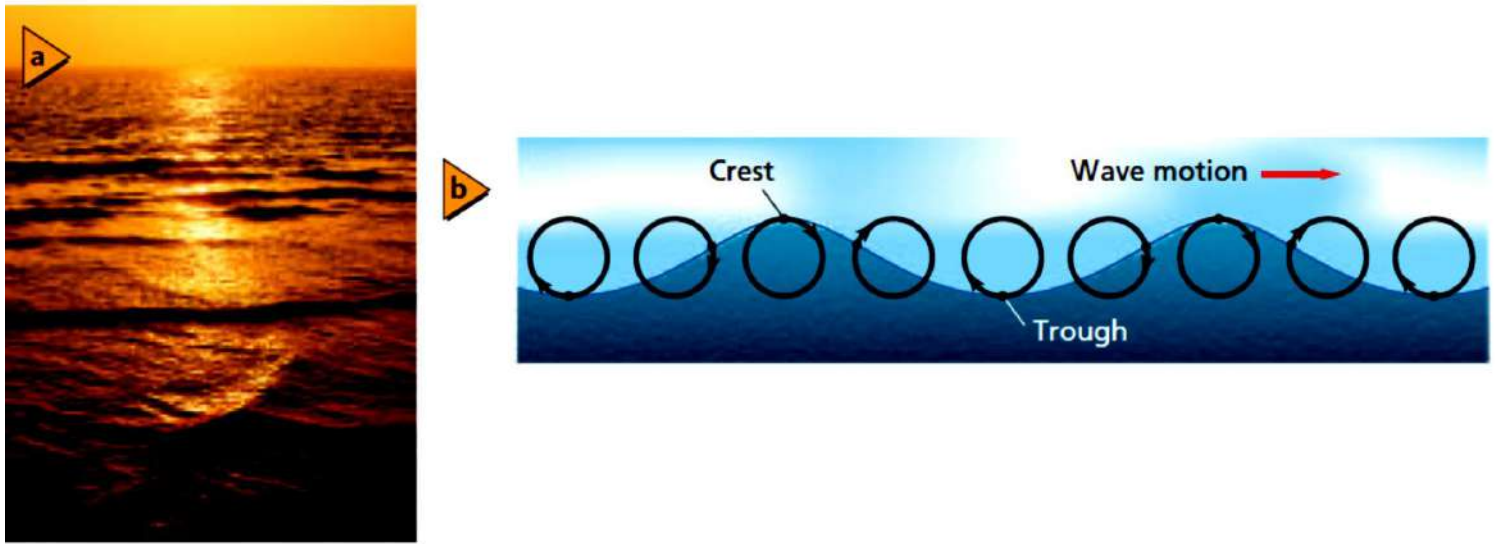


Figure 14-6 Surface waves have properties of both transverse and longitudinal waves (a).The paths of the individual particles are circular (b).

9.3 Measuring a Wave

There are many ways to describe or measure a wave. Some characteristics depend on how the wave is produced, whereas others depend on the medium through which the wave travels.

9.3 a Speed How fast does a wave move? The speed of the pulse shown in **Figure 14-7** can be found in the same way, as the speed of a moving car is determined. First, measure the displacement of the wave peak, Δd , then divide this by the time interval, Δt , to find the speed, given by $v = \Delta d / \Delta t$. The speed of a periodic wave can be found in the same way. For most mechanical waves, both transverse and longitudinal, the speed depends only on the medium through which the waves move.

9.3 b Amplitude how does the pulse generated by gently shaking a rope differ from the pulse produced by a violent shake? The difference is similar to the difference between a ripple in a pond and an ocean breaker: they have different amplitudes. You have learned that the amplitude of a wave is the maximum displacement of the wave from its position of rest, or equilibrium. Two similar waves having different amplitudes are shown in **Figure 14-8**. A wave's amplitude depends on how it is generated, but not on its speed. More work must be done to generate a wave with a greater amplitude. For example, strong winds produce larger water waves than those formed by gentle breezes. Waves with greater amplitudes transfer more energy.

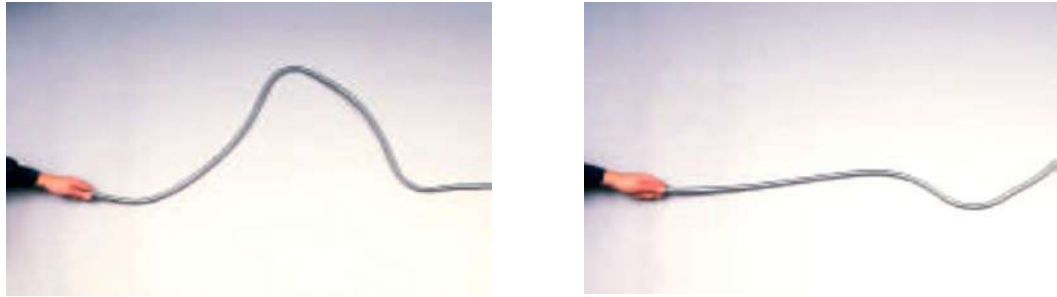


Figure 14-7 these two photographs were taken 0.20 s apart. During that time, the crest moved 0.80 m. The velocity of the wave is 4.0 m/s.

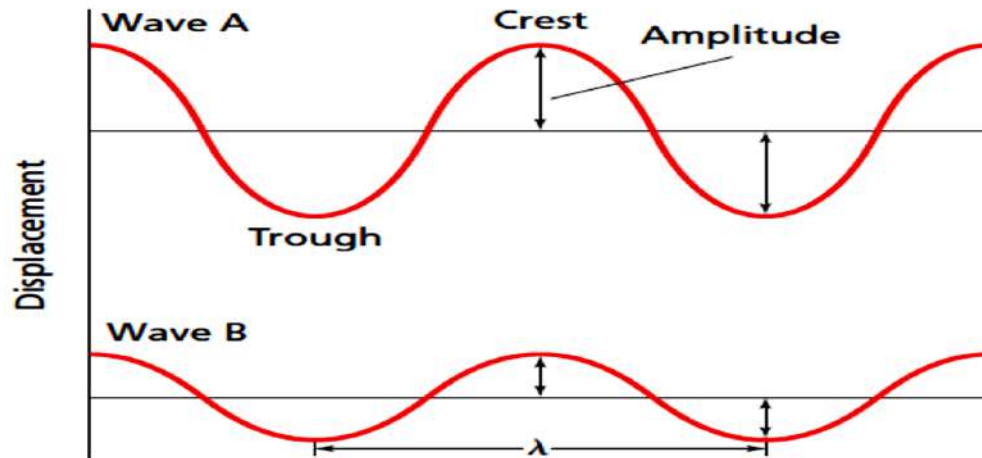


Figure 14-8 The amplitude of wave A is larger than that of wave B.

Whereas a small wave might move sand on a beach a few centimeters, a giant wave can uproot and move a tree. For waves that move at the same speed, the rate at which energy is transferred is proportional to the square of the amplitude. Thus, doubling the amplitude of a wave increases the amount of energy it transfers each second by a factor of 4.

9.3 c Wavelength Rather than focusing on one point on a wave, imagine taking a snapshot of the wave so that you can see the whole wave at one instant in time. Figure 14-8 shows each low point, called a **trough**, and each high point, called a **crest**, of a wave. The shortest distance between points where the wave pattern repeats itself is called the **wavelength**. Crests are spaced by one wavelength. Each trough also is one wavelength from the next. The Greek letter lambda, λ , represents wavelength.

9.3 d Phase any two points on a wave that are one or more whole wavelengths apart are in phase. Particles in the medium are said to be in phase with one another when they have the same displacement from equilibrium and the same velocity. Particles in the medium with opposite displacements and velocities are 180° out of phase. A crest and a trough, for example, are 180° out of phase with each other. Two particles in a wave can be anywhere from 0° to 180° out of phase with one another.

9.4 e Period and frequency although wave speed and amplitude can describe both pulses and periodic waves, period, T , and frequency, f , apply only to periodic waves. You have learned that the period of a simple harmonic oscillator, such as a pendulum, is the time it takes for the motion of the oscillator to complete one cycle. Such an oscillator is usually the source, or cause, of a periodic wave. The period of a wave is equal to the period of the source. In **Figures 14-9a** through **14-9d**, the period, T , equals 0.04 s, which is the time it takes the source to complete one cycle. The same time is taken by P, a point on the rope, to return to its initial phase.

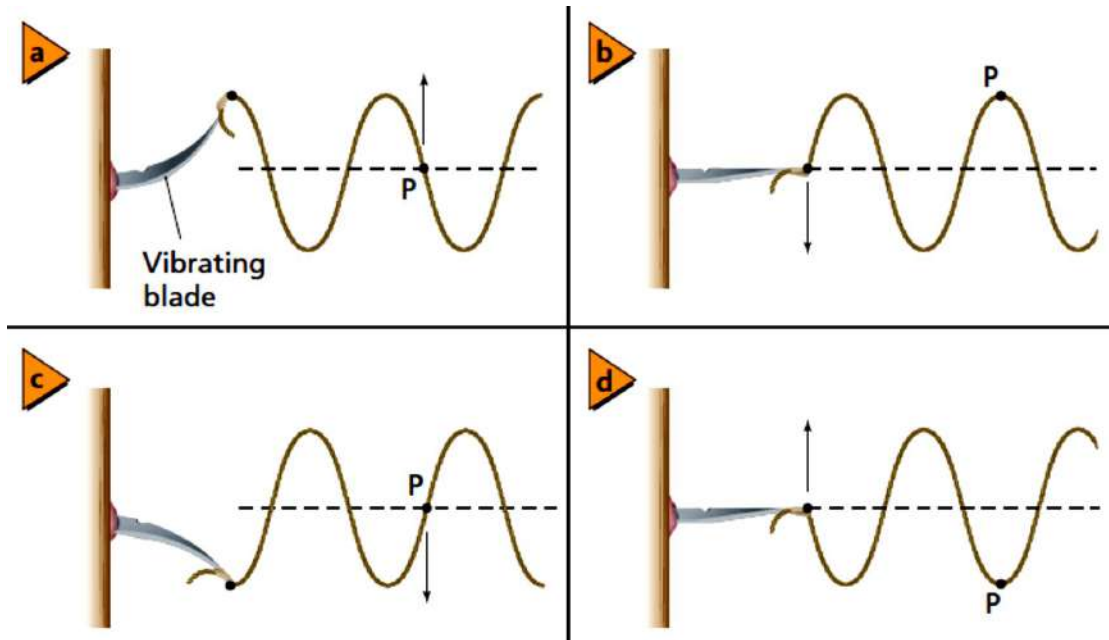


Figure 14-9 one end of a string, with a piece of tape at point P, is attached to a blade vibrating 25 times per second. Note the change in position of point P over time.

9.3 f Frequency of a wave, f , is the number of complete oscillations it makes each second. Frequency is measured in hertz. One hertz (Hz) is one oscillation per second. The frequency and period of a wave are related by the following equation.

$$\text{Frequency of a Wave } f = \frac{1}{T}$$

The frequency of a wave is equal to the reciprocal of the period.

Both the period and the frequency of a wave depend only on its source. They do not depend on the wave's speed or the medium. Although you can directly measure a wavelength, the wavelength depends on both the frequency of the oscillator and the speed of the wave. In the time interval of one period, a wave moves one wavelength. Therefore, the wavelength of a

wave is the speed multiplied by the period, $\lambda = vT$. Because the frequency is usually more easily found than the period, this equation is most often written in the following way.

Wavelength $\lambda = \frac{v}{f}$

The wavelength of a wave is equal to the velocity divided by the frequency.



Ministry of Higher Education and Scientific Research

Middle Technical University

Electrical Engineering Technical College

Department of Cybersecurity Technology Engineering



General Physics

Lecture Ten

First Stage

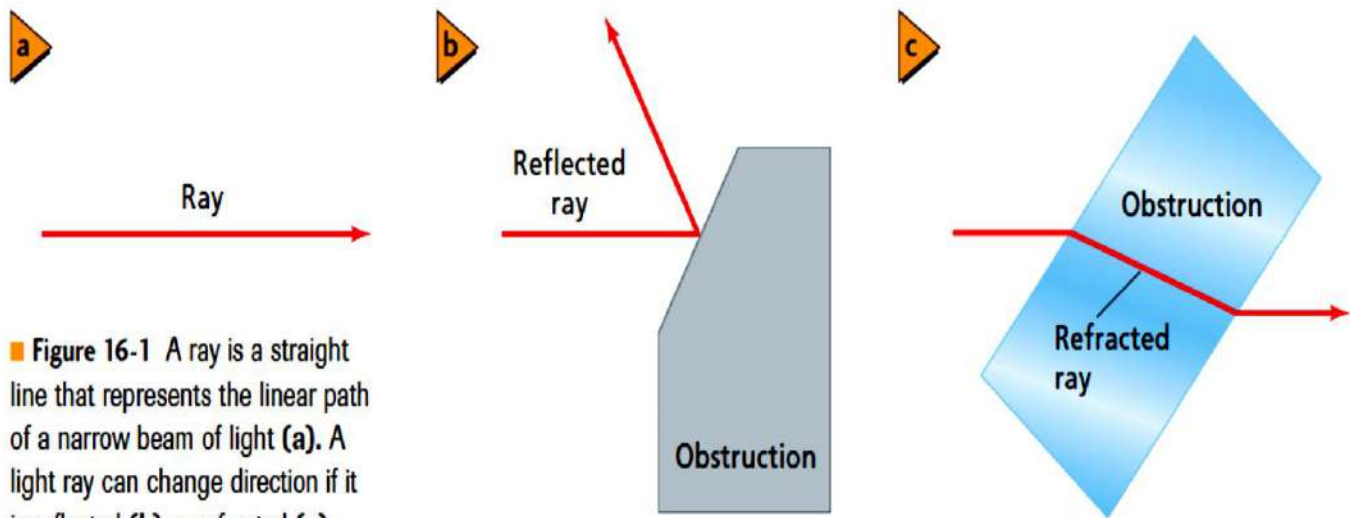
The Light

By

Lecturer: Dr. Thamir. Al-hamadany

10. The Light

Light and sound are two methods by which you can receive information. Of the two, light seems to provide the greater variety of information. The human eye can detect tiny changes in the size, position, brightness, and color of an object. Our eyes usually can distinguish shadows from solid objects and sometimes distinguish reflections of objects from the objects themselves. In this section, you will learn where light comes from and how it illuminates the universe around you. One of the first things that you ever discovered about light, although you may not have been conscious of your discovery, is that it travels in a straight line. How do you know this? When a narrow beam of light, such as that of a flashlight or sunlight streaming through a small window, is made visible by dust particles in the air, you see the path of the light as a straight line. When your body blocks sunlight, you see your outline in a shadow. In addition, whenever you locate an object with your eyes and walk toward it, you most likely walk in a straight path. These things are possible only because light travels in straight lines. Based on this knowledge of how light travels, models have been developed that describe how light works. Isaac Newton, whose laws of motion that you studied previously, believed that light is a stream of fast-moving, unimaginably tiny particles, which he called corpuscles. However, his model could not explain all of the properties of light. Experiments showed that light also behaves like a wave. In the **ray model of light**, light is represented as a ray that travels in a straight path, the direction of which can be changed only by placing an obstruction in the path, as shown in **Figure 16-1**. The ray model of light was introduced, as a way to study how light interacts with matter, regardless of whether light is a particle or a wave. This study of light is called ray optics or geometric optics.



■ **Figure 16-1** A ray is a straight line that represents the linear path of a narrow beam of light (a). A light ray can change direction if it is reflected (b) or refracted (c).

10.1 Definition of Light:

Light is a form of electromagnetic radiation that behaves both as particles (photons) and as waves. It is the visible and perceptible part of the electromagnetic spectrum.

10.2 Nature of Light:

Light is a transverse wave, which means that it oscillates perpendicular to its direction of travel. It travels at the speed of approximately 299,792 kilometers per second in a vacuum, denoted as 'c.'

10.3 Properties of Light

- **Speed and Velocity:** The speed of light in a vacuum is a fundamental constant and serves as the speed limit for the universe. Light slows down when it passes through other materials, which results in phenomena like refraction.
- **Reflection and Refraction:** Light can be reflected from surfaces, following the law of reflection. It can also be bent as it passes from one medium to another, following the law of refraction.
- **Dispersion:** White light can be separated into its constituent colors by passing through a prism, demonstrating the phenomenon of dispersion.
- **Absorption and Transmission:** Different materials absorb and transmit light differently, giving rise to the colors and transparency of substances.
- **Interference and Diffraction:** Light can create interference patterns when two or more waves overlap, and it can diffract, producing patterns that depend on its wavelength and the size of the aperture.
- **Wave-Particle Duality:** Light exhibits both wave-like and particle-like properties. This phenomenon is known as wave-particle duality.

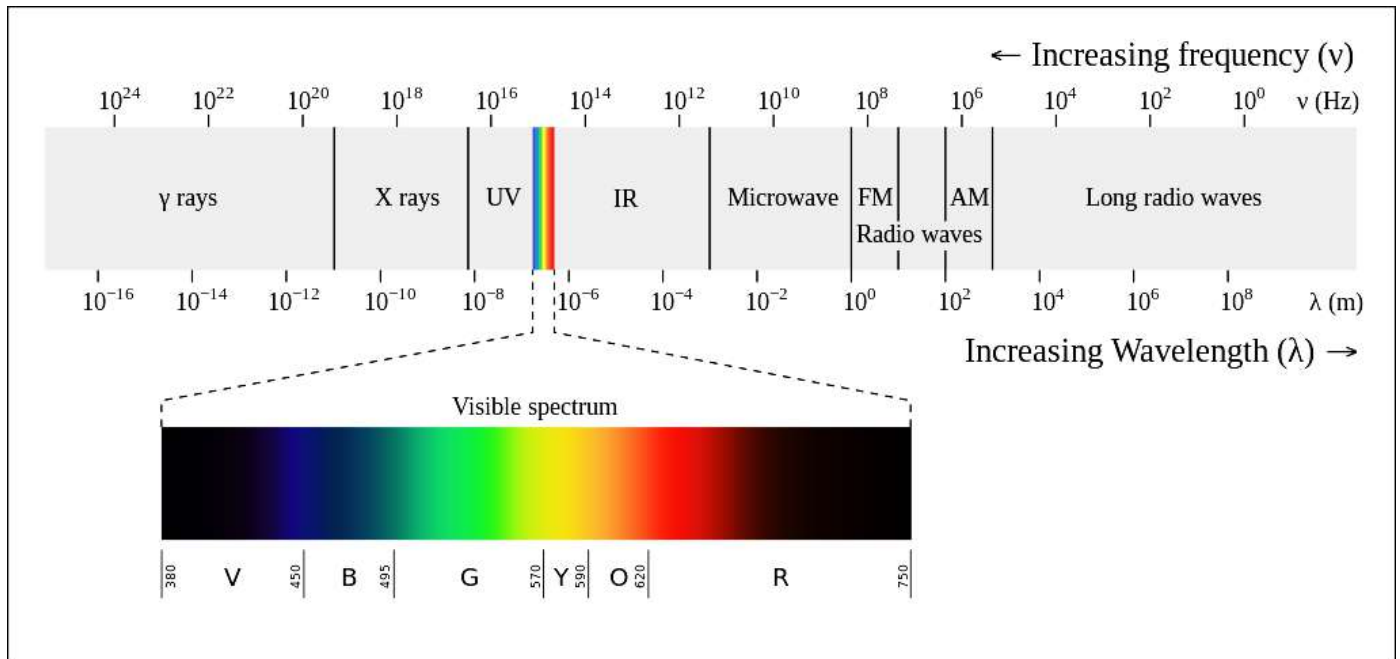
11.4 Applications of Light

We are aware of the health benefits of vitamin D, which is easily available to us through sunlight. Sun is, therefore, the best source of this vitamin for obtaining this energy and preventing multiple diseases.

- We can use the light energy we receive from the sun for domestic tasks like running solar cookers, and the sunlight is also used for solar water heaters.
- The only thing that can help us see the beautiful things around us is light.
- Visible light is electromagnetic radiation that falls inside the human eye's visible spectrum.
- Light is what makes it all possible. We use spectroscopy to investigate various chemicals in both quantitative and qualitative ways. Spectroscopy takes different forms depending on the wavelength of light that is used in each.
- The UV portion of the light is a powerful sterilizer. You may have noticed that since UV lamps are the most effective sterilizers, hospitals, and retail establishments use them.
- Due to the UV rays' ability to destroy airborne microorganisms, sanitary conditions in public places like supermarkets and hospitals are maintained.
- As a result, UV light is thought of as a dry form of sterilization.

Summary:

The primary application of light is that it enables us to see everything around us and study distant objects in space. Moreover, it is one of the most efficient sterilizing agents, and sunlight helps eliminate all the microorganisms found in hospital mattresses or other bedding. Solar energy can also help us power electronic devices and is an unlimited source of energy.



The electromagnetic spectrum, with the visible portion highlighted.



Ministry of Higher Education and Scientific Research

Middle Technical University

Electrical Engineering Technical College

Department of Cybersecurity Technology Engineering



General Physics

Lecture Eleven

First Stage

Electromagnetic Spectrum

By

Lecturer: Dr. Thamir. Al-hamadany

11. Electromagnetic spectrum

The electromagnetic (EM) spectrum is the range of all types of EM radiation. Radiation is energy that travels and spreads out as it goes the visible light that comes from a lamp in your house and the radio waves that come from a radio station are two types of electromagnetic radiation. The other types of EM radiation that make up the electromagnetic spectrum are microwaves, infrared light, ultraviolet light, X-rays and gamma-rays.

11.1 Type of Electromagnetic spectrum

- Radio: Your radio captures radio waves emitted by radio stations, bringing your favorite tunes. Radio waves are also emitted by stars and gases in space.
- Microwave: Microwave radiation will cook your popcorn in just a few minutes, but is also used by astronomers to learn about the structure of nearby galaxies.
- Infrared: Night vision goggles pick up the infrared light emitted by our skin and objects with heat. In space, infrared light helps us map the dust between stars.
- Visible: Our eyes detect visible light. Fireflies, light bulbs, and stars all emit visible light.
- Ultraviolet: ultraviolet radiation is emitted by the Sun, are the reason skin tans, and burns. "Hot" objects in space emit UV radiation as well.

- X-ray: A dentist uses X-rays to image your teeth, and airport security uses them to see through your bag. Hot gases in the Universe also emit X-rays.
- Gamma ray: Doctors use gamma-ray imaging to see inside your body. The biggest gamma-ray generator of all is the Universe.

Type of EM wave	Production	Applications	Life sciences aspect	Issues
Radio & TV	Accelerating charges	Communications Remote controls	MRI	Requires controls for band use
Microwaves	Accelerating charges & thermal agitation	Communications Ovens Radar	Deep heating	Cell phone use
Infrared	Thermal agitations & electronic transitions	Thermal imaging Heating	Absorbed by atmosphere	Greenhouse effect
Visible light	Thermal agitations & electronic transitions	All pervasive	Photosynthesis Human vision	
Ultraviolet	Thermal agitations & electronic transitions	Sterilization Cancer control	Vitamin D production	Ozone depletion Cancer causing
X-rays	Inner electronic transitions and fast collisions	Medical Security	Medical diagnosis Cancer therapy	Cancer causing
Gamma rays	Nuclear decay	Nuclear medicine Security	Medical diagnosis Cancer therapy	Cancer causing Radiation damage

As noted before, an electromagnetic wave has a frequency and a wavelength associated with it and travels at the speed of light, or. The relationship among these wave characteristics can be described by $v_w = f\lambda$, where v_w is the propagation speed of the wave, f is the frequency, and λ is the wavelength. Here $v_w = c$, so that for all electromagnetic waves,

$$c = f\lambda.$$

Thus, for all electromagnetic waves, the greater the frequency, the smaller the wavelength. Figure below shows how the various types of electromagnetic waves are categorized according to their wavelengths and frequencies that is, it shows the electromagnetic spectrum. Many of the characteristics of the various types of electromagnetic waves are related to their frequencies and wavelengths, as we shall see.

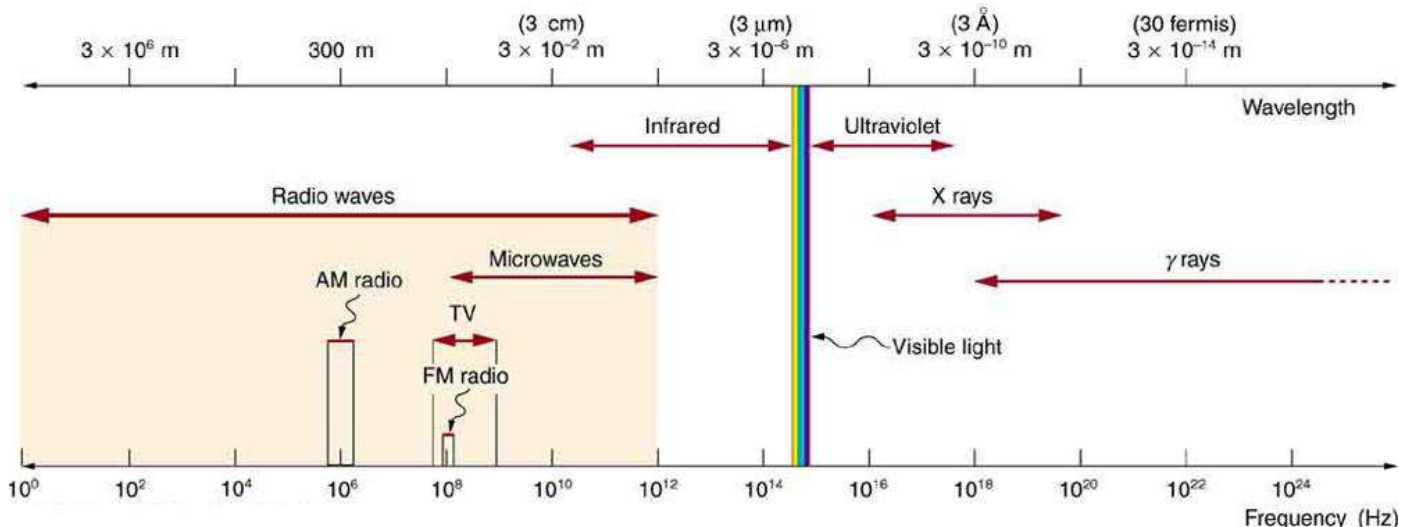


Figure show the electromagnetic spectrum, showing the major categories of electromagnetic waves. The range of frequencies and wavelengths is remarkable. The dividing line between some categories is distinct, whereas other categories overlap.

11.2 Detecting Electromagnetic Waves from Space

Deep space communication antennas and receivers are capable of detecting many different kinds of natural emitters of electromagnetic radiation, including the stars, the Sun, molecular clouds, and gas giant planets such as Jupiter. A final note on star gazing. Researchers for investigating stars, space, and time use the entire electromagnetic spectrum. As noted earlier, Penzias and Wilson detected microwaves to identify the background radiation originating from the Big Bang. Radio telescopes such as the Arecibo Radio Telescope in Puerto Rico and Parkes Observatory in Australia were designed to detect radio waves. Infrared telescopes need to have their detectors cooled by liquid nitrogen to be able to gather useful signals. Since infrared radiation is predominantly from thermal agitation, if the detectors were not cooled, the vibrations of the molecules in the antenna would be stronger than the signal being collected. The most famous of these infrared sensitive telescopes is the James Clerk Maxwell Telescope in Hawaii. The earliest telescopes, developed in the seventeenth century, were optical telescopes, collecting visible light. Telescopes in the ultraviolet, X-ray, and γ -ray regions are placed outside the atmosphere on satellites orbiting the Earth. The Hubble Space Telescope (launched in 1990) gathers ultraviolet radiation as well as visible light. In the X-ray region, there is the Chandra X-ray Observatory (launched in 1999), and in the γ -ray region, there is the new Fermi Gamma-ray Space Telescope (launched in 2008—taking the place of the Compton Gamma Ray Observatory, 1991–2000.).

11.3 Energy in Electromagnetic Waves

Anyone who has used a microwave oven knows there is energy in electromagnetic waves. Sometimes this energy is obvious, such as in the warmth of the summer sun. Other times it is subtle, such as the unfelt energy of gamma rays, which can destroy living cells.

Electromagnetic waves can bring energy into a system by virtue of their **electric and magnetic fields**. These fields can exert forces and move charges in the system and, thus, do work on them. If the frequency of the electromagnetic wave is the same as the natural frequencies of the system (such as microwaves at the resonant frequency of water molecules), the transfer of energy is much more efficient. However, there is energy in an electromagnetic wave, whether it is absorbed or not. Once created, the fields carry energy away from a source. If absorbed, the field strengths are diminished and anything left travels on. Clearly, the larger the strength of the electric and magnetic fields, the more work they can do and the greater the energy the electromagnetic wave carries. A wave's energy is proportional to its **amplitude squared** (E^2 or B^2). This is true for waves on guitar strings, for water waves, and for sound waves, where amplitude is proportional to pressure. In electromagnetic waves, the amplitude is the **maximum field strength** of the electric and magnetic fields. (See Figure below). Thus, the energy carried and the **intensity** of an electromagnetic wave is proportional to E^2 and B^2 . In fact, for a continuous sinusoidal electromagnetic wave, the average intensity I_{avg} is given by:

$$I_{ave} = \frac{c\epsilon_0 E_0^2}{2},$$

Where C is the speed of light, ϵ_0 is the permittivity of free space, and E_0 is the maximum electric field strength; intensity, as always, is power per unit area (here in W/ m²).

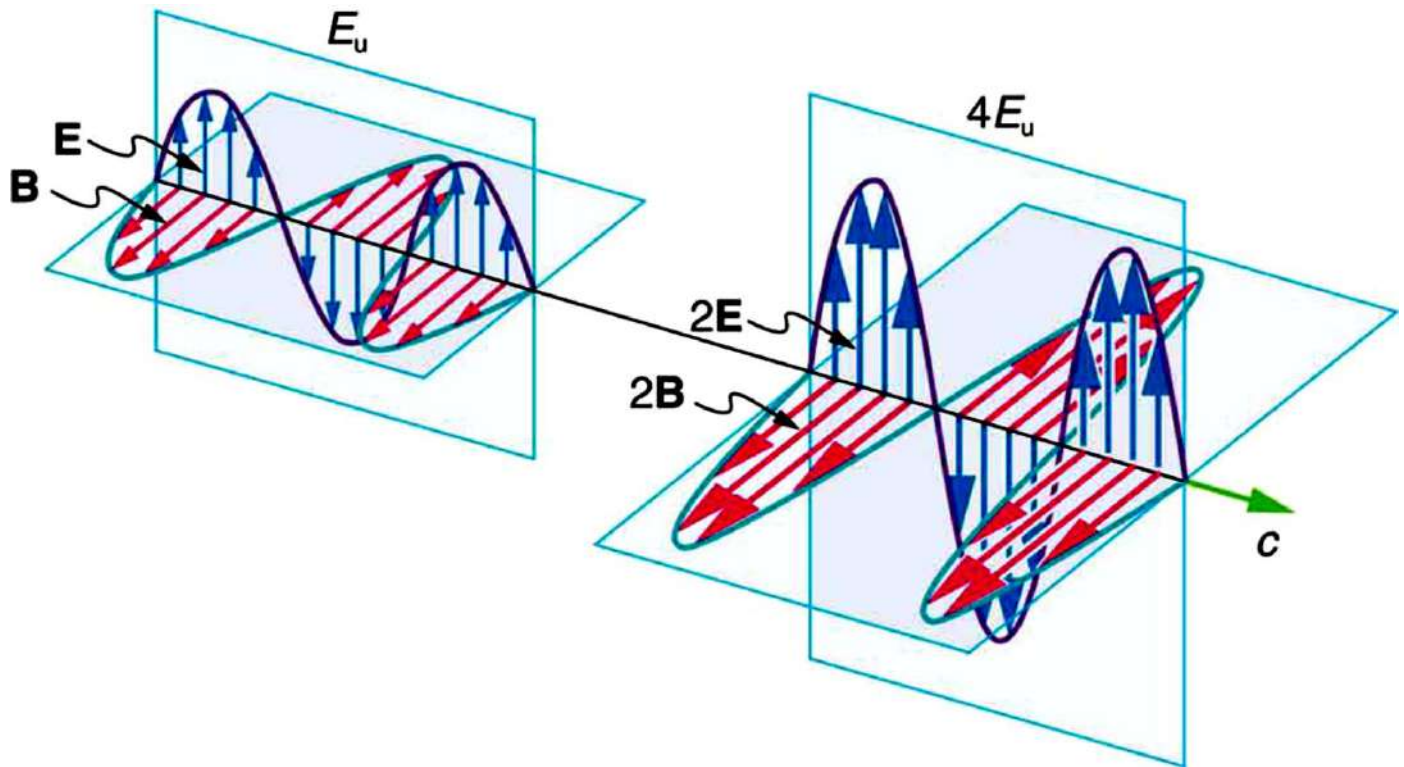


Figure shows: Energy carried by a wave is proportional to its amplitude squared. With electromagnetic waves, larger E -fields and B -fields exert larger forces and can do more work.

The average intensity of an electromagnetic wave I_{avg} can also be expressed in terms of the magnetic field strength by using the relationship $B = E/c$ and the fact that $\epsilon_0 = 1/\mu_0 c^2$ where μ_0 is the permeability of free space. Algebraic manipulation produces the relationship

$$I_{ave} = \frac{cB_0^2}{2\mu_0},$$

Where B_0 is the maximum magnetic field strength.

One more expression for I_{avg} in terms of both electric and magnetic field strengths is useful. Substituting the fact that $\mathbf{C} \cdot \mathbf{B} = E_c$, the previous expression becomes

$$I_{ave} = \frac{E_0 B_0}{2\mu_0}.$$

Whichever of the three preceding equations is most convenient can be used, since they are really just different versions of the same principle: Energy in a wave is related to amplitude squared. Furthermore, since these equations are based on the assumption that the electromagnetic waves are sinusoidal, peak intensity is twice the average; that is, $I_0 = 2I_{ave}$.

Calculate Microwave Intensities and Fields

On its highest power setting, a certain microwave oven projects 1.00 kW of microwaves onto a 30.0 by 40.0 cm area. (a) What is the intensity in W/m^2 ? (b) Calculate the peak electric field strength E_0 in these waves. (c) What is the peak magnetic field strength B_0 ?

Solution for (a)

Entering the given power into the definition of intensity, and noting the area is 0.300 by 0.400 m, yields

$$I = \frac{P}{A} = \frac{1.00 \text{ kW}}{0.300 \text{ m} \times 0.400 \text{ m}}.$$

Here $I = I_{\text{ave}}$, so that

$$I_{\text{ave}} = \frac{1000 \text{ W}}{0.120 \text{ m}^2} = 8.33 \times 10^3 \text{ W/m}^2.$$

Note that the peak intensity is twice the average:

$$I_0 = 2I_{\text{ave}} = 1.67 \times 10^4 \text{ W/m}^2.$$

Solution for (b)

To find E_0 , we can rearrange the first equation given above for I_{ave} to give

$$E_0 = \left(\frac{2I_{\text{ave}}}{c\epsilon_0} \right)^{1/2}.$$

Entering known values gives

$$\begin{aligned} E_0 &= \sqrt{\frac{2(8.33 \times 10^3 \text{ W/m}^2)}{(3.00 \times 10^8 \text{ m/s})(8.85 \times 10^{-12} \text{ C}^2/\text{N}\cdot\text{m}^2)}} \\ &= 2.51 \times 10^3 \text{ V/m}. \end{aligned}$$

Solution for (c)

Perhaps the easiest way to find magnetic field strength, now that the electric field strength is known, is to use given by

$$B_0 = \frac{E_0}{c}.$$

Entering known values gives

$$\begin{aligned} B_0 &= \frac{2.51 \times 10^3 \text{ V/m}}{3.0 \times 10^8 \text{ m/s}} \\ &= 8.35 \times 10^{-6} \text{ T}. \end{aligned}$$

Exercise:

- What is the maximum electric field strength in an electromagnetic wave that has a maximum magnetic field strength of 5.00×10^{-4} T (about 10 times the Earth's)?
- The maximum magnetic field strength of an electromagnetic field is 5×10^{-6} T. Calculate the maximum electric field strength if the wave is traveling in a medium in which the speed of the wave is $0.75c$.
- Find the frequency range of visible light, given that it encompasses wavelengths from 380 to 760 nm.
- Electromagnetic radiation having a $15.0 \mu\text{m}$ wavelength is classified as infrared radiation. What is its frequency?
- Determine the amount of time it takes for X-rays of frequency 3×10^{18} to travel (a) 1 mm and (b) 1 cm.
- If you wish to detect details of the size of atoms (about 1×10^{-10} m) with electromagnetic radiation, it must have a wavelength of about this size. (a) What is its frequency? (b) What type of electromagnetic radiation might this be?
- If the Sun suddenly turned off, we would not know it until its light stopped coming. How long would that be, given that the Sun is 1.50×10^{11} m away?



Ministry of Higher Education and Scientific Research
Middle Technical University
Electrical Engineering Technical College
Department of Cybersecurity Technology Engineering



General Physics

Lecture Twelve

First Stage

Sounds and Hearing

By

Lecturer: Dr. Thamir. Al-hamadany

12. Sounds and Hearing

Sound can be used as a familiar illustration of waves. Because hearing is one of our most important senses, it is interesting to see how the physical properties of sound correspond to our perceptions of it. **Hearing** is the perception of sound, just as vision is the perception of visible light. However, sound has important applications beyond hearing. Ultrasound, for example, is not heard but can be employed to form medical images and is also used in treatment. The physical phenomenon of **sound** is defined to be a disturbance of matter that is transmitted from its source outward. Sound is a wave. On the atomic scale, a disturbance of atoms is far more ordered than their thermal motions. In many instances, sound is a periodic wave, and the atoms undergo simple harmonic motion. In this text, we shall explore such periodic sound waves. A vibrating string produces a sound wave as illustrated in [Figure 17.3](#), [Figure 17.4](#), and [Figure 17.5](#). As the string oscillates back and forth, it transfers energy to the air, mostly as thermal energy created by turbulence. Nevertheless, a small part of the string's energy goes into compressing and expanding the surrounding air, creating slightly higher and lower local pressures. These compressions (high-pressure regions) and rarefactions (low-pressure regions) move out as longitudinal pressure waves having the same frequency as the string they are the disturbance that is a sound wave. (Sound waves in air and most fluids are longitudinal, because fluids have almost no shear strength. In solids, sound waves can be both transverse and longitudinal.) [Figure 17.5](#) shows a graph of gauge pressure versus distance from the vibrating string.

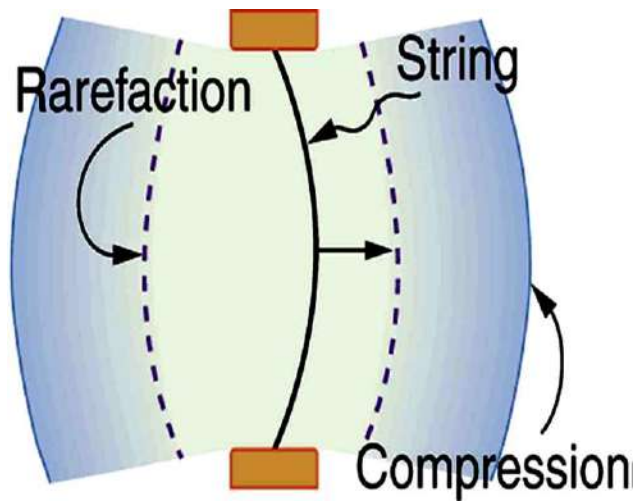


Figure 17.3 A vibrating string moving to the right compresses the air in front of it and expands the air behind it.

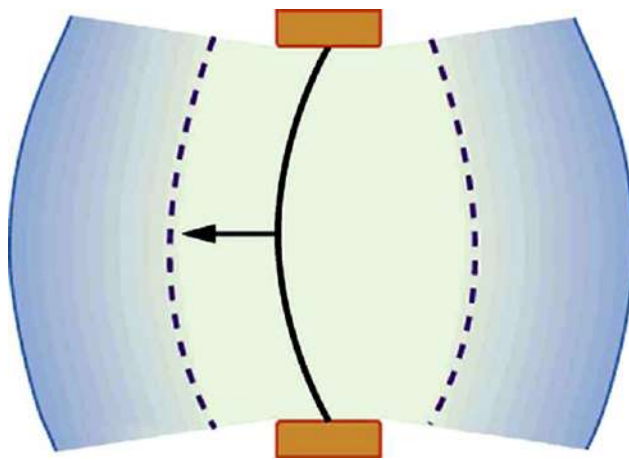


Figure 17.4 As the string moves to the left, it creates another compression and rarefaction as the ones on the right move away from the string.

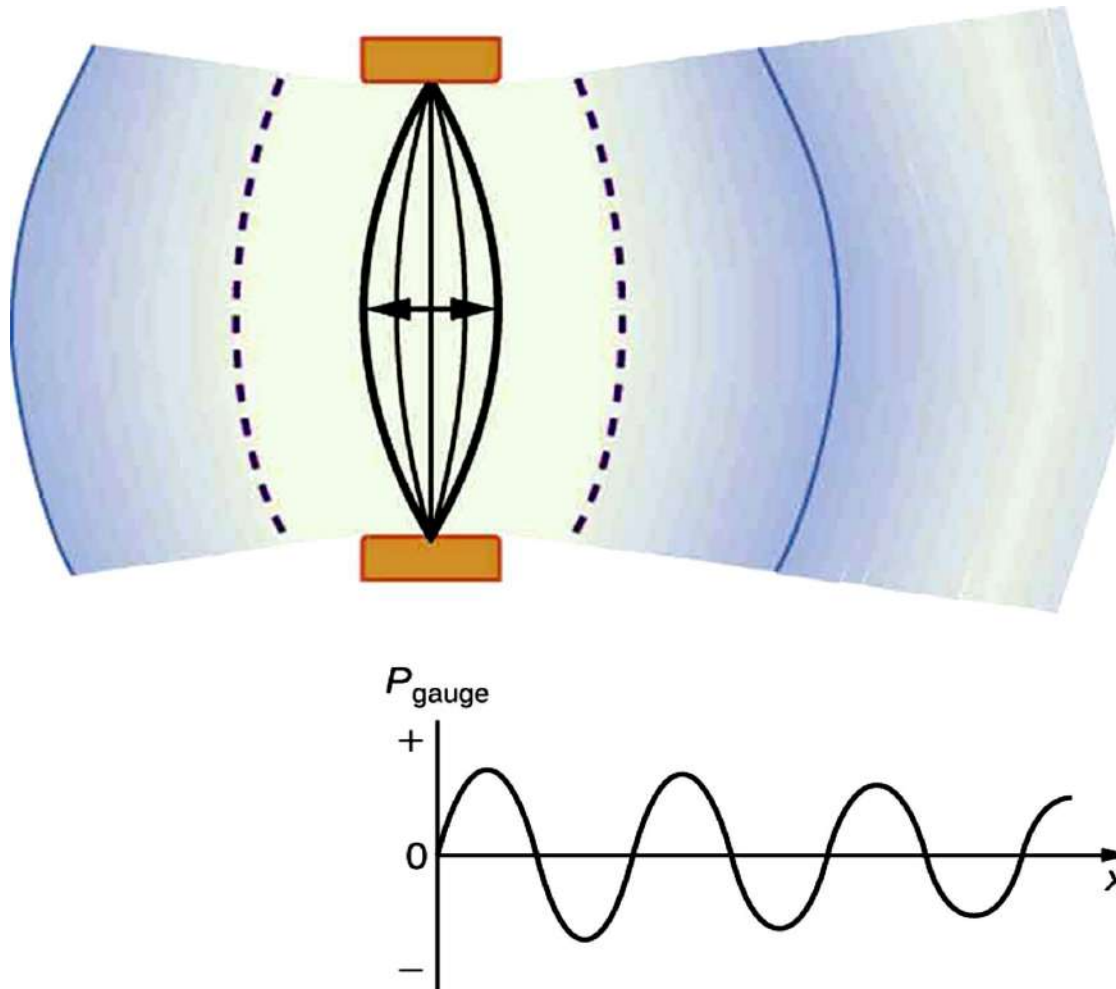


Figure 17.5 After many vibrations, there are a series of compressions and rarefactions moving out from the string as a sound wave. The graph shows gauge pressure versus distance from the source. Pressures vary only slightly from atmospheric for ordinary sounds.

The amplitude of a sound wave decreases with distance from its source, because the energy of the wave is spread over a larger and larger area. However, it is also absorbed by objects, such as the eardrum in [Figure 17.6](#), and converted to thermal energy by the viscosity of air.

In addition, during each compression a little heat transfers to the air and during each rarefaction even less heat transfers from the air, so that the heat transfer reduces the organized disturbance into random thermal motions. (These processes can be viewed as a manifestation of the second law of thermodynamics presented in [Introduction to the Second Law of Thermodynamics: Heat Engines and Their Efficiency](#).) Whether the heat transfer from compression to rarefaction is significant depends on how far apart they are that is, it depends on wavelength. Wavelength, frequency, amplitude, and speed of propagation are important for sound, as they are for all waves.

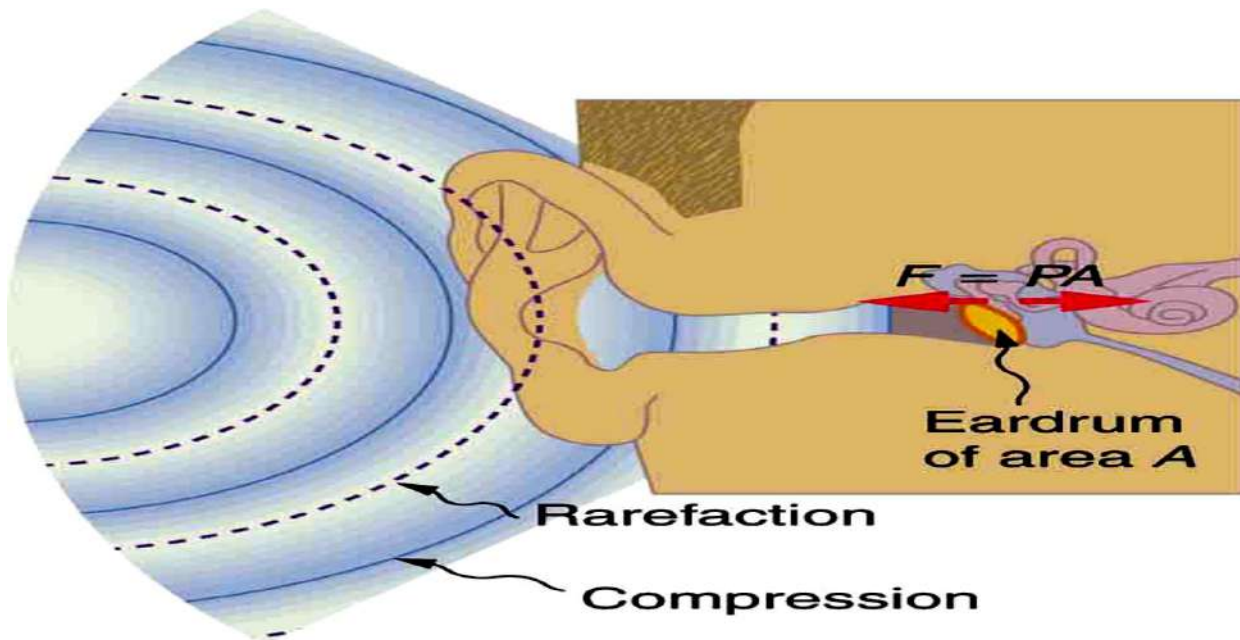


Figure 17.6 Sound wave compressions and rarefactions travel up the ear canal and force the eardrum to vibrate. There is a net force on the eardrum, since the sound wave pressures differ from the atmospheric pressure found behind the eardrum. A complicated mechanism converts the vibrations to nerve impulses, which are perceived by the person.

12.2 Speed of Sound, Frequency, and Wavelength

Sound, like all waves, travels at a certain speed and has the properties of frequency and wavelength. You can observe direct evidence of the speed of sound while watching a fireworks display. The flash of an explosion is seen well before its sound is heard, implying both that sound travels at a finite speed and that it is much slower than light. You can also directly sense the frequency of a sound. Perception of frequency is called **pitch**. The wavelength of sound is not directly sensed, but indirect evidence is found in the correlation of the size of musical instruments with their pitch. Small instruments, such as a piccolo, typically make high-pitch sounds, while large instruments, such as a tuba, typically make low-pitch sounds. High pitch means small wavelength, and the size of a musical instrument is directly related to the wavelengths of sound it produces. Therefore, a small instrument creates short-wavelength sounds. Similar arguments hold that a large instrument creates long-wavelength sounds. The relationship of the speed of sound, its frequency, and wavelength is the same as for all waves:

$$v_w = f\lambda,$$

where v_w is the speed of sound, f is its frequency, and λ is its wavelength. The wavelength of a sound is the distance between adjacent identical parts of a wave—for example, between adjacent compressions as illustrated in Figure below. The frequency is the same as that of the source and is the number of waves that pass a point per unit time.

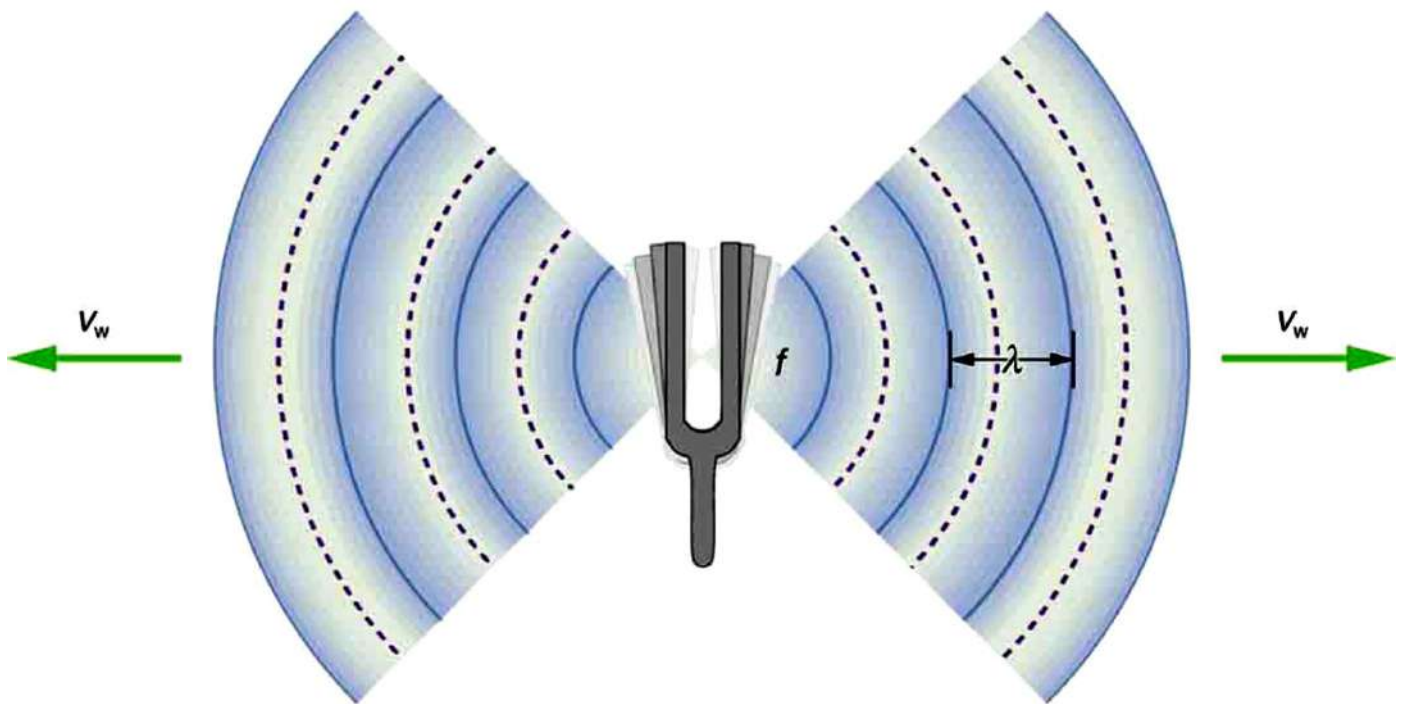


Figure shows A sound wave emanates from a source vibrating at a frequency f , propagates at v_w , and has a wavelength λ .

Table below makes it apparent that the speed of sound varies greatly in different media. A combination of the medium's rigidity (or compressibility in gases) and its density determine the speed of sound in a medium. The more rigid (or less compressible) the medium, the faster the speed of sound. For materials that have similar rigidities, sound will travel faster through the one with the lower density because the sound energy is more easily transferred from particle to particle. The speed of sound in air is low, because air is compressible. Because liquids and solids are relatively rigid and very difficult to compress, the speed of sound in such media is generally greater than in gases.

Medium	$v_w(\text{m/s})$
<i>Gases at 0°C</i>	
Air	331
Carbon dioxide	259
Oxygen	316
Helium	965
Hydrogen	1290
<i>Liquids at 20°C</i>	
Ethanol	1160
Mercury	1450
Water, fresh	1480
Sea water	1540
Human tissue	1540
<i>Solids (longitudinal or bulk)</i>	
Vulcanized rubber	54

Table 1. Speed of Sound in Various Media

Medium	$v_w(\text{m/s})$
Polyethylene	920
Marble	3810
Glass, Pyrex	5640
Lead	1960
Aluminum	5120
Steel	5960

Table 2. Speed of Sound in Various Media

Earthquakes, essentially sound waves in Earth's crust, are an interesting example of how the speed of sound depends on the rigidity of the medium. Earthquakes have both longitudinal

and transverse components, and these travel at different speeds. The bulk modulus of granite is greater than its shear modulus. For that reason, the speed of longitudinal or pressure waves (P-waves) in earthquakes in granite is significantly higher than the speed of transverse or shear waves (S-waves). Both components of earthquakes travel slower in less rigid material, such as sediments. P-waves have speeds of 4 to 7 km/s, and S-waves correspondingly range in speed from 2 to 5 km/s, both being faster in more rigid material. The P-wave gets progressively farther ahead of the S-wave as they travel through Earth's crust. The time between the P- and S-waves is routinely used to determine the distance to their source, the epicenter of the earthquake. The speed of sound is affected by temperature in a given medium. For air at sea level, the speed of sound is given by:

$$v_w = (331 \text{ m/s}) \sqrt{\frac{T}{273 \text{ K}}},$$

Where the temperature (denoted as T) is in units of kelvin. The speed of sound in gases is related to the average speed of particles in the gas, v_w , and that:

$$v_{\text{rms}} = \sqrt{\frac{3kT}{m}},$$

Where k is the Boltzmann constant ($1.38 \times 10^{-23} \text{ J/k}$) and m is the mass of each (identical) particle in the gas. Therefore, it is reasonable that the speed of sound in air and other gases should depend on the square root of temperature. While not negligible, this is not a strong

dependence. At 0°C , the speed of sound is 331 m/s , whereas at 20.0°C it is 343 m/s , less than a 4% increase. Figure below shows a use of the speed of sound by a bat to sense distances. Echoes are also used in medical imaging.

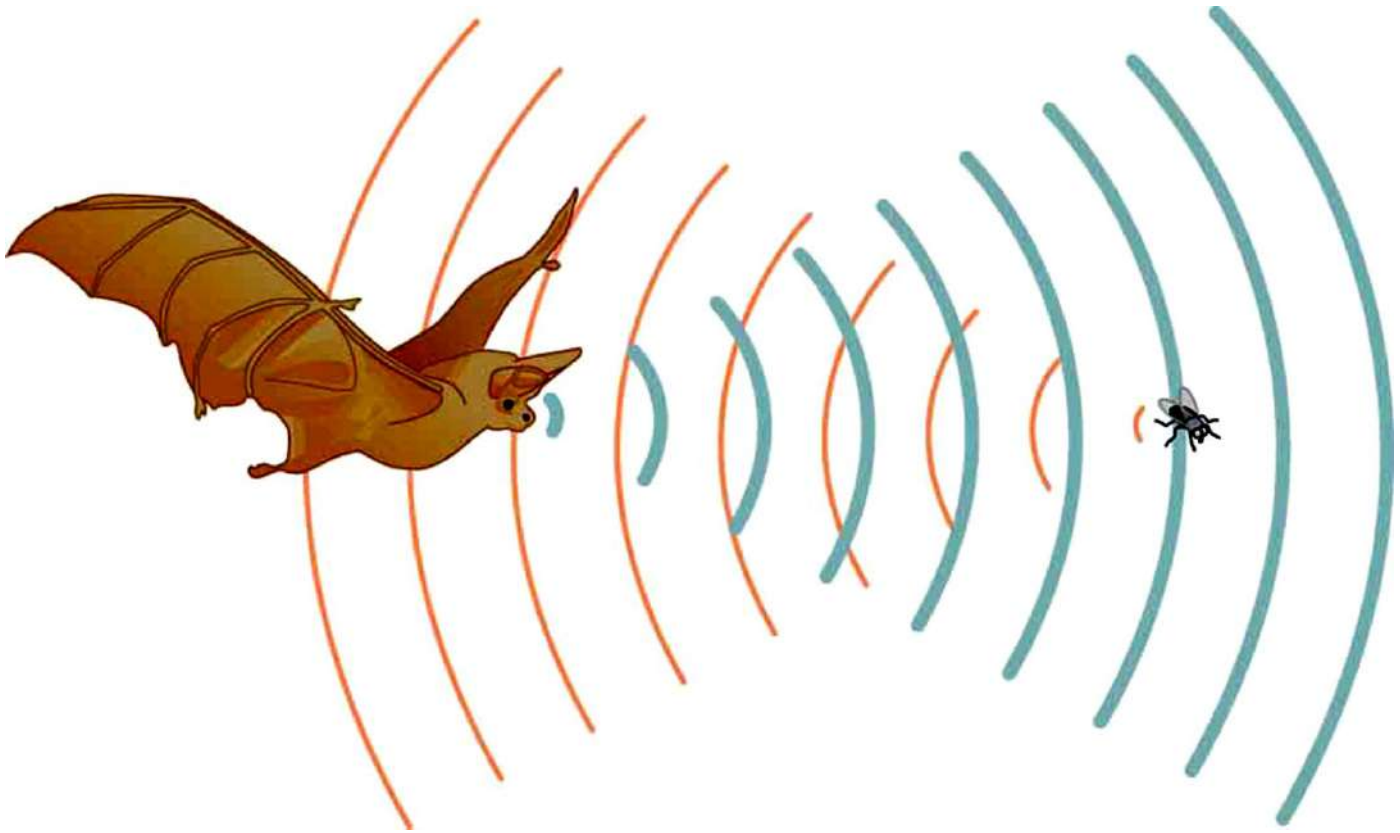


Figure 17.9 A bat uses sound echoes to find its way about and to catch prey. The time for the echo to return is directly proportional to the distance.

One of the more important properties of sound is that its speed is nearly independent of frequency. This independence is certainly true in open air for sounds in the audible range of 20 to $20,000\text{ Hz}$. If this independence were not true, you would certainly notice it for music

played by a marching band in a football stadium, for example. Suppose that high frequency sounds traveled faster— then the farther you were from the band, the more the sound from the low-pitch instruments would lag that from the high-pitch ones. However, the music from all instruments arrives in cadence independent of distance, and so all frequencies must travel at nearly the same speed. Recall that

$$v_w = f\lambda$$

In a given medium under fixed conditions, v_w is constant, so that there is a relationship between f and λ ; the higher the frequency, the smaller the wavelength. See [Figure below](#) and consider the following example.

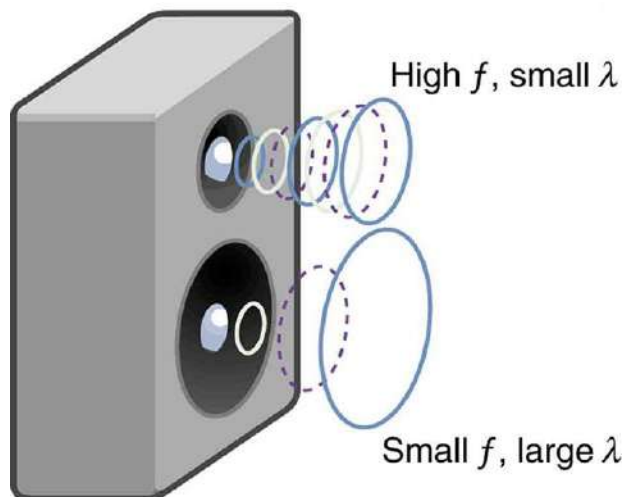


Figure shows because they travel at the same speed in a given medium, low-frequency sounds must have a greater wavelength than high frequency sounds. Here, the lower-frequency sounds are emitted by the large speaker, called a woofer, while the small speaker emits the higher-frequency sounds, called a tweeter.

Example: Calculating Wavelengths: What Are the Wavelengths of Audible Sounds?

Calculate the wavelengths of sounds at the extremes of the audible range, 20 and 20,000 Hz, in 30.0°C air. (Assume that the frequency values are accurate to two significant figures.)

Strategy

To find wavelength from frequency, we can use $v_w = f\lambda$.

Solution

1. Identify knowns. The value for v_w , is given by

$$v_w = (331 \text{ m/s})\sqrt{\frac{T}{273 \text{ K}}}.$$

2. Convert the temperature into kelvin and then enter the temperature into the equation

$$v_w = (331 \text{ m/s})\sqrt{\frac{303 \text{ K}}{273 \text{ K}}} = 348.7 \text{ m/s}.$$

3. Solve the relationship between speed and wavelength for λ :

$$\lambda = \frac{v_w}{f}.$$

4. Enter the speed and the minimum frequency to give the maximum wavelength:

$$\lambda_{\max} = \frac{348.7 \text{ m/s}}{20 \text{ Hz}} = 17 \text{ m}.$$

5. Enter the speed and the maximum frequency to give the minimum wavelength:

$$\lambda_{\min} = \frac{348.7 \text{ m/s}}{20,000 \text{ Hz}} = 0.017 \text{ m} = 1.7 \text{ cm}.$$

CHECK YOUR UNDERSTANDING

- Imagine you observe two fireworks explode. You hear the explosion of one as soon as you see it. However, you see the other firework for several milliseconds before you hear the explosion. Explain why this is so.

Your answer:

- You observe two musical instruments that you cannot identify. One plays high-pitch sounds and the other plays low-pitch sounds. How could you determine which is which without hearing either of them play?

Your answer:

12.3 Sound Intensity and Sound Level

In a quiet forest, you can sometimes hear a single leaf fall to the ground. After settling into bed, you may hear your blood pulsing through your ears. However, when a passing motorist has his stereo turned up, you cannot even hear what the person next to you in your car is saying. We are all very familiar with the loudness of sounds and aware that they are related to how energetically the source is vibrating. In cartoons depicting a screaming person (or an animal making a loud noise), the cartoonist often shows an open mouth with a vibrating uvula, the hanging tissue at the back of the mouth, to suggest a loud sound coming from the throat. Figure below shows that. High noise exposure is hazardous to hearing, and it is common for musicians to have hearing losses that are sufficiently severe that they interfere with the musicians' abilities to perform. The relevant physical quantity is sound intensity, a concept that is valid for all sounds whether they are in the audible range. Intensity is defined to be the

power per unit area carried by a wave. Power is the rate at which energy is transferred by the wave. In equation form, **intensity** is

$$I = \frac{P}{A},$$

Where ***P*** is the power through an area ***A***. The SI unit for ***I*** is W / m². The intensity of a sound wave is related to its amplitude squared by the following relationship:

$$I = \frac{(\Delta p)^2}{2\rho v_w}.$$

Here Δp is the pressure variation or pressure amplitude (half the difference between the maximum and minimum pressure in the sound wave) in units of pascals (Pa) or N / m². (We are using a lower case ***p*** for pressure to distinguish it from power, denoted ***P*** by above.) The energy (as kinetic energy $\frac{1}{2}mv^2$) of an oscillating element of air due to a traveling sound wave is proportional to its amplitude squared. In this equation, ***ρ*** is the density of the material in which the sound wave travels, in units of kg / m³, and ***v_w*** is the speed of sound in the medium, in units of m/s. The pressure variation is proportional to the amplitude of the oscillation, and so ***I*** varies as $(\Delta p)^2$ (Figure below). This relationship is consistent with the fact that the sound wave is produced by some vibration; the greater its pressure amplitude, the more the air is compressed in the sound it creates.

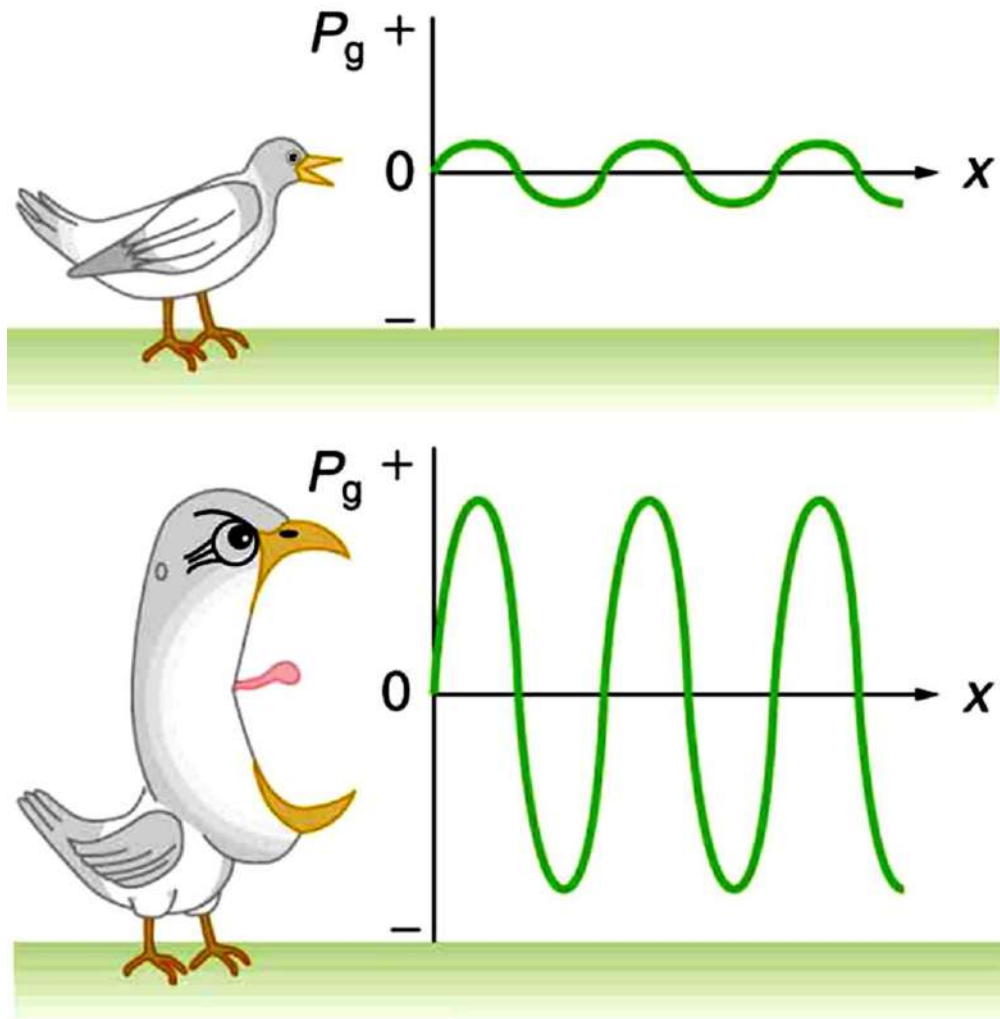


Figure reveals Graphs of the gauge pressures in two sound waves of different intensities. The more intense sound is produced by a source that has larger-amplitude oscillations and has greater pressure maxima and minima. Because pressures are higher in the greater-intensity sound, it can exert larger forces on the objects it encounters.

Sound intensity levels are quoted in decibels (dB) much more often than sound intensities in watts per meter squared. Decibels are the unit of choice in the scientific literature as well as in the popular media. The reasons for this choice of units are related to how we perceive sounds. How our ears perceive sound can be more accurately described by the logarithm of the intensity rather than directly to the intensity. The **sound intensity level** β in decibels of a sound having an intensity I in watts per meter squared is defined to be

$$\beta \text{ (dB)} = 10 \log_{10} \left(\frac{I}{I_0} \right),$$

Where $I_0=10^{-12} \text{ W/m}^2$ is a reference intensity. In particular, I_0 is the lowest or threshold intensity of sound a person with normal hearing can perceive at a frequency of 1000 Hz. Sound intensity level is not the same as intensity. Because β is defined in terms of a ratio, it is a unitless quantity telling you the level of the sound relative to a fixed standard (10^{-12} W/m^2 , in this case). The units of decibels (dB) are used to indicate this ratio is multiplied by 10 in its definition. The bel, upon which the decibel is based, is named for Alexander Graham Bell, the inventor of the telephone.

Table shows Sound Intensity Levels and Intensities.

Sound intensity level β (dB)	Intensity I (W/m ²)	Example/effect
0	1×10^{-12}	Threshold of hearing at 1000 Hz
10	1×10^{-11}	Rustle of leaves
20	1×10^{-10}	Whisper at 1 m distance
30	1×10^{-9}	Quiet home
40	1×10^{-8}	Average home
50	1×10^{-7}	Average office, soft music
60	1×10^{-6}	Normal conversation
70	1×10^{-5}	Noisy office, busy traffic
80	1×10^{-4}	Loud radio, classroom lecture
90	1×10^{-3}	Inside a heavy truck; damage from prolonged exposure ¹
100	1×10^{-2}	Noisy factory, siren at 30 m; damage from 8 h per day exposure
110	1×10^{-1}	Damage from 30 min per day exposure
120	1	Loud rock concert, pneumatic chipper at 2 m; threshold of pain
140	1×10^2	Jet airplane at 30 m; severe pain, damage in seconds
160	1×10^4	Bursting of eardrums

The decibel level of a sound having the threshold intensity of 10^{-12} W/m^2 is $\beta = 0 \text{ dB}$, because $\log_{10} 1 = 0$. That is, the threshold of hearing is 0 decibels. [Table above](#) gives levels in decibels and intensities in watts per meter squared for some familiar sounds.

One of the more striking things about the intensities in [Table above](#) is that the intensity in watts per meter squared is quite small for most sounds. The ear is sensitive to as little as a trillionth of a watt per meter squared –even more impressive when you realize that the area of the eardrum is only about 1 cm^2 , so that only 10^{-16} W falls on it at the threshold of hearing! Air molecules in a sound wave of this intensity vibrate over a distance of less than one molecular diameter, and the gauge pressures involved are less than 10^{-9} atm . Another impressive feature of the sounds in [Table above](#) is their numerical range. Sound intensity varies by a factor of from threshold to a sound that causes damage in seconds. You are unaware of this tremendous range in sound intensity because how your ears respond can be described approximately as the logarithm of intensity. Thus, sound intensity levels in decibels fit your experience better than intensities in watts per meter squared. The decibel scale is also easier to relate to because most people are more accustomed to dealing with numbers such as 0, 53, or 120 than numbers such as 1.00×10^{-11} . One more observation readily verified by examining [Table above](#) or using $(I = (\Delta p)^2 / 2\rho v_w)$ is that each factor of 10 in intensity corresponds to 10 dB. For example, a 90 dB sound compared with a 60 dB sound is 30 dB greater, or three factors of 10 (that is, 10^3 times) as intense. Another example is that if one sound is 10^7 as intense as another as, it is 70 dB higher. See [Table below](#).

Table reveals Ratios of Intensities and Corresponding Differences in Sound Intensity Levels.

I_2/I_1	$\beta_2 - \beta_1$
2.0	3.0 dB
5.0	7.0 dB
10.0	10.0 dB

Example: Calculating Sound Intensity Levels: Sound Waves

Calculate the sound intensity level in decibels for a sound wave traveling in air at 0 °C and having a pressure amplitude of 0.656 Pa.

Strategy

We are given Δp , so we can calculate I using the equation $I = (\Delta p)^2 / (2\rho v_w)^2$. Using I , we can calculate β straight from its definition in $\beta \text{ (dB)} = 10 \log_{10}(I/I_0)$.

Solution

(1) Identify knowns:

Sound travels at 331 m/s in air at 0°C.

Air has a density of 1.29 kg/m³ at atmospheric pressure and 0°C.

(2) Enter these values and the pressure amplitude into $I = (\Delta p)^2 / (2\rho v_w)$:

$$I = \frac{(\Delta p)^2}{2\rho v_w} = \frac{(0.656 \text{ Pa})^2}{2 (1.29 \text{ kg/m}^3) (331 \text{ m/s})} = 5.04 \times 10^{-4} \text{ W/m}^2.$$

(3) Enter the value for I and the known value for I_0 into $\beta \text{ (dB)} = 10 \log_{10}(I/I_0)$. Calculate to find the sound intensity level in decibels:

$$10 \log_{10}(5.04 \times 10^8) = 10 (8.70) \text{ dB} = 87 \text{ dB}.$$

Example: Change Intensity Levels of a Sound: What Happens to the Decibel Level?

Show that if one sound is twice as intense as another, it has a sound level about 3 dB higher.

Strategy

You are given that the ratio of two intensities is 2 to 1, and are then asked to find the difference in their sound levels in dB. You can solve this problem using the properties of logarithms.

Solution

(1) Identify knowns:

The ratio of the two intensities is 2 to 1, or:

$$\frac{I_2}{I_1} = 2.00.$$

We wish to show that the difference in sound levels is about 3 dB. That is, we want to show:

$$\beta_2 - \beta_1 = 3 \text{ dB}.$$

Note that:

$$\log_{10} b - \log_{10} a = \log_{10} \left(\frac{b}{a} \right).$$

(2) Use the definition of β to get:

$$\beta_2 - \beta_1 = 10 \log_{10} \left(\frac{I_2}{I_1} \right) = 10 \log_{10} 2.00 = 10 (0.301) \text{ dB}.$$

Thus,

$$\beta_2 - \beta_1 = 3.01 \text{ dB}.$$

It should be noted at this point that there is another decibel scale in use, called the **sound pressure level**, based on the ratio of the pressure amplitude to a reference pressure. This scale is used particularly in applications where sound travels in water. It is beyond the scope of most introductory texts to treat this scale because it is not commonly used for sounds in air, but it is important to note that very different decibel levels may be encountered when sound

pressure levels are quoted. For example, ocean noise pollution produced by ships may be as great as 200 dB expressed in the sound pressure level, where the more familiar sound intensity level we use here would be something under 140 dB for the same sound.

CHECK YOUR UNDERSTANDING

- Describe how amplitude is related to the loudness of a sound.
- Identify common sounds at the levels of 10 dB, 50 dB, and 100 dB.

12.4 Hearing

The human ear has a tremendous range and sensitivity. It can give us a wealth of simple information—such as pitch, loudness, and direction. In addition, from its input we can detect musical quality and nuances of voiced emotion. How is our hearing related to the physical qualities of sound, and how does the hearing mechanism work?

Hearing is the perception of sound. (Perception is commonly defined to be awareness through the senses, a typically circular definition of higher-level processes in living organisms.) Normal human hearing encompasses frequencies from 20 to 20,000 Hz, an impressive range. Sounds below 20 Hz are called **infrasound**, whereas those above 20,000 Hz are **ultrasound**. Neither is perceived by the ear, although infrasound can sometimes be felt as vibrations. When we do hear low-frequency vibrations, such as the sounds of a diving board, we hear the individual vibrations only because there are higher-frequency sounds in each. Other animals have hearing ranges different

from that of humans. Dogs can hear sounds as high as 30,000 Hz, whereas bats and dolphins can hear up to 100,000-Hz sounds. You may have noticed that dogs respond to the sound of a dog whistle, which produces sound out of the range of human hearing. Elephants are known to respond to frequencies below 20 Hz. The perception of frequency is called **pitch**. Most of us have excellent relative pitch, which means that we can tell whether one sound has a different frequency from another. Typically, we can discriminate between two sounds if their frequencies differ by 0.3% or more. For example, 500.0 and 501.5 Hz are noticeably different. Pitch perception is directly related to frequency and is not greatly affected by other physical quantities such as intensity. Musical **notes** are particular sounds that can be produced by most instruments and in Western music have particular names. Combinations of notes constitute music. Some people can identify musical notes, such as A-sharp, C, or E-flat, just by listening to them. This uncommon ability is called perfect pitch. The ear is remarkably sensitive to low-intensity sounds. The lowest audible intensity or threshold is about 0 dB. Sounds as much as more intense can be briefly tolerated. Very few measuring devices are capable of observations over a range of a trillion. The perception of intensity is called **loudness**. At a given frequency, it is possible to discern differences of about 1 dB, and a change of 3 dB is easily noticed. However, loudness is not related to intensity alone. Frequency has a major effect on

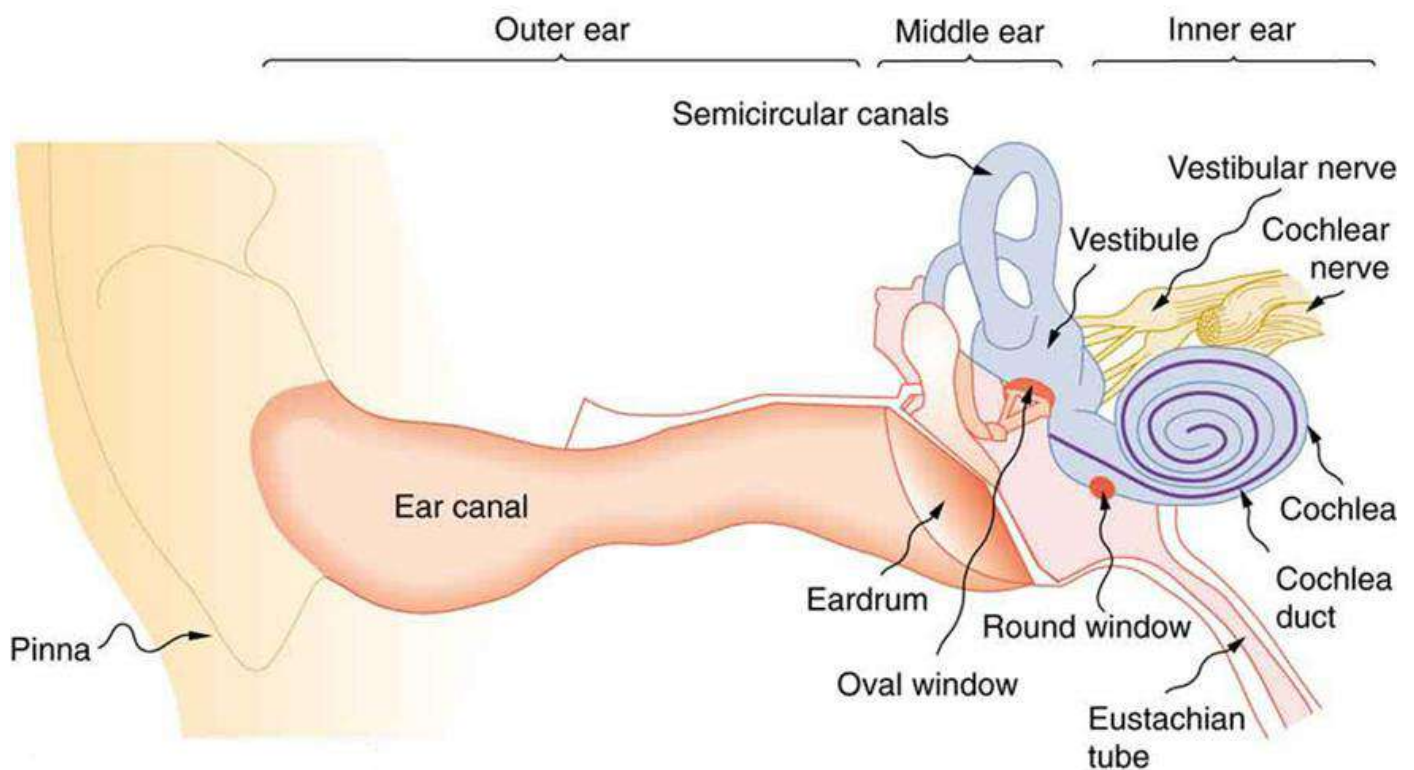
how loud a sound seems. The ear has its maximum sensitivity to frequencies in the range of 2000 to 5000 Hz, so that sounds in this range are perceived as being louder than, say, those at 500 or 10,000 Hz, even when they all have the same intensity. Sounds near the high- and low-frequency extremes of the hearing range seem even less loud, because the ear is even less sensitive at those frequencies. [Table below](#) gives the dependence of certain human hearing perceptions on physical quantities.

-Table Reveals Sound Perceptions

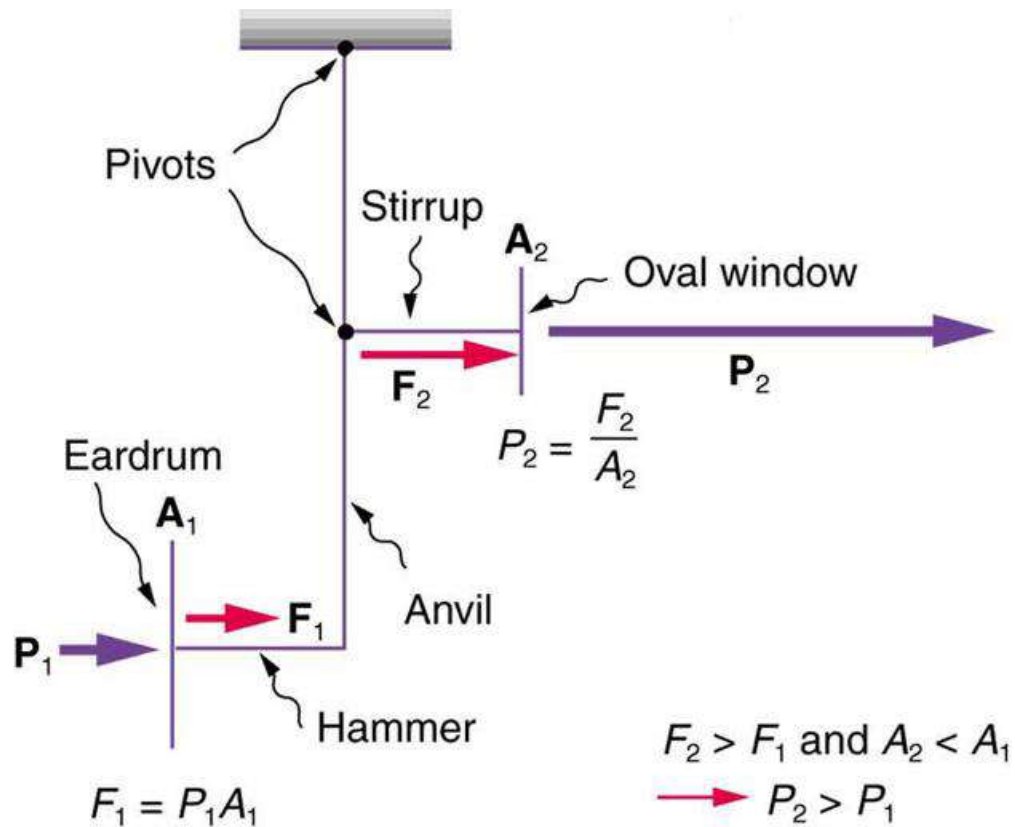
Perception	Physical quantity
Pitch	Frequency
Loudness	Intensity and Frequency
Timbre	Number and relative intensity of multiple frequencies. Subtle craftsmanship leads to non-linear effects and more detail.
Note	Basic unit of music with specific names, combined to generate tunes
Tone	Number and relative intensity of multiple frequencies.

12.5 The Hearing Mechanism

The hearing mechanism involves some interesting physics. The sound wave that impinges upon our ear is a pressure wave. The ear is a transducer that converts sound waves into electrical nerve impulses in a manner much more sophisticated than, but analogous to, a microphone. [Figure below](#) shows the gross anatomy of the ear with its division into three parts: the outer ear or ear canal; the middle ear, which runs from the eardrum to the cochlea; and the inner ear, which is the cochlea itself. The body part normally referred to as the ear is technically called the pinna.



The outer ear, or ear canal, carries sound to the recessed protected eardrum. The air column in the ear canal resonates and is partially responsible for the sensitivity of the ear to sounds in the 2000 to 5000 Hz range. The middle ear converts sound into mechanical vibrations and applies these vibrations to the cochlea. The lever system of the middle ear takes the force exerted on the eardrum by sound pressure variations, amplifies it and transmits it to the inner ear via the oval window, creating pressure waves in the cochlea approximately 40 times greater than those impinging on the eardrum. (See [Figure below](#).) Two muscles in the middle ear (not shown) protect the inner ear from very intense sounds. They react to intense sound in a few milliseconds and reduce the force transmitted to the cochlea. This protective reaction can also be triggered by your own voice, so that humming while shooting a gun, for example, can reduce noise damage.



This schematic shows the middle ear's system for converting sound pressure into force, increasing that force through a lever system, and applying the increased force to a small area of the cochlea, thereby creating a pressure about 40 times that in the original sound wave. A protective muscle reaction to intense sounds greatly reduces the mechanical advantage of the lever system.

Figure below shows the middle and inner ear in greater detail. Pressure waves moving through the cochlea cause the tectorial membrane to vibrate, rubbing cilia (called hair cells), which stimulate nerves that send electrical signals to the brain. The membrane resonates at different positions for different frequencies, with high frequencies stimulating nerves at the near end and low frequencies at the far end. The complete operation of the cochlea is still not understood, but several mechanisms for sending information to the brain are known to be involved. For sounds below about 1000 Hz, the nerves send signals at the same frequency as the sound. For frequencies greater than about 1000 Hz, the nerves signal frequency by position. There is a structure to the cilia, and there are connections between nerve cells that perform signal processing before information is sent to the brain. Intensity information is partly indicated by the number of nerve signals and by volleys of signals. The brain processes the cochlear nerve signals to provide additional information such as source direction (based on time and intensity comparisons of sounds from both ears). Higher-level processing produces many nuances, such as music appreciation.

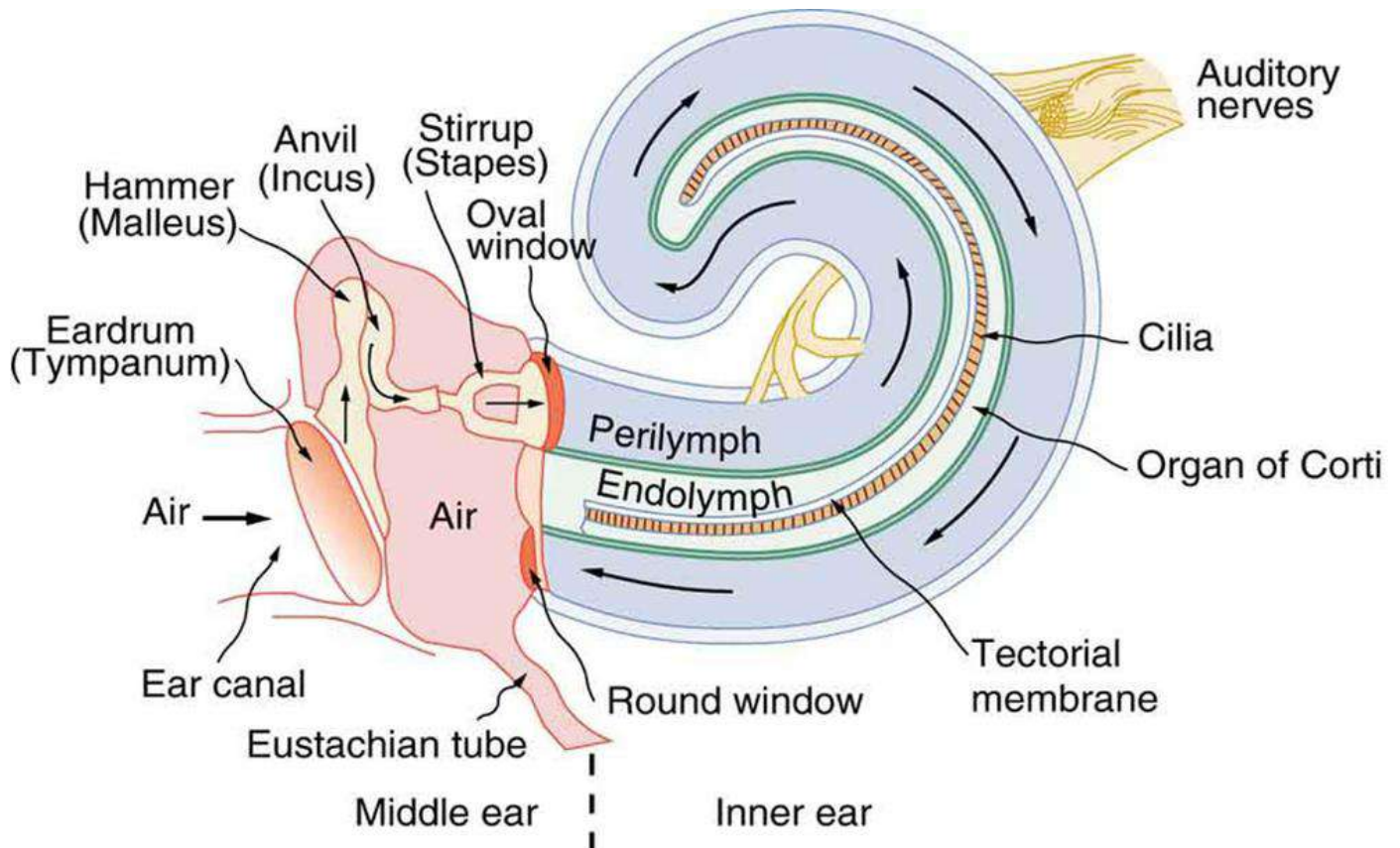


Figure shows: The inner ear, or cochlea, is a coiled tube about 3 mm in diameter and 3 cm in length if uncoiled. When the oval window is forced inward, as shown, a pressure wave travels through the perilymph in the direction of the arrows, stimulating nerves at the base of cilia in the organ of Corti.

Hearing losses can occur because of problems in the middle or inner ear. Conductive losses in the middle ear can be partially overcome by sending sound vibrations to the cochlea through the skull. Hearing aids for this purpose usually press against the bone behind the ear, rather than simply

amplifying the sound sent into the ear canal as many hearing aids do. Damage to the nerves in the cochlea is not repairable, but amplification can partially compensate. There is a risk that amplification will produce further damage. Another common failure in the cochlea is damage or loss of the cilia but with nerves remaining functional. Cochlear implants that stimulate the nerves directly are now available and widely accepted. Over 100,000 implants are in use, in about equal numbers of adults and children. The cochlear implant was pioneered in Melbourne, Australia, by Graeme Clark in the 1970s for his deaf father. The implant consists of three external components and two internal components. The external components are a microphone for picking up sound and converting it into an electrical signal, a speech processor to select certain frequencies and a transmitter to transfer the signal to the internal components through electromagnetic induction. The internal components consist of a receiver/ transmitter secured in the bone beneath the skin, which converts the signals into electric impulses and sends them through an internal cable to the cochlea and an array of about 24 electrodes wound through the cochlea. These electrodes in turn send the impulses directly into the brain. The electrodes emulate the cilia.

Check Your Understanding

- Are ultrasound and infrasound imperceptible to all hearing organisms? Explain your answer.



*Ministry of Higher Education and Scientific Research
Middle Technical University
Electrical Engineering Technical College
Department of Cybersecurity Technology Engineering*



General Physics

Lecture Thirteen

First Stage

Magnetism

By

Lecturer: Dr. Thamir. Al-hamadany

13. Magnetism

13.1 Introduction to Magnetism:

People have been aware of magnets and magnetism for thousands of years. The earliest records date to well before the time of Christ, particularly in a region of Asia Minor called Magnesia (the name of this region is the source of words like magnetic). Magnetic rocks found in Magnesia, which is now part of western Turkey, stimulated interest during ancient times. A practical application for magnets was found later, when they were employed as navigational compasses. The use of magnets in compasses resulted not only in improved long-distance sailing, but also in the names of “north” and “south” being given to the two types of magnetic poles. Today magnetism plays many important roles in our lives. Physicists’ understanding of magnetism has enabled the development of technologies that affect our everyday lives. The iPod in your purse or backpack, for example, wouldn’t have been possible without the applications of magnetism and electricity on a small scale. The discovery that weak changes in a magnetic field in a thin film of iron and chromium could bring about much larger changes in electrical resistance was one of the first large successes of nanotechnology. The 2007 Nobel Prize in Physics went to Albert Fert from France and Peter Grunberg from Germany for this discovery of giant magnetoresistance and its applications to computer memory. All electric motors, with uses as diverse as powering refrigerators, starting cars, and moving elevators, contain magnets. Magnetism is used to explain atomic energy levels, cosmic rays, and charged particles trapped in the Van Allen belts. Once again, we will find all these disparate phenomena are linked by a small number of underlying physical principles.

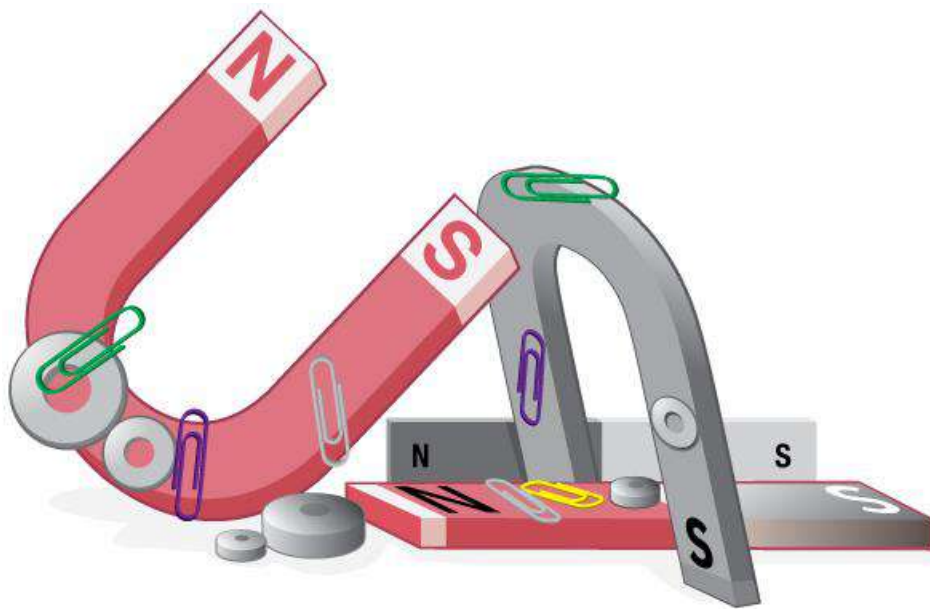


Figure shows Magnets come in various shapes, sizes, and strengths. All have both a north pole and a south pole. There is never an isolated pole (a monopole).

All magnets attract iron, such as that in a refrigerator door. However, magnets may attract or repel other magnets. Experimentation shows that all magnets have two poles. If freely suspended, one pole will point toward the north. The two poles are thus named the **north magnetic pole** and the **south magnetic pole** (or more properly, north-seeking and south seeking poles, for the attractions in those directions).

14.2 Current: The Source of All Magnetism

An electromagnet creates magnetism with an electric current. In later sections, we explore this more quantitatively, finding the strength and direction of magnetic fields created by various currents. However, what about ferromagnets? [Figure below a&b](#) shows models of how electric currents create magnetism at the submicroscopic level. (Note that we cannot directly observe the paths of individual electrons about atoms, and so a model or visual image, consistent with all direct observations, is made. We can directly observe the electron's orbital angular momentum, its spin momentum, and subsequent magnetic moments, all of which are explained with electric-current-creating subatomic magnetism.) Currents, including those associated with other submicroscopic particles like protons; allow us to explain ferromagnetism and all other magnetic effects. Ferromagnetism, for example, results from an internal cooperative alignment of electron spins, possible in some materials but not in others. Crucial to the statement that electric current is the source of all magnetism is the fact that it is impossible to separate north and south magnetic poles. (This is far different from the case of positive and negative charges, which are easily separated.) A current loop always produces a magnetic dipole—that is, a magnetic field that acts like a North Pole and South Pole pair. Since isolated north and south magnetic poles, called **magnetic monopoles**, are not observed, currents are used to explain all magnetic effects. If magnetic monopoles did exist, then we would have to modify this underlying connection that all magnetism is due to electrical current. There is no known reason that magnetic monopoles should not exist—they are simply never observed—and so searches at the subnuclear level continue. If they do not

exist, we would like to find out why not. If they do exist, we would like to see evidence of them.



Electric current is the source of all magnetism.

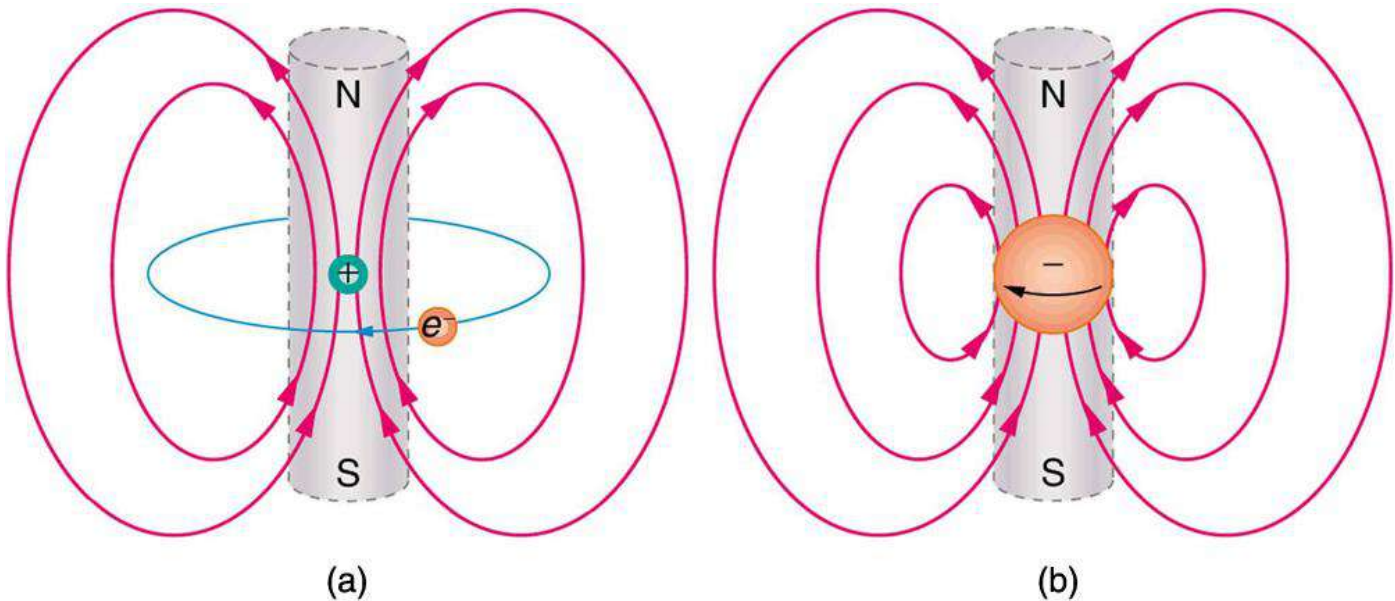


Figure display (a) In the planetary model of the atom, an electron orbits a nucleus, forming a closed-current loop and producing a magnetic field with a north pole and a south pole. (b) Electrons have spin and can be crudely pictured as rotating charge, forming a current that produces a magnetic field with a north pole and a south pole. Neither the planetary model nor the image of a spinning electron is completely consistent with modern physics. However, they do provide a useful way of understanding phenomena.

13.3 Universal Characteristics of Magnets and Magnetic Poles

It is a universal characteristic of all magnets that like poles repel and unlike poles attract. (Note the similarity with electrostatics: unlike charges attract and like charges repel.) Further experimentation shows that it is impossible to separate north and south poles in the manner that + and – charges can be separated.

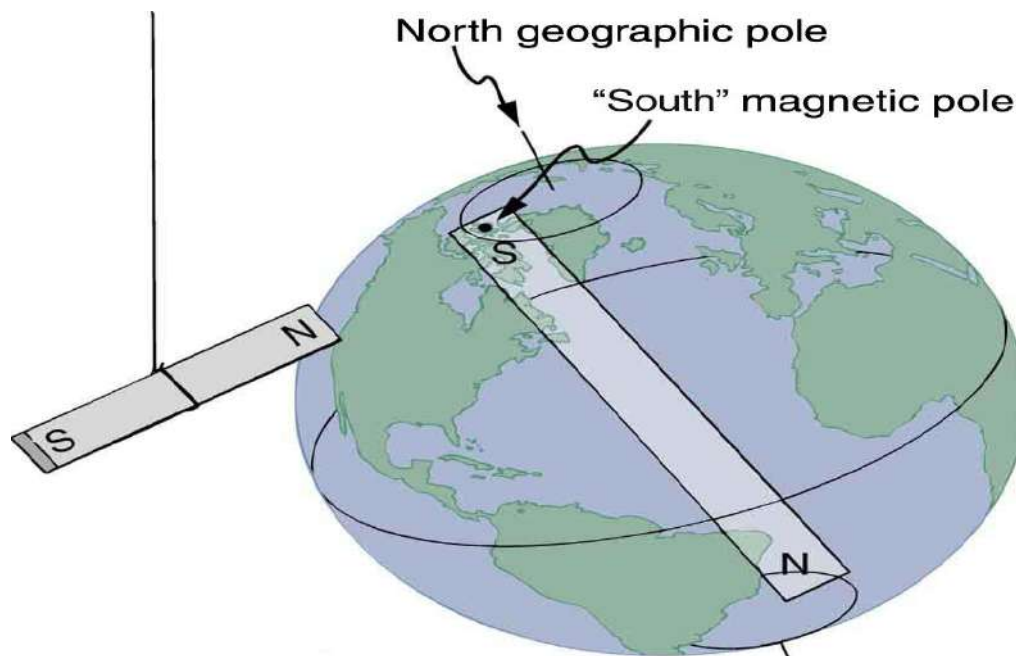


Figure display One end of a bar magnet is suspended from a thread that points toward north. The magnet's two poles are labeled N and S for north-seeking and south-seeking poles, respectively.

Earth acts like a very large bar magnet with its south-seeking pole near the geographic North Pole. That is why the north pole of your compass is attracted toward the geographic north pole of Earth—because the magnetic pole that is near the geographic North Pole is actually a south magnetic pole! Confusion arises because the geographic term “North Pole” has come to be used (incorrectly) for the magnetic pole that is near the North Pole. Thus, “north magnetic pole” is actually a misnomer—it should be called the south magnetic pole.

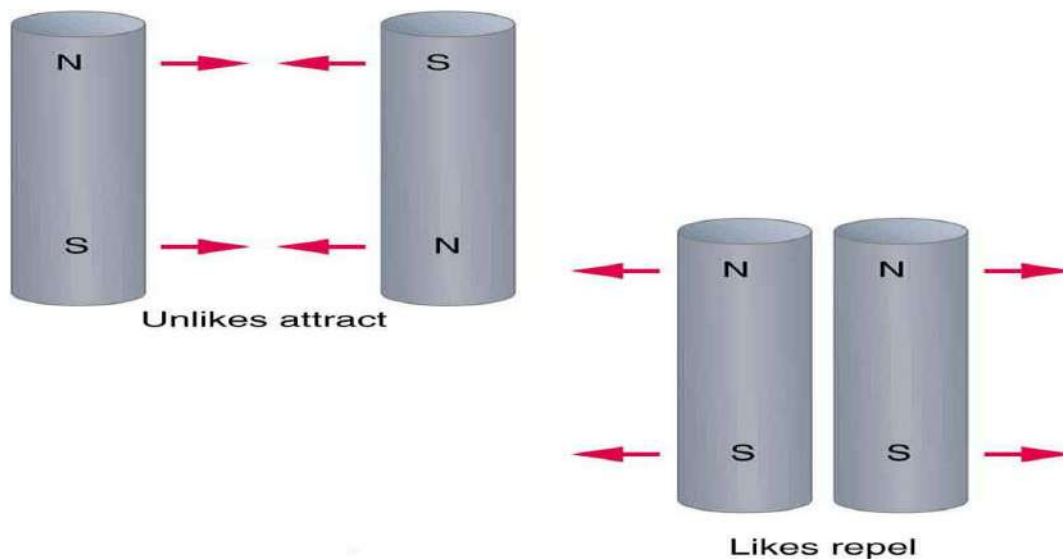


Figure reveals that unlike poles attract, whereas like poles repel.

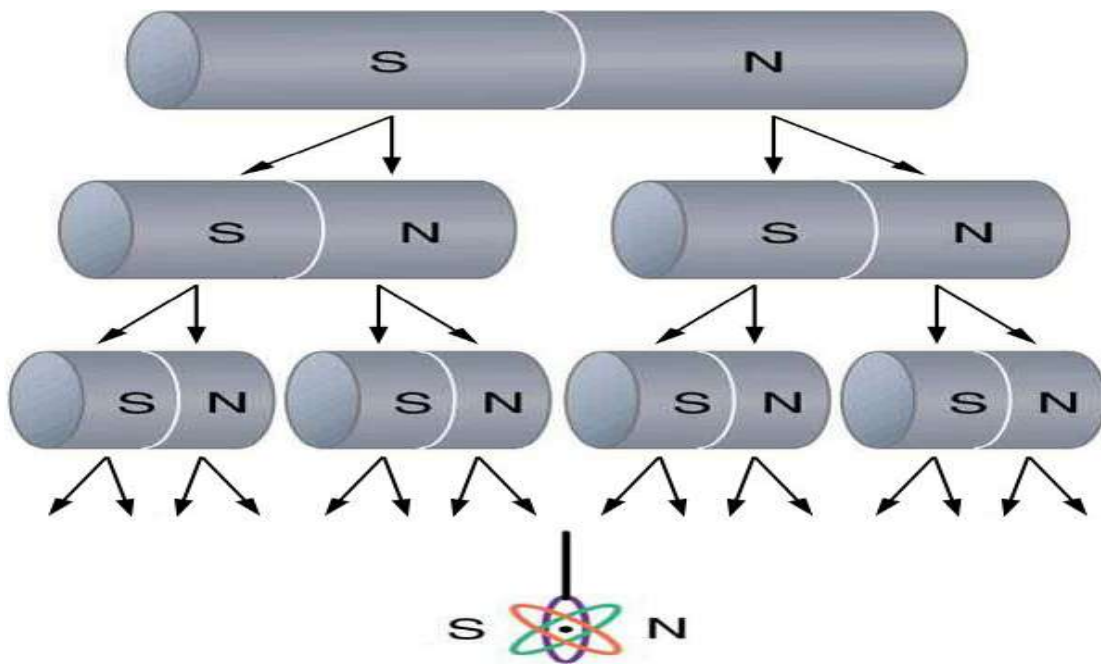


Figure shows North and south poles always occur in pairs. Attempts to separate them result in more pairs of poles. If we continue to split the magnet, we will eventually get down to an iron atom with a north pole and a south pole—these, too, cannot be separated.

The fact that magnetic poles always occur in pairs of north and south is true from the very large scale—for example, sunspots always occur in pairs that are north and south magnetic poles—all the way down to the very small scale. Magnetic atoms have both a north pole and a south pole, as do many types of subatomic particles, such as electrons, protons, and neutrons.

13.4 Types of Magnetism: There are Four Different Kinds of Magnetism:

13.4.1. Diamagnetism

A phenomenon in some materials in which the susceptibility is negative, i.e. the magnetization opposed the magnetizing force. It arises from the precession of spinning charges in a magnetic field. The magnetization is in the **opposite** direction to that of the applied field, i.e. the magnetic susceptibility is negative ($\chi_m < 0$). Although **all** substances are diamagnetic, it is a weak form of magnetism and may be masked by other, stronger forms. Diamagnetism results from changes induced in the orbits of electrons in the atoms of a substance by the applied field, the direction of change (in accordance with Lenz's law) opposing the applied magnetic flux. There is thus a weak negative susceptibility ($\chi_m < 0$) and a relative magnetic permeability of slightly less than one (i.e. $\mu_r = \mu / \mu_0 = (1 + \chi_m) < 1$). **Diamagnetism** is a very weak form of magnetism that is only exhibited in the presence of an external magnetic field. It is the result of changes in the orbital motion of electrons due to the external magnetic field. The induced magnetic moment is very small and in a direction opposite to that of the applied field. When placed between the poles of a strong electromagnet, diamagnetic materials are attracted towards regions where the magnetic field is weak. Diamagnetism is found in all materials; however, because it is so weak it can only be observed in materials that do not exhibit other forms of magnetism. In addition, diamagnetism is found in elements with paired electrons. Oxygen was once thought to be diamagnetic, but a new revised molecular orbital (MO) model confirmed oxygen's paramagnetic nature. An exception to the "weak" nature of diamagnetism occurs with the rather large number of materials that become superconducting, something that usually happens at lowered temperatures. Superconductors are perfect diamagnets and

when placed in an external magnetic field expel the field lines from their interiors (depending on field intensity and temperature). Superconductors also have zero electrical resistance, a consequence of their diamagnetism. Superconducting structures have been known to tear themselves apart with astonishing force in their attempt to escape an external field. Superconducting magnets are the major component of most magnetic resonance imaging systems, perhaps the only important application of diamagnetism. Perhaps the substance that displays the strongest diamagnetism is bismuth, which is used in guns. Melting down bismuth and then molding, it is a very efficient way of capturing the diamagnetic properties. A thin slice of pyrolytic graphite, which is an unusually strongly diamagnetic material, can be stably floated on a magnetic field, such as that from rare earth permanent magnets. This can be done with all components at room temperature, making a visually effective demonstration of diamagnetism. The Radboud University Nijmegen has conducted experiments where they have successfully levitated water and a live frog. Diamagnetic materials have a relative magnetic permeability that is less than 1, and a magnetic susceptibility that is less than 0. (i.e. $\mu_r = \mu / \mu_0 = (1 + \chi_m) < 1$ and $\chi_m < 0$). Diamagnetism was discovered and named in September 1845 by Michael Faraday.

13.4. 2. Paramagnetism

In paramagnetism, the atoms or molecules of the substance have net orbital or spin magnetic moments that are capable of being aligned in the direction of the applied field. They therefore have a positive (but small) susceptibility and a relative permeability slightly in excess of one. Paramagnetism occurs in all atoms and molecules with unpaired electrons; e.g. free atoms, free radicals, and compounds of transition metals containing ions with unfilled electron shells.

It also occurs in metals as a result of the magnetic moments associated with the spins of the conducting electrons. **Paramagnetism** is the tendency of the atomic magnetic dipoles, due to quantum-mechanical spin angular momentum, in a material that is otherwise non-magnetic to align with an external magnetic field. This alignment of the atomic dipoles with the magnetic field tends to strengthen it, and is described by a relative magnetic permeability, μ_r greater than unity (or, equivalently, a small positive magnetic susceptibility greater than zero), i.e. (i.e. $\mu_r = \mu / \mu_0 = (1 + \chi_m) > 1$ and $\chi_m > 0$). In pure paramagnetism, the external magnetic field acts on each atomic dipole independently and there are no interactions between individual atomic dipoles. Such paramagnetic behavior can also be observed in ferromagnetic materials that are above their Curie temperature. Paramagnetic materials attract and repel like normal magnets when subject to a magnetic field. Under relatively low magnetic field saturation when the majority of the atomic dipoles are not aligned with the field, paramagnetic materials exhibit magnetization according to **Curie's Law**: $M = C \cdot B/T$, where **M** is the resulting magnetization (magnetic dipole moment/unit volume), measured in Amps/meter. **B** is the magnetic flux density of the applied field, measured in Teslas. **T** is absolute temperature, measured in Kelvins. **C** is a material-specific Curie constant this law indicates that paramagnetic materials tend to become increasingly magnetic, as the applied magnetic field is increased, but less magnetic as temperature is increased. Curie's law is incomplete because it fails to predict what will happen when most of the little magnets are aligned (after everything is aligned, increasing the external field will not increase the total magnetization) so Curie's Constant really should be expressed as a function of how much of the material is already aligned.

Paramagnetic materials in magnetic fields will act like magnets but when the field is removed, thermal motion will quickly disrupt the magnetic alignment. In general, paramagnetic effects are small (magnetic susceptibility of the order of $\chi_m \sim 10^{-3}$ to 10^{-5}). Ferromagnetic materials above the Curie temperature become paramagnetic.

Paramagnetic Materials

- Aluminum
- Barium
- Calcium
- Liquid Oxygen
- Platinum
- Sodium
- Strontium
- Uranium

13.4.3. Ferromagnetism

A phenomenon in some magnetically ordered materials in which there is a bulk magnetic moment and the magnetization is large. The electron spins of the atoms in the microscopic regions, domains, are aligned. In the presence of an external magnetic field, the domains oriented favorably with respect to the field grow at the expense of the others and the magnetization of the domains tends to align with the field. Above the Curie temperature, the thermal motion is sufficient to offset the aligning force and the material becomes paramagnetic. Certain elements (iron, nickel and cobalt), and alloys with other elements (titanium, aluminium) exhibit relative magnetic permeabilities up to 10^4 (ferromagnetic materials). Some show marked hysteresis and are used for permanent magnets, magnetic amplifiers etc. In ferromagnetic substances, within a certain temperature range, there are net

atomic magnetic moments, which line up in such a way that magnetization persists after the removal of the applied field. Below a certain temperature, called the Curie point (or Curie temperature) an increasing magnetic field applied to a ferromagnetic substance will cause increasing magnetization to a high value called the saturation magnetization. This is because a ferromagnetic substance consists of small-magnetized regions called domains. The total magnetic moment of a sample of the substance is the vector sum of the magnetic moments of the component domains. **Ferromagnetism** is a phenomenon by which a material can exhibit a spontaneous magnetization, and is one of the strongest forms of magnetism. It is responsible for most of the magnetic behavior encountered in everyday life, and is the basis for all permanent magnets (as well as the metals that are noticeably attracted to them).

Ferromagnetic materials

There are a number of crystalline materials that exhibit ferromagnetism. We list a representative selection of them here (Kittel, p. 449), along with their Curie temperatures, the temperature above which they cease to be ferromagnetic (see right hand table - A selection of crystalline ferromagnetic materials, along with their Curie temperatures in kelvins (K).) One can also make amorphous (non-crystalline) ferromagnetic metallic alloys by very rapid quenching (cooling) of a liquid alloy. These have the advantage that their properties are nearly isotropic (not aligned along a crystal axis); this results in low coercivity, low hysteresis loss, high permeability, and high electrical resistivity. A typical such material is a transition metal-metalloid alloy, made from about 80% transition metal (usually Fe, Co, or Ni) and a metalloid component (B, C, Si, P, or Al) that lowers the melting point.

One example of such an amorphous alloy is $\text{Fe}_{80}\text{B}_{20}$ (Metglas 2605) which has a Curie temperature of 647 K and a room temperature (300 K) saturation magnetization of 125.7 milliteslas (1257 gauss), compared with 1043 K and 170.7 mT (1707 gauss) for pure iron from above. The melting point, or more precisely the glass transition temperature, is only 714 K for the alloy versus 1811 K for pure iron.

Physical origin

The property of ferromagnetism is due to the direct influence of two effects from quantum mechanics: spin and the Pauli Exclusion Principle. The intrinsic spin angular momentum of an electron has a magnetic dipole moment associated with it, and it creates a magnetic dipole field. (The classical analog of quantum-mechanical spin is a spinning ball of charge; however the quantum version has distinct differences, such as the fact that it has discrete up/down states that are not described by a vector.) In many materials (specifically those with a filled electron shell), the electrons come in pairs of opposite spin, which cancel one another's dipole moments. Only atoms with unpaired electrons (partially filled shells) can experience a net magnetic moment from spin. A ferromagnetic material has many such electrons, and if they are aligned, they create a measurable macroscopic field. The spins/dipoles tend to align in parallel to an external magnetic field, an effect called paramagnetism. (A similar effect due to the orbital motion of the electrons, which effectively forms a microscopic current loop that also has a magnetic dipole moment, is called diamagnetism.) Ferromagnetism involves an additional phenomenon, however: the spins tend to *align spontaneously*, without any applied field. This is a purely quantum-mechanical effect. According to classical electromagnetism, two nearby magnetic dipoles will tend to align in *opposite* directions (which would create an

antiferromagnetic material). In a ferromagnet, however, they tend to align in the *same* direction because of the Pauli principle: two electrons with the same spin cannot lie at the same position, and thus feel an effective additional repulsion that lowers their electrostatic energy. This difference in energy is called the *exchange energy* and induces nearby electrons to align. At long distances (after many thousands of ions), the exchange energy advantage is overtaken by the classical tendency of dipoles to anti-align. This is why, in an equilibrated (non-magnetized) ferromagnetic material, the spins in the whole material are not aligned. Rather, they organize into *domains* that are aligned (magnetized) at short range, but at long range, adjacent domains are anti-aligned. The transition between two domains, where the magnetization flips, is called a Bloch wall, and is a gradual transition on the atomic scale (covering a distance of about 300 ions for iron). Thus, an ordinary piece of iron generally has little or no net magnetic moment. However, if it is placed in a strong enough external magnetic field, the domains will re-orient in parallel with that field, and will remain re-oriented when the field is turned off, thus creating a "permanent" magnet. This magnetization as a function of the external field is described by a hysteresis curve. Although this state of aligned domains is not a minimal-energy configuration, it is extremely stable and has been observed to persist for millions of years in seafloor magnetite aligned by the Earth's magnetic field (whose poles can thereby be seen to flip at long intervals). The net magnetization can be destroyed by heating and then cooling (*annealing*) the material without an external field, however. As the temperature increases, thermal oscillation, or entropy, competes with the ferromagnetic tendency for spins to align. When the temperature rises beyond a certain point, called the **Curie temperature**, there is a second-order phase transition and the system

can no longer maintain a spontaneous magnetization, although it still responds paramagnetically to an external field. Below that temperature, there is a spontaneous symmetry breaking and random domains form (in the absence of an external field). The Curie temperature itself is a critical point, where the magnetic susceptibility is theoretically infinite and, although there is no net magnetization, domain-like spin correlations fluctuate at all length scales. The study of ferromagnetic phase transitions, especially via the simplified Ising spin model, had an important impact on the development of statistical physics. There, it was first clearly shown that mean field theory approaches failed to predict the correct behavior at the critical point (which was found to fall under a *universality class* that includes many other systems, such as liquid-gas transitions), and had to be replaced by renormalization group theory.

Unusual Ferromagnetism

In 2004, it was discovered that a certain allotrope of carbon, nanofoam, exhibited ferromagnetism. The effect dissipates after a few hours at room temperature, but lasts longer at cold temperatures. The material is also a semiconductor. It is thought that other similarly formed materials, of boron and nitrogen, may also be ferromagnetic.

Material	Curie temp. (K)
Fe	1043
Co	1388
Ni	627
Gd	292
Dy	88
MnAs	318
MnBi	630
MnSb	587
CrO ₂	386
MnOFe ₂ O ₃	573
FeOFe ₂ O ₃	858
NiOFe ₂ O ₃	858
CuOFe ₂ O ₃	728
MgOFe ₂ O ₃	713
EuO	69
Y ₃ Fe ₅ O ₁₂	560

13.4.4 Antiferromagnetism

Phenomenon in some magnetically ordered materials in which there is an anti-parallel alignment of spins in two interpenetrating structures so that there is no overall bulk spontaneous magnetization. In ferromagnetic materials, it is energetically favorable for the spins atomic to align, leading to spontaneous magnetization. However, in antiferromagnetic materials, the conditions are such that it is energetically favorable for the spins to oppose, leading to no overall magnetization. In materials that exhibit **antiferromagnetism**, the spins of magnetic electrons align in a regular pattern with neighboring spins pointing in opposite directions. This is the opposite of ferromagnetism. Generally, antiferromagnetic materials exhibit antiferromagnetism at a low temperature, and become disordered above a certain temperature; the transition temperature is called the Neel temperature. Above the Neel temperature, the material is typically paramagnetic. The antiferromagnetic behaviour at low temperature usually results in diamagnetic properties, but can sometimes display ferrimagnetic behaviour, which in many physically observable properties are more similar to ferromagnetic interactions. The magnetic susceptibility, χ_m of an antiferromagnetic material will appear to go through a maximum as the temperature is lowered; in contrast, that of a paramagnet will continually increase with decreasing temperature. Antiferromagnetic materials have a negative coupling between adjacent moments and low frustration.

Antiferromagnetic materials are relatively uncommon. An example is the heavy-fermion superconductor URu_2Si_2 . There are also numerous examples among high nuclearity metal clusters. Antiferromagnetism, also known as Ferrimagnetism is a property exhibited by materials whose atoms or ions tend to assume an ordered but nonparallel arrangement in

zero applied field below a certain characteristic temperature known as the Neel temperature. In the usual case, within a magnetic domain, a substantial net magnetization results from the antiparallel alignment of neighboring nonequivalent sublattices. The macroscopic behavior is similar to ferromagnetism. Above the Neel temperature, these substances become paramagnetic.



Ministry of Higher Education and Scientific Research

Middle Technical University

Electrical Engineering Technical College

Department of Cybersecurity Technology Engineering



General Physics

Lecture Fourteen

First Stage

Electromagnetism

By

Lecturer: Dr. Thamir. Al-hamadany

14. Electromagnetism

14.1 Brief History of the Science of Electromagnetism

Electric and magnetic phenomena have been known for millennia. The earliest examples were forces produced by static electricity and by ferromagnetism. Of course, there are other phenomena that we recognize today to be of electromagnetic origin, which have been observed since the beginning of time. For example, lightning is an electric discharge. In addition, light consists of electromagnetic waves, or, in the quantum theory, photons. However, before the scientific revolution it was not recognized that these varied phenomena have a common origin. During the 19th century a unified theory was constructed, describing electricity, magnetism, and optics, based on the electromagnetic field. The word "electric" was coined by William Gilbert as a name for the phenomena of static electricity. The origin of the word is from the Greek "electron" (ηλεκτροο) which means amber. Amber is a fossilized plant resin, an electrical insulator which, when rubbed with animal fur, acquires the ability to attract small bits of matter such as straw or paper. The philosophers of ancient Greece knew this primitive example of static electricity. The word "magnetic" was also coined by Gilbert as a name for the phenomena of ferromagnetism. The origin of this word is the Greek place-name "Magnesia," which was a region of ancient Greece where magnetic iron ore (magnetite) occurs in the Earth. The forces between natural magnets were also known to the philosophers of ancient Greece. These earliest observed electric and magnetic phenomena must have seemed interesting (in the way of a mystery) to philosophers, but they had no practical importance. In contrast, the scientific study of electricity and magnetism during the past 400 years has dispelled much of the mystery, and created technological power that the ancients could not have imagined. The study of electricity and magnetism was an early part of the scientific revolution.

In 1600, Gilbert published an important book, *De Magnete*, one of the first scientific works on electricity and magnetism. It was Gilbert who demonstrated that a compass needle points north because the Earth itself is a large magnet exerting a magnetic force on the magnetized needle. Regarding electricity, Gilbert discovered that many substances besides amber exhibit static electricity from friction, including glass, sulphur, sealing wax, and gemstones.

14.2 Unification of Electricity and Magnetism

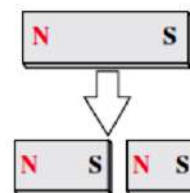
In 1820, during a public lecture on electric and magnet Oersted discovered that electric current in a metal wire produces deflection of a compass needle. This accidental discovery was the first indication that electricity and magnetism are related. In modern terms, Oersted had found that an electric current produces a magnetic field. Within a few days of the discovery, Biot and Savart had reproduced the effect in their laboratory, and found that the strength of the magnetic field varies inversely with the distance from the wire. As to direction, by tracing the magnetic force in the region of the wire they showed that the magnetic field curls around the current. Ampere studied in great detail how electric currents produce magnetic effects. His ultimate result was an equation for the action of one current element $I_1 d\ell_1$ on another $I_2 d\ell_2$; the force on the first element has the form

$$d\mathbf{F}_1 = C I_1 d\ell_1 \times (I_2 d\ell_2 \times \hat{\mathbf{r}}) / r^2$$

Where C is a constant that depends on the system of units. $\hat{\mathbf{r}}$ is the unit vector in the direction from $d\ell_2$ to $d\ell_1$ and r is the distance between them.) The student may recall that the cross product of two vectors points in a direction perpendicular to both. Ampere's result, which involves two cross products, anticipates that magnetic fields and forces may be a little tricky to figure out. If electric current produces magnetism, then a natural question is whether magnetism can produce an electric current. Faraday began experiments on electricity and magnetism in 1821, and he discovered electromagnetic induction in 1831. Magnetism does produce an electromotive force when the magnetic field is changing in time. The same effect was discovered at about the same time by Henry in America. After these observations relating electricity and magnetism, the challenge was to develop a theory that could account for all the effects in a unified way. The theory that finally succeeded was field theory.

14.3 Magnetic Poles versus Electric Charge

The concept of magnetic poles may appear similar to that of electric charge, and north and south poles may seem analogous to positive and negative charge. However, the analogy can be misleading. While isolated positive and negative charges exist, there is *no* experimental evidence that a single isolated magnetic pole exists; poles always appear in pairs. If a bar magnet is broken in two, each broken end becomes a pole as shown in figure below. The existence of an isolated magnetic pole, or **magnetic monopole**, would have sweeping implications for theoretical physics. Extensive searches for magnetic monopoles have been carried out, but so far without success.



14.4 Magnetic Field

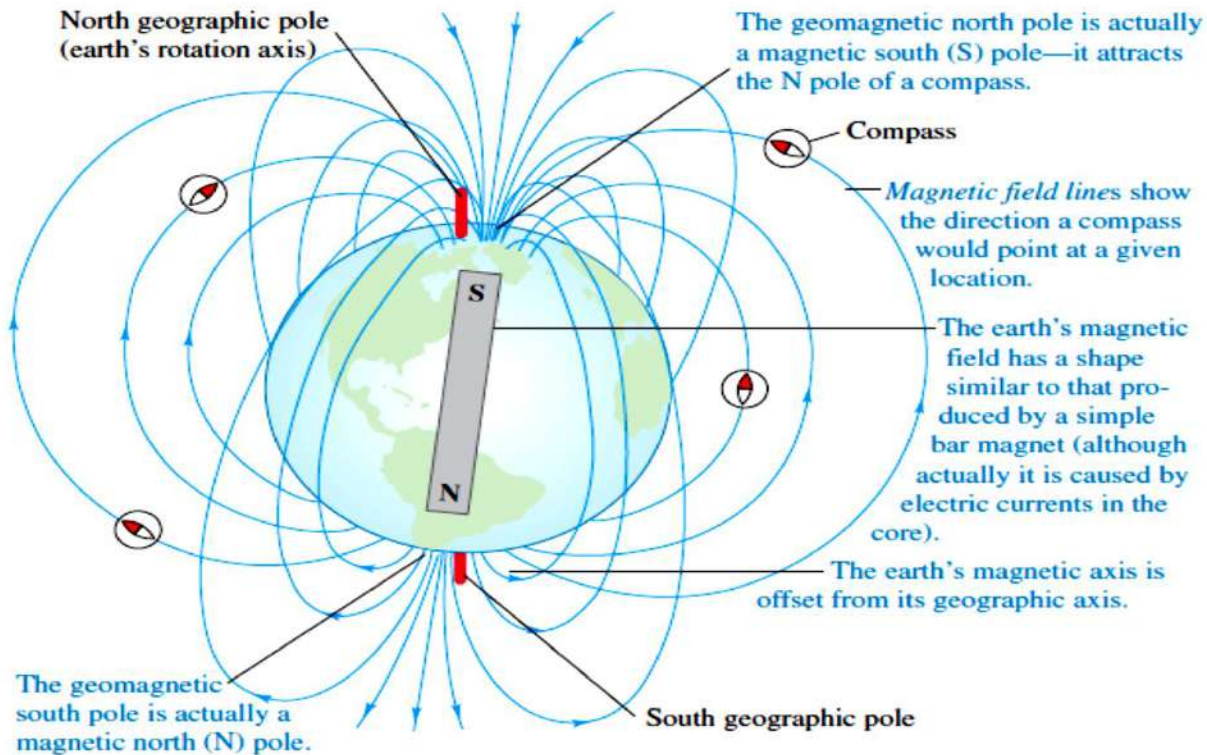
To introduce the concept of magnetic field properly, let's review our formulation of *electric* interactions, where we introduced the concept of *electric* field. We represented electric interactions in two steps:

1. A distribution of electric charge at rest creates an electric field in the surrounding space.
2. The electric field exerts a force $\vec{F} = q\vec{E}$ on any other charge that is present in the field.

We can describe magnetic interactions in a similar way:

1. A moving charge or a current creates a **magnetic field** in the surrounding space (in addition to its *electric* field).
2. The magnetic field exerts a force $\vec{F}$ on any other moving charge or current that is present in the field.

Like electric field, magnetic field is a **vector field**—that is, a vector quantity associated with each point in space. We will use the symbol $\vec{B}$ for magnetic field. At any position, the direction of $\vec{B}$ is defined as the direction in which the north pole of a compass needle tends to point. The arrows in Fig. below suggest the direction of the earth's magnetic field; for any magnet, $\vec{B}$ points out of its north pole and into its south pole. A sketch of the earth's magnetic field. The field, which is caused by currents in the earth's molten core, changes with time; geologic evidence shows that it reverses direction entirely at irregular intervals of 10^4 to 10^6 years.



14.5 Magnetic Forces on Moving Charges

There are four key characteristics of the magnetic force on a moving charge. **First**, its magnitude is proportional to the magnitude of the charge. If a $1\text{-}\mu\text{C}$ charge and a $2\text{-}\mu\text{C}$ charge move through a given magnetic field with the same velocity, experiments show that the force on the $2\text{-}\mu\text{C}$ charge is twice as great as the force on the $1\text{-}\mu\text{C}$ charge. **Second**, the magnitude of the force is also proportional to the magnitude, or “strength,” of the field; if we double the magnitude of the field (for example, by using two identical bar magnets instead of one) without changing the charge or its velocity, the force doubles. A **third** characteristic is that the magnetic force depends on the particle's velocity. This is quite different from the electric-field force, which is the same whether the charge is moving or not. A charged particle at rest

experiences *no* magnetic force, and [fourth](#), we find by experiment that the magnetic force $\vec{F}$ *does not* have the same direction as the magnetic field $\vec{B}$ but instead is always *perpendicular* to both $\vec{B}$ and the velocity $\vec{v}$. The magnitude F of the force is found to be proportional to the component of $\vec{v}$ perpendicular to the field; when that component is zero (that is, when $\vec{v}$ and $\vec{B}$ are parallel or antiparallel), the force is zero. Figure below shows these relationships. The direction of $\vec{F}$ is always perpendicular to the plane containing $\vec{v}$ and $\vec{B}$.